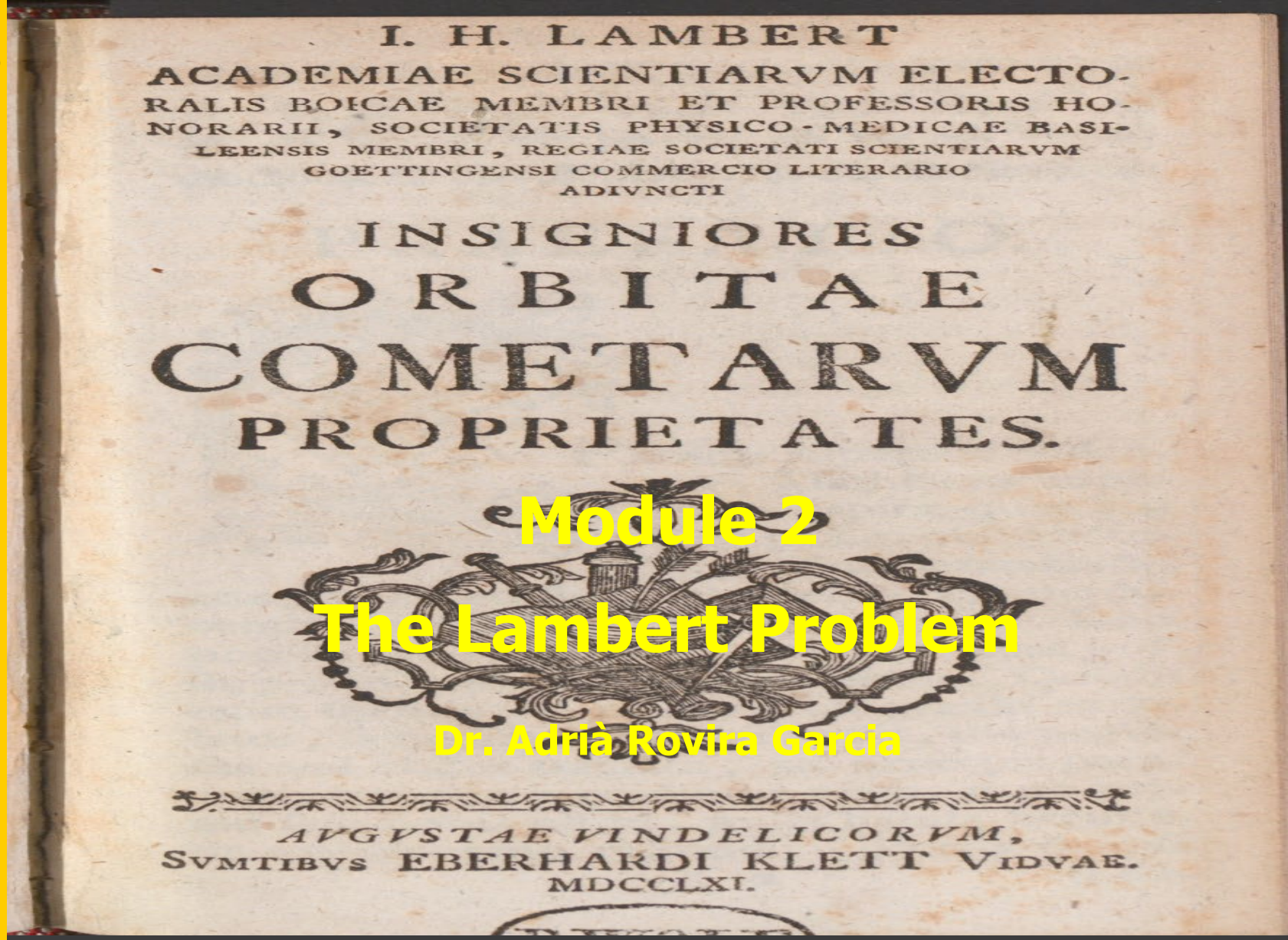
Escola Superior d'Enginyeries Industrial, Aeroespacial i Audiovisual de Terrassa (**ESEIAAT**)



Module 2
The Lambert Problem

Dr. Adrià Rovira Garcia

Johann Heinrich Lambert (1761) Insigniores Orbitae Cometarum Proprietates - Notable properties of comets' orbits (URL)

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Acknowledgement

The author expresses the gratitude on the tireless efforts of Dr. De la Torre in raising the education on the art and science of orbital mechanics by, among others, the design and implementation of extensive educational tools.

Module 2 Requirements

Objectives:

- Problem definition, statement of the theorem of Lambert, sketch of proof
- Numerical aspects of the problem
- Classical methods of solution (Lagrange, Gauss)
- Introduction to modern methods
- Pork Chop Plot (PCP)
- Application to the problem of interplanetary transfer

Time Allocation:

- 8 hours

Previous: the Lagrange Coefficients

If the position $\mathbf{r_0}$ and velocity $\mathbf{v_0}$ of an orbiting body are known at a given instant, the position $\mathbf{r}$ and velocity $\mathbf{v}$ at any later time are found in terms of:

$$\mathbf{r} = f \cdot \mathbf{r_0} + g \cdot \mathbf{v_0}$$
$$\mathbf{v} = \dot{f} \cdot \mathbf{r_0} + \dot{g} \cdot \mathbf{v_0}$$

where the **scalar Lagrange coefficients** $f, \dot{f}, g$ **and** $\dot{g}$ can be expressed in terms of the change in true anomaly $\Delta\theta$:

$$g = \frac{r r_0}{h} \sin \Delta\theta$$
$$\dot{g} = 1 - \frac{\mu r_0}{h^2}(1 - \cos \Delta\theta)$$

Remark: Only 3 out of 4 coefficients are independent

$$f = 1 - \frac{\mu r}{h^2}(1 - \cos \Delta\theta)$$
$$\dot{f} = \frac{\mu}{h} \frac{1 - \cos \Delta\theta}{\sin \Delta\theta} \left[\frac{\mu}{h^2}(1 - \cos \Delta\theta) - \frac{1}{r_0} - \frac{1}{r} \right]$$

We do not need to know the type of orbit (ellipse, parabola or hyperbola)

$$r = \frac{h^2}{\mu} \frac{1}{1 + \left(\frac{h^2}{\mu r_0} - 1 \right) \cos \Delta\theta - \frac{h}{\mu} \frac{\mathbf{v_0} \cdot \mathbf{r_0}}{r_0} \sin \Delta\theta}$$

$$h = |\mathbf{h}| = |\mathbf{r_0} \times \mathbf{v_0}|$$

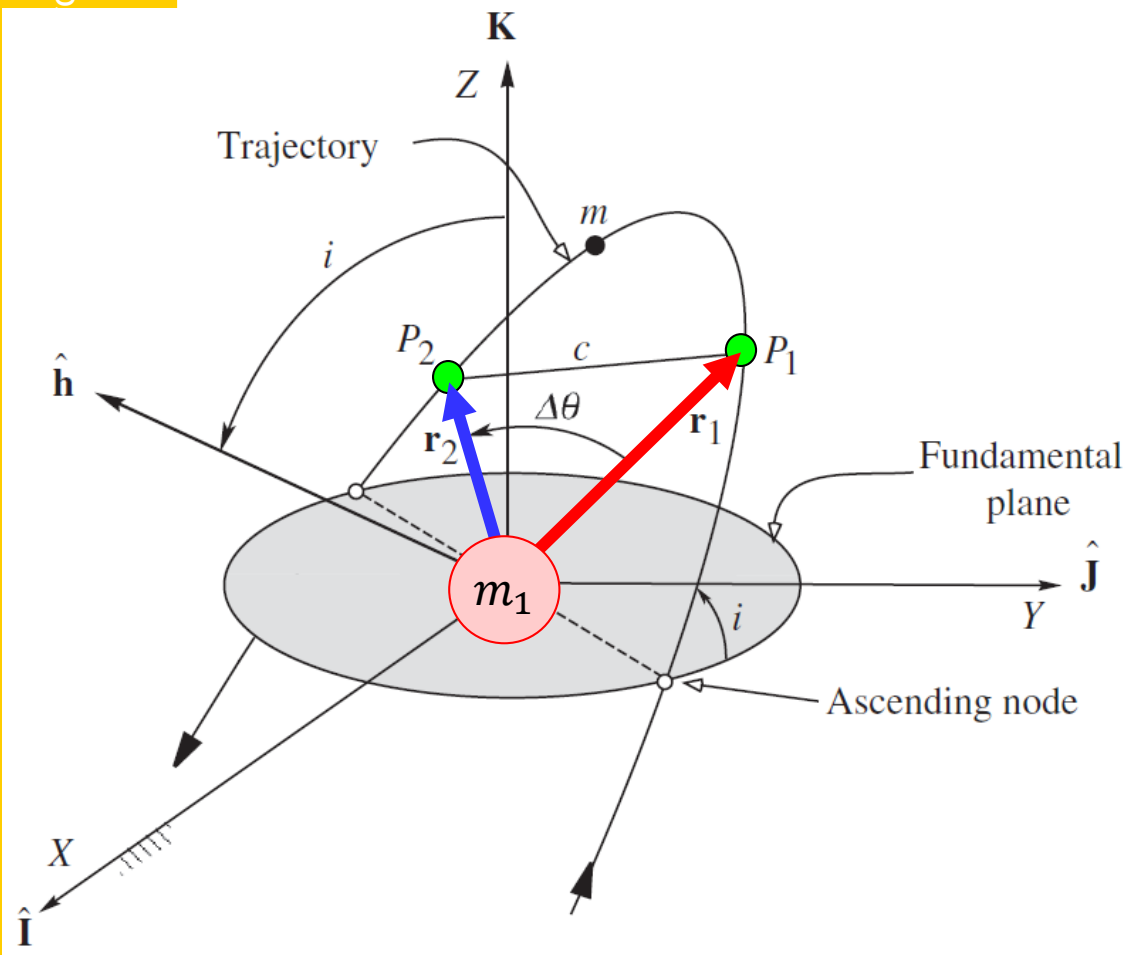
… alternatively, compute osculating elements from $\mathbf{r_0}, \mathbf{v_0}$ then $(a, e, \Omega, i, \omega, \theta(t))$

Source: Curtis HD (2010) "Orbital Mechanics for Engineering Students" (3rd Edition) Elsevier.

Definition of the Lambert Problem

We know the transfer time Δt from the position $\mathbf{r_1}$ to the position $\mathbf{r_2}$.

Lambert problem is to determine the orbital segment joining $\mathbf{r_1}$ and $\mathbf{r_2}$ in such Δt



Lambert proposed that the orbital transfer time Δt from $\mathbf{P_1}$ to $\mathbf{P_2}$:

$$\sqrt{\mu}\Delta t = f(a, r_1 + r_2, c)$$

- depends only of:
  - semi-major axis a of the transfer orbit (unknown)
  - $\mathrm{r}_1 + \mathrm{r}_2$: the sum of the magnitudes of the position vectors $\mathbf{r_1}$ and $\mathbf{r_2}$
  - the length c of the chord joining $\mathbf{P_1}$ and $\mathbf{P_2}$

$$c = \sqrt{r_1^2 + r_2^2 - 2\mathrm{r}_1\mathrm{r}_2 \cos\Delta\theta}$$

- is independent of the transfer orbit eccentricity

J. H. Lambert (1728–1777) was a French-born German astronomer, physicist, and mathematician
Source: Fig. 5.3 Curtis HD (2010) "Orbital Mechanics for Engineering Students" (3rd Edition) Elsevier

The Lambert Problem

We need to solve the second-order differential equation of motion

$$\ddot{\mathbf{r}} + \frac{\mu}{r^3}\mathbf{r} = 0$$

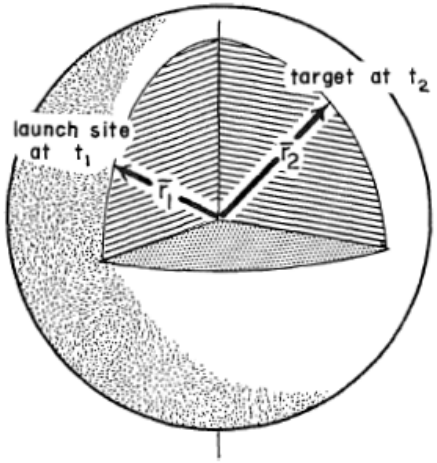
with boundary conditions: 2 vectors and 2 scalars:

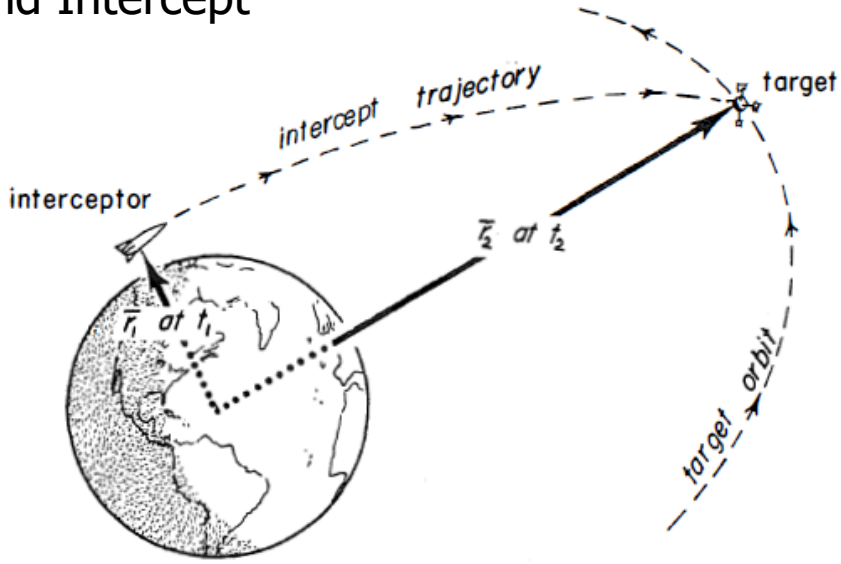
$\mathbf{r}(t_1) = \boldsymbol{r_1}$ $\quad\quad$ $\Delta t = t_2 - t_1$

$\mathbf{r}(t_2) = \boldsymbol{r_2}$ $\quad\quad$ +specification for a long or short path orbit

Thus, Lambert is a *Orbital Boundary Value Problem*, which is a special case of the mathematical *Two Point Boundary Value Problem* applied to:

- Mission design: Interplanetary transfers
- Orbital determination: when the (initial) orbit is not known
- Orbital transfer: Rendezvous and Intercept
- Ballistic missile targeting





Source: Walter U (2019) "Astronautics. The Physics of Space Flight" 3rd Edition Springer
Fig. 5.7-1 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

Preliminary concepts

We need to know the angle $\Delta\theta$ trough which the transfer takes place

$$\cos\Delta\theta = \frac{\boldsymbol{r_1}\cdot\boldsymbol{r_2}}{r_1\cdot r_2}$$

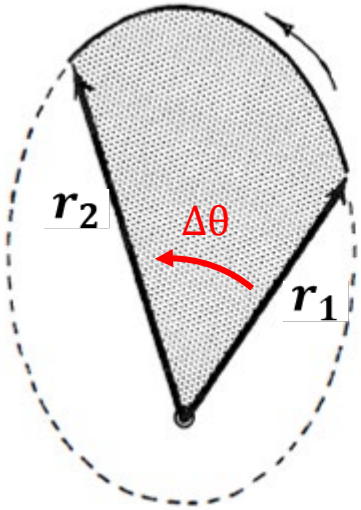
We must assume a direction of transfer:

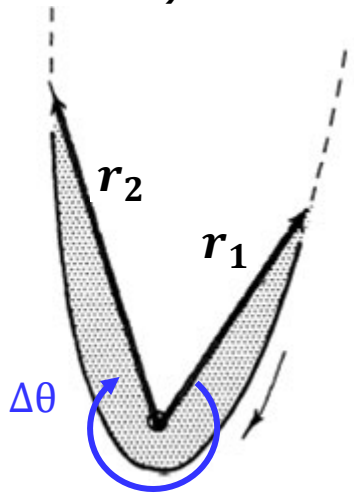
$$sin\,\Delta\theta = t_m\sqrt{1-cos^2\Delta\theta}$$

Infinite orbits pass from $\boldsymbol{r_1}$ to $\boldsymbol{r_2}$. Only two orbits have the specified Δt; once we specify the transfer method t_m :

- Short way $\Delta\theta < 180°$ with $t_m = +1$ (counter clock wise)
- Long way $\Delta\theta > 180°$ with $t_m = -1$ (clock wise)

$\Delta\theta < 180°$
$t_m = +1$



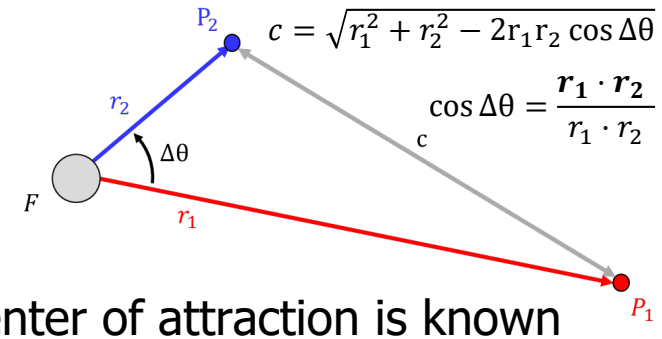


$\Delta\theta > 180°$
$t_m = -1$

Note: $\Delta t_{SW} = \Delta t_{LW}$ but very different orbit from $\boldsymbol{r_1}$ to $\boldsymbol{r_2}$

Source: Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition
Fig. 5.2-1 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

Preliminary concepts



A conic has two foci (F, F').

For a given transfer $(a, r_1 + r_2, c)$:

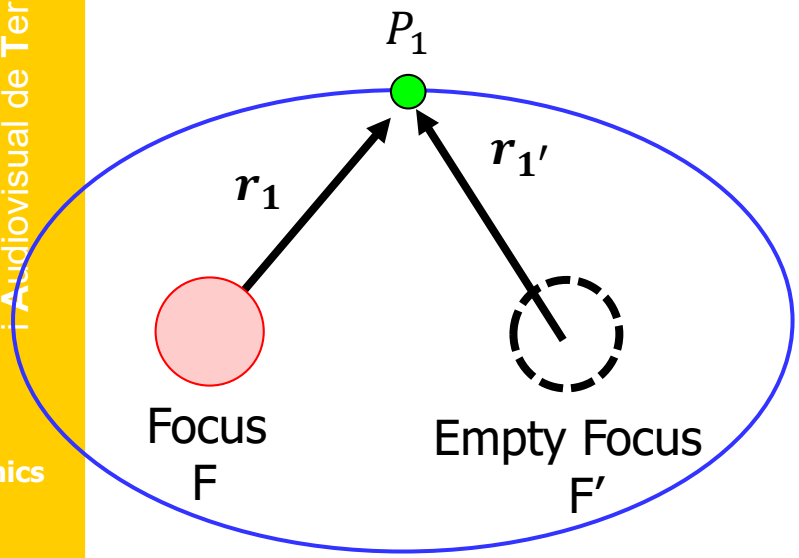
- the position of the focus F containing the center of attraction is known
- the position of the empty focus F' is unknown

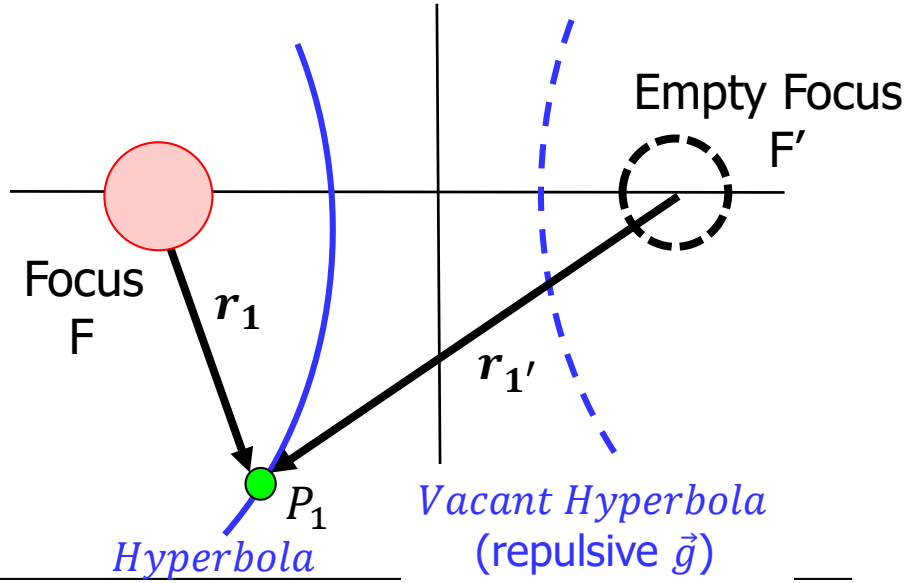
Ellipses and hyperbolas require, by definition:

$$P_1F + P_1F' = P_2F + P_2F' = 2a \quad \rightarrow \quad P_1F' = 2a - r_1 \quad \text{(ellipse)}$$

$$P_1F - P_1F' = P_2F - P_2F' = -2a \quad \rightarrow \quad P_1F' = 2a + r_1 \quad \text{(hyperbola, } a > 0\text{)}$$

($P_1F = r_1$)





Source: Figure 3-2 Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition, Space Fig 2.25 Curtis HD (2010) "Orbital Mechanics for Engineering Students" (3rd Ed) Elsevier.

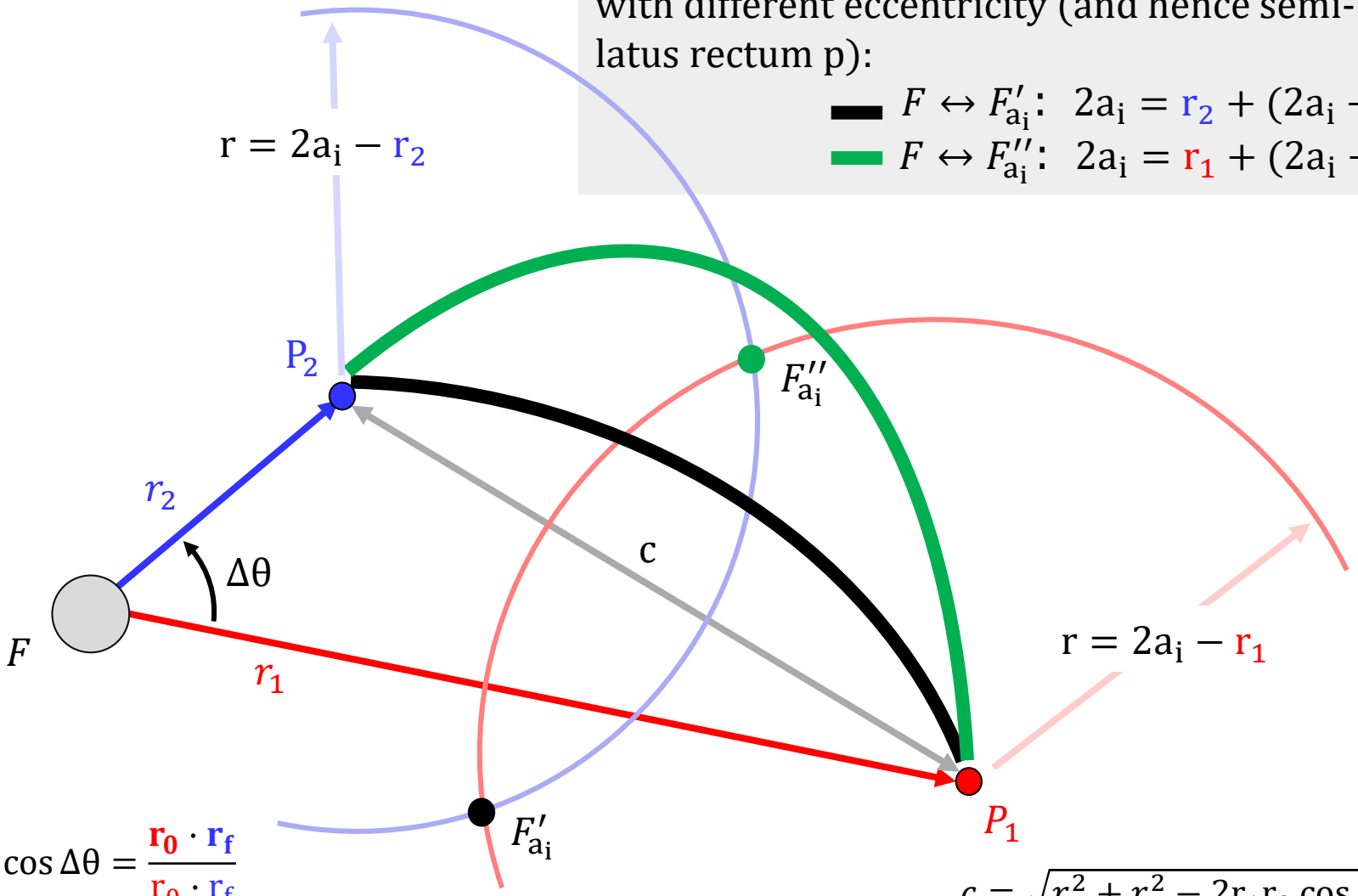
Lambert Transfers – Ellipses

Knowing the semi-major of the orbit a_i, we can locate the vacant focus.

2 possible foci → 2 possible (conjugate) ellipses with different eccentricity (and hence semi-latus rectum p):

$F \leftrightarrow F'_{a_i}$: $2a_i = r_2 + (2a_i - r_2)$

$F \leftrightarrow F''_{a_i}$: $2a_i = r_1 + (2a_i - r_1)$

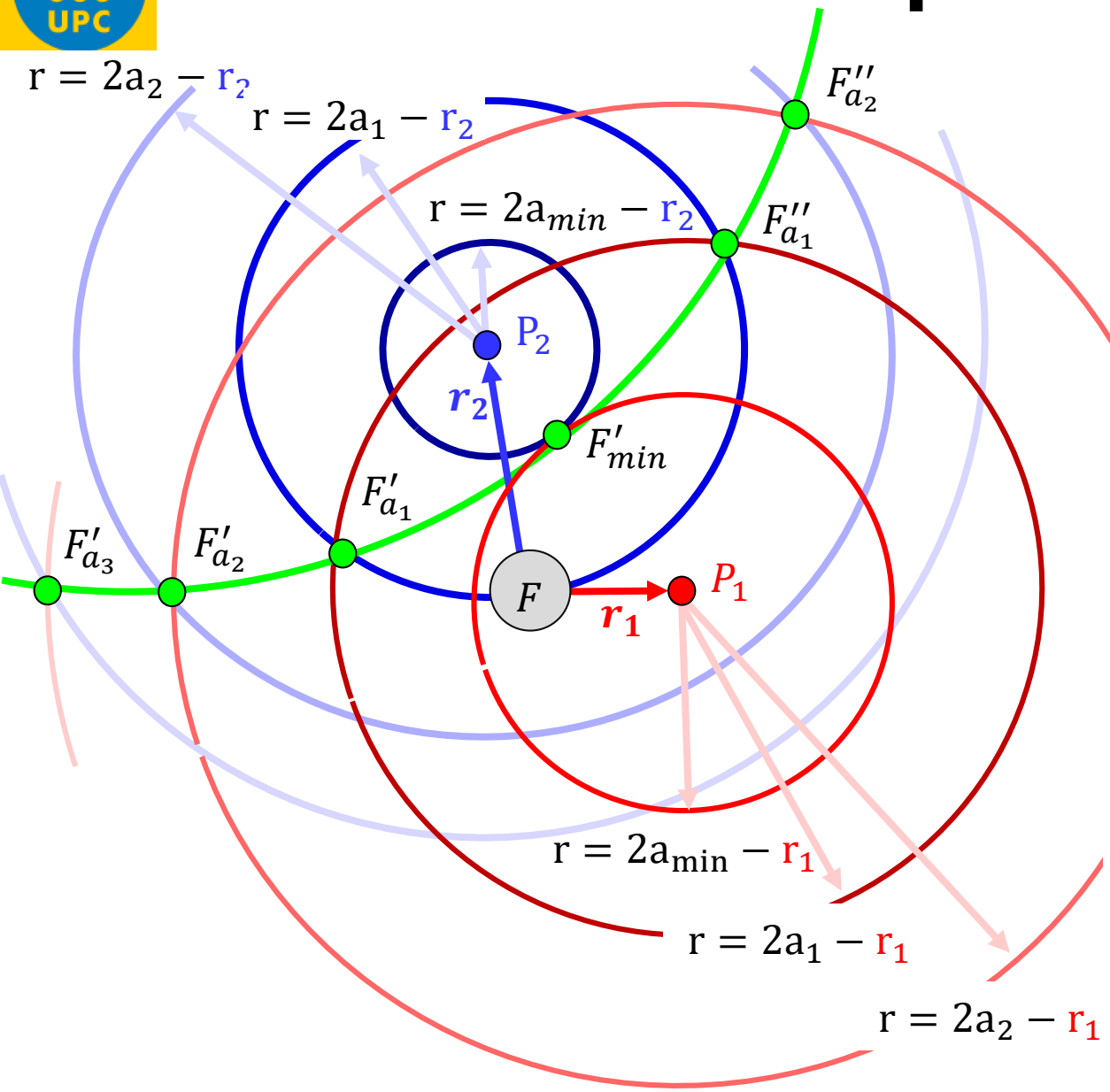


$$\cos\Delta\theta = \frac{\mathbf{r_0} \cdot \mathbf{r_f}}{r_0 \cdot r_f}$$

$$c = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\Delta\theta}$$

Source: Fig 8.6 Walter U (2019) "Astronautics. The Physics of Space Flight" 3rd Edition Springer

Locus of vacant Elliptical foci (for every a_i)



- Given a_i we determine the two conjugate Foci intersecting two circles centered at P_1 and P_2 with radii $2a_i - r_1$ and $2a_i - r_2$:

$$FF'_{a_i}: \quad 2a_i = r_2 + (2a_i - r_2)$$
$$FF''_{a_i}: \quad 2a_i = r_1 + (2a_i - r_1)$$

- The locus of the vacant foci lies on the branch of a hyperbola with eccentricity and semi-major axis:

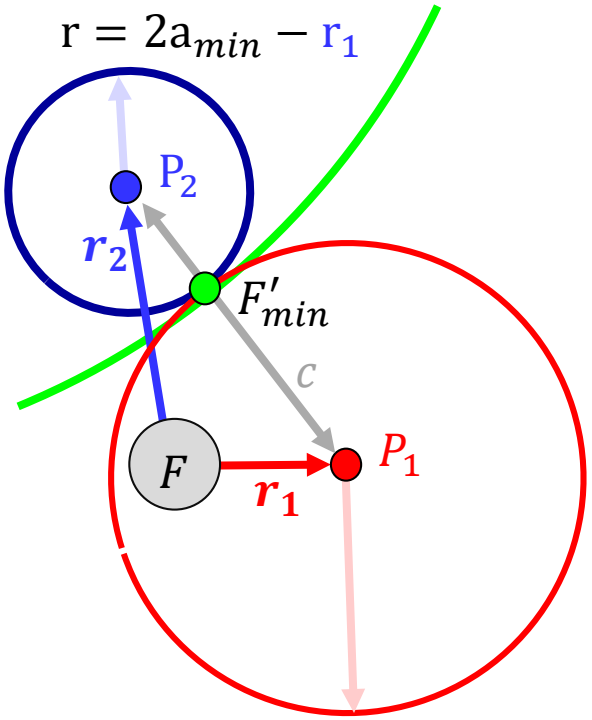
$$e = \frac{r_2 - r_1}{c}$$
$$a = \frac{r_1 - r_2}{2}$$

- There is only one a_i (i.e. one pair of conjugate F'_{a_i}, F''_{a_i}) for which the transfer time is Δt

- There is only one secondary Focus F'_{min} in the minimum-energy solution: where the circles first touch

Source: Fig 6.17 Battin RH (1999) "An introduction to the mathematics and methods of astrodynamics" 1st Ed, AIAA

Minimum Energy Transfer Orbit



- The energy of an elliptical orbit

$$\varepsilon = -\frac{\mu}{2a} < 0$$

- From all ellipses, there is only one that has minimum energy:

$$FF'P_1: \quad 2a_{min} = r_1 + F'_{min}P_1$$
$$FF'P_2: \quad 2a_{min} = r_2 + F'_{min}P_2 \quad +$$
$$4a_{min} = r_1 + r_2 + F'_{min}P_1 + F'_{min}P_2$$

- Given $r_1 + r_2$, a becomes minimum if $F'_{min}P_1 + F'_{min}P_2$ becomes minimum → Vacant focus F'_{min} must lie in the chord $c = F'_{min}P_1 + F'_{min}P_2$

- Then, the minimum energy transfer orbit is derived as

$$a_{min} = \frac{r_1 + r_2 + c}{4}$$

$$p_{min} = \frac{h_{min}^2}{\mu} = \frac{r_1 r_2}{c}(1 - cos\Delta\theta) = \frac{c}{2}\left[1 - \left(\frac{r_1 - r_2}{c}\right)^2\right]$$

$$e_{min} = \sqrt{1 - p_{min}/a_{min}}$$

- Then, flight time Δt is no longer freely chosen. The Δt of the Minimum Energy Transfer Orbit becomes

$$\Delta t_{min} = \sqrt{\frac{a_{min}^3}{\mu}}\left[\pi - arc\,cos\left(\frac{c}{a_{min}} - 1\right) + \sqrt{\frac{c}{a_{min}}\left(2 - \frac{c}{a_{min}}\right)}\right]$$

Remind: this is for ellipses! In a parabola with $v \to \infty$ the $\Delta t \to 0$

Source: Fig 8.6 Walter U (2019) "Astronautics. The Physics of Space Flight" 3rd Edition Springer

Minimum Energy Transfer Orbit

If $\Delta\theta = 180$, the Minimum Energy Transfer Orbit becomes the well-known Hohmann Transfer as a particular case of the Lambert Transfer:

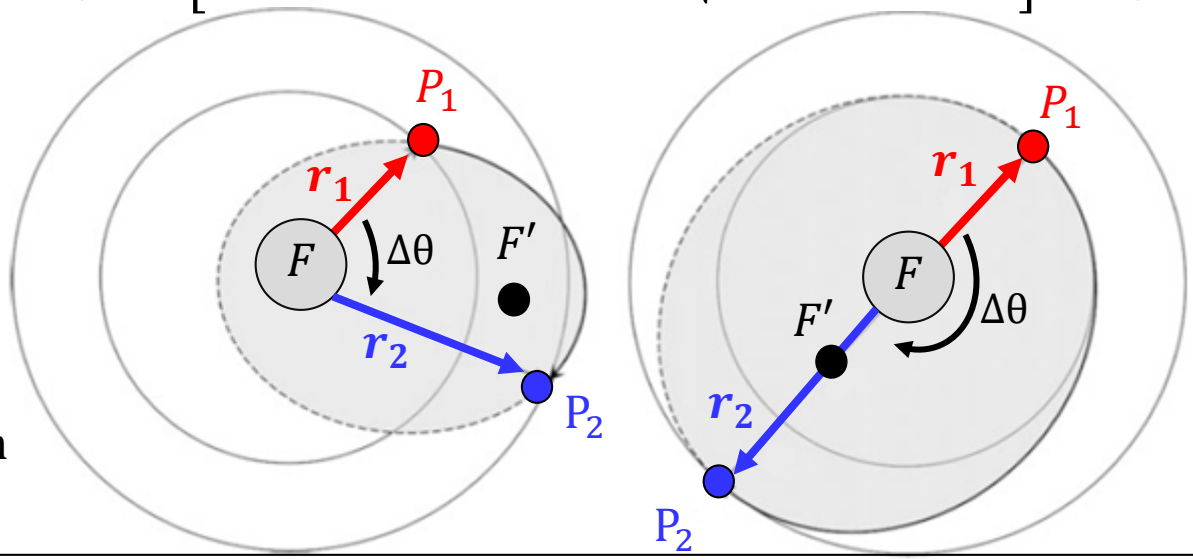
$$c = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\Delta\theta} = r_1 + r_2$$

$$a_{min} = \frac{r_1 + r_2 + c}{4} = \frac{r_1 + r_2}{2} = \frac{c}{2}$$

$$p_{min} = \frac{h_{min}^2}{\mu} = \frac{r_1 r_2}{c}(1 - cos\Delta\theta) = 2\frac{r_1 r_2}{r_1 + r_2}$$

$$e_{min} = \sqrt{1 - p_{min}/a_{min}} = \sqrt{1 - 4\frac{r_1 r_2}{(r_1 + r_2)^2}}$$

$$\Delta t_{min} = \sqrt{\frac{a_{min}^3}{\mu}}\left[\pi - arc\,cos\left(\frac{c}{a_{min}} - 1\right) + \sqrt{\frac{c}{a_{min}}\left(2 - \frac{c}{a_{min}}\right)}\right] = \pi\sqrt{\frac{a_{min}^3}{\mu}}$$



CW motion
$t_m = -1$

CW motion
$t_m = -1$

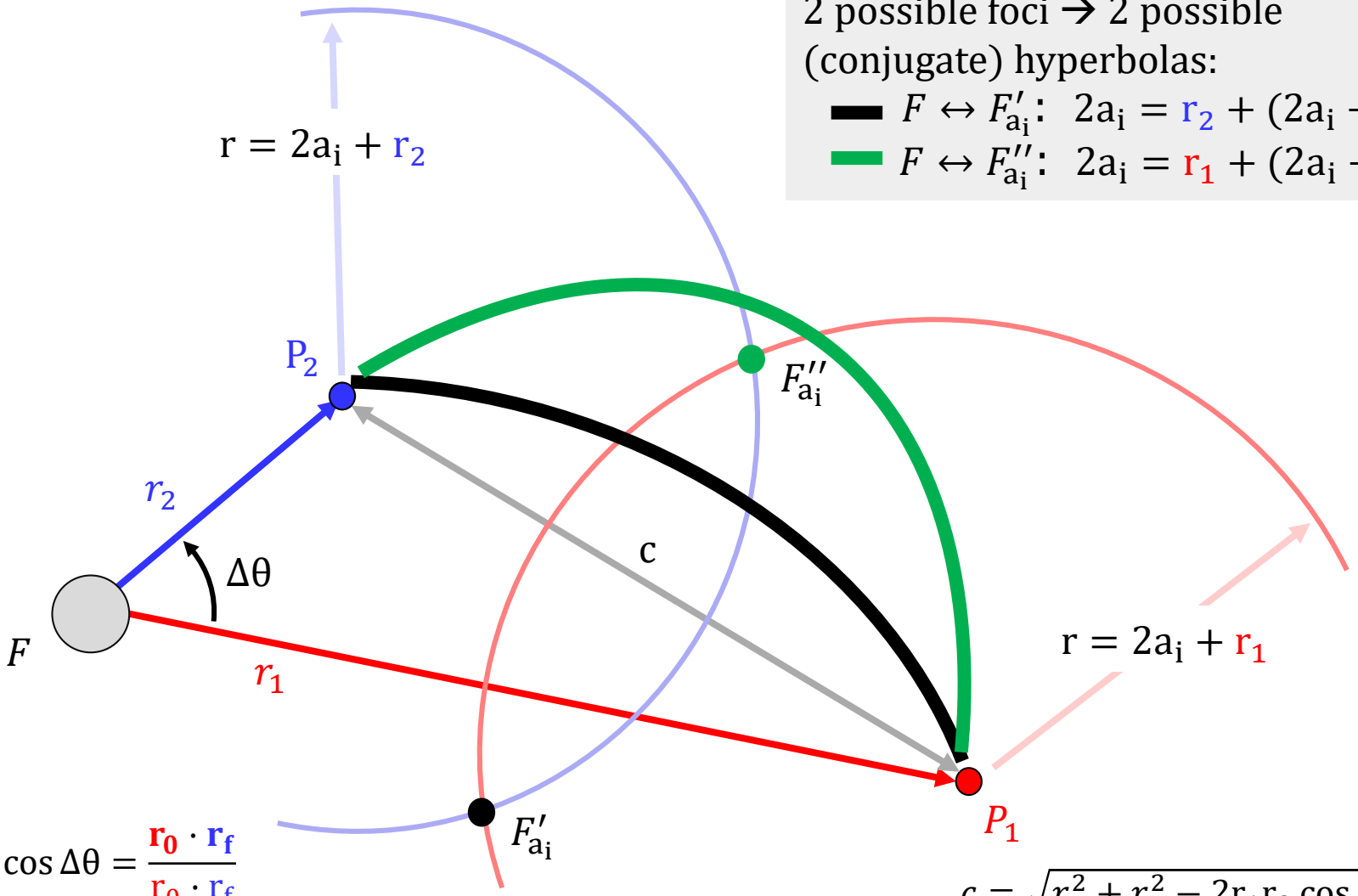
Source: Fig 8.14 Walter U (2019) "Astronautics. The Physics of Space Flight" 3rd Edition Springer

Lambert Transfers – Hyperbolas

Knowing the semi-major of the orbit a_i, we can locate the vacant focus. In an hyperbola, the difference of the distances from the foci to any point equals minus two times the semi-major axis:

2 possible foci → 2 possible (conjugate) hyperbolas:

- $F \leftrightarrow F'_{a_i}$: $2a_i = r_2 + (2a_i + r_2)$
- $F \leftrightarrow F''_{a_i}$: $2a_i = r_1 + (2a_i + r_1)$

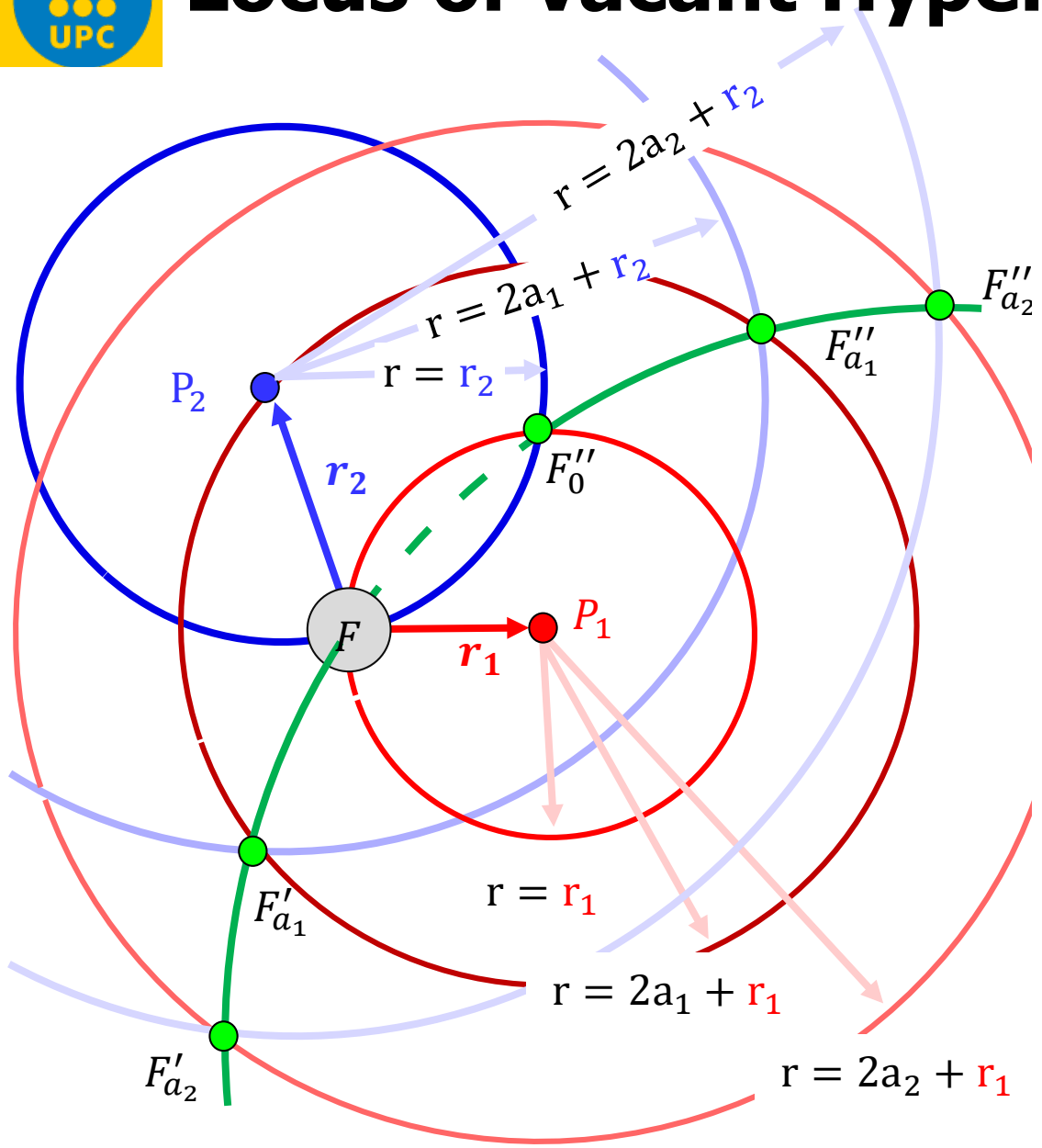


$$\cos\Delta\theta = \frac{\mathbf{r_0} \cdot \mathbf{r_f}}{r_0 \cdot r_f}$$

$$c = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos\Delta\theta}$$

Source: Fig 8.6 Walter U (2019) "Astronautics. The Physics of Space Flight" 3rd Edition Springer

Locus of vacant Hyperbolic foci (for every a_i)



- Given a_i we determine the two conjugate Foci intersecting two circles centered at P_1 and P_2 with radii $2a_i + r_1$ and $2a_i + r_2$:

  FF'_{a_i}: $2a_i = r_2 + (2a_i + r_2)$

  FF''_{a_i}: $2a_i = r_1 + (2a_i + r_1)$

- The locus of the vacant foci lies on the branch of a hyperbola with eccentricity and semi-major axis:

$$e = \frac{r_2 - r_1}{c}$$

$$a = \frac{r_1 - r_2}{2}$$

- There is only one a_i (i.e one pair of conjugate F'_{a_i}, F''_{a_i}) for which the transfer time is Δt

- Intersection point must lie outside the circles with radius r_1 and r_2 centered at P_1 and P_2

- when $a_0 = 0$: $F'_0 = F, F''_0$ orbit degenerates into a straight line with $v \to \infty$

Source: Fig 6.18 Battin RH (1999) "An introduction to the mathematics and methods of astrodynamics" 1st Ed, AIAA

Module 2 Requirements

Objectives:

- **Problem definition, statement of the theorem of Lambert, sketch of proof**
- **Numerical aspects of the problem**
- **Classical methods of solution (Lagrange, Gauss)**
- **Introduction to modern methods**
- **Application to Optimization, Rendezvouz, Pork Chop Plots**
- **Application to the problem of interplanetary transfer**

Time Allocation:

- **8 hours**

Lambert Theorem

Lambert conjectured that *the transfer time of a body moving between two points on a conic trajectory is a function only of the sum of the distances of the two points from the origin of force, the length c of the chord joining these two positions, the semi-major axis of the conic and the direction of motion*

If $t_2 - t_1$ is the time to describe the arc from P_1 to P_2, then Lambert's theorem states that:

$$\sqrt{\mu}(t_2 - t_1) = f(a, r_1 + r_2, c)$$

Alternatively, Lambert's theorem can be stated as

$$(t_2 - t_1) = \Delta t = f(r_1, r_2, a, c, \Delta\theta)$$

Remarks

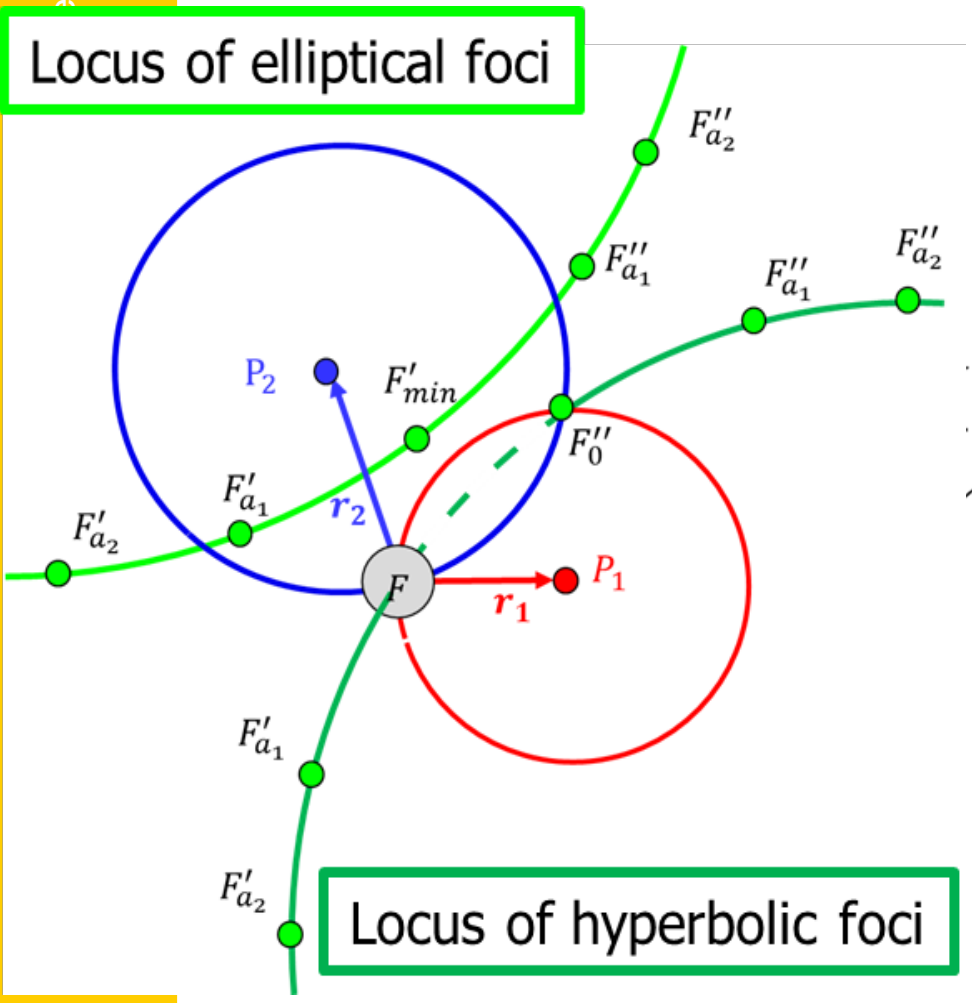
- True for any type of conic
- Δt is independent of the individual locations of the starting or arrival points P_1 and P_2
- Δt does not depend on the true anomaly of the starting θ_1 or arrival points θ_2, but its difference $\Delta\theta$
- Δt is independent of the transfer orbit eccentricity

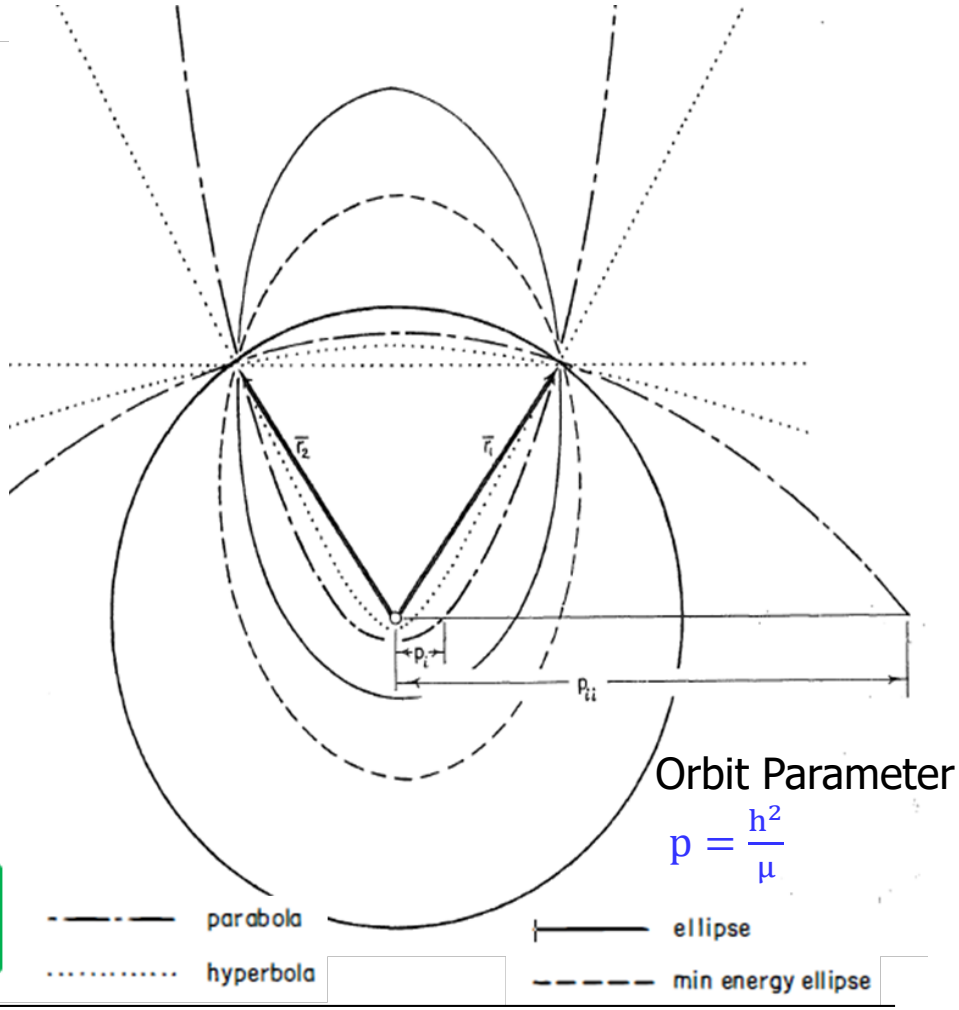
Source: Battin RH (1999) "An introduction to the mathematics and methods of astrodynamics" 1st Ed, AIAA
Walter U (2019) "Astronautics. The Physics of Space Flight" 3rd Edition Springer

Solving the Lambert Problem

We can determine the trajectory connecting P_1 and P_2, <u>given a flight time</u> Δt

$$\sqrt{\mu}\Delta t = f(a, r_1 + r_2, c)$$





Orbit Parameter

$$p = \frac{h^2}{\mu}$$

Source: Fig 7.11 Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition, Space Technology Figure 5.4-2 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

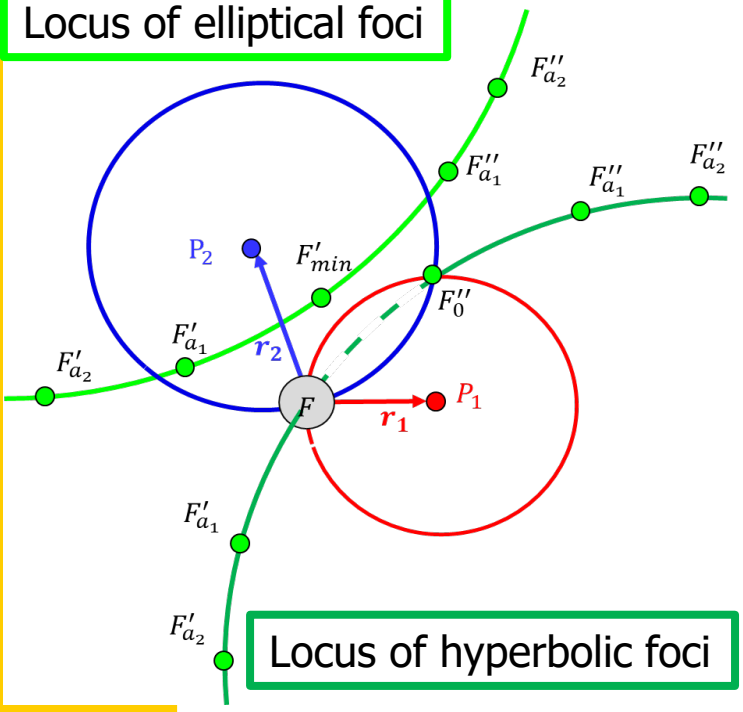
Solving the Lambert Problem

The amount of time required for the transfer can be determined for elliptical and hyperbolic orbits with the **Lagrange time equation:**

$$\Delta t = \sqrt{\frac{a^3}{\mu}}\left[\alpha_e - sin(\alpha_e) - t_m\left(\beta_e - sin(\beta_e)\right) + 2\pi \cdot n_{rev}\right] \quad \text{ellipses}$$

$$= \sqrt{\frac{-a^3}{\mu}}\left[\alpha_h + sinh(\alpha_h) - t_m(sinh(\beta_h) - \beta_h)\right] \quad \text{hyperbolas}$$

Locus of elliptical foci



Locus of hyperbolic foci

where t_m is the transfer method:

- $t_m = +1$ (counter clock wise)
- $t_m = -1$ (clock wise)

and:

$$sin\left(\frac{\alpha_e}{2}\right) = \sqrt{\frac{r_1 + r_2 + c}{4a}} \qquad sin\left(\frac{\beta_e}{2}\right) = \sqrt{\frac{r_1 + r_2 - c}{4a}}$$

$$sinh\left(\frac{\alpha_h}{2}\right) = \sqrt{\frac{r_1 + r_2 + c}{-4a}} \qquad sinh\left(\frac{\beta_h}{2}\right) = \sqrt{\frac{r_1 + r_2 - c}{-4a}}$$

Source: Fig 7.11 Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition, Space Technology

Module 2 Requirements

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- Application to the problem of interplanetary transfer

Time Allocation:

- 8 hours

Proof of Lambert's Theorem (1788)

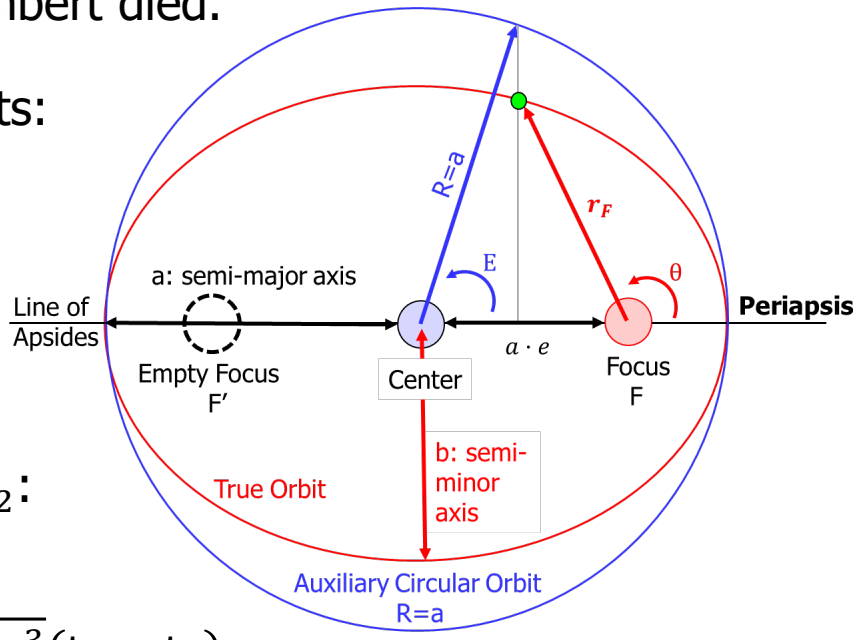
Lagrange was first to supply the analytic proof of Lambert's theorem for **elliptic** orbits just one year before Lambert died.

Recall Kepler's equation for elliptic orbits:

$$M_E = E - e \sin E$$

$$M_E = M_{E_0} + n(t - t_0)$$

$$n = \frac{2\pi}{T} = \sqrt{\mu/a^3}$$



Then, the time to traverse from P_1 to P_2:

$$M_{E_2} = M_{E_1} + \sqrt{\mu/a^3}(t_2 - t_1)$$

$$E_2 - e \sin E_2 = E_1 - e \sin E_1 + \sqrt{\mu/a^3}(t_2 - t_1)$$

$$\sqrt{\mu/a^3}(t_2 - t_1) = (E_2 - E_1) - e(\sin E_2 - \sin E_1)$$

$$\sqrt{\mu}(t_2 - t_1) = 2\sqrt{a^3}\left[\frac{1}{2}(E_2 - E_1) - e\left(\sin\frac{1}{2}(E_2 - E_1)\cos\frac{1}{2}(E_2 + E_1)\right)\right]$$

$$\sqrt{\mu}(t_2 - t_1) = 2\sqrt{a^3}[\psi - \sin\psi\cos\phi\,] \text{ with } \begin{cases} \psi = \frac{1}{2}(E_2 - E_1) \\ \cos\phi = e\cos\frac{1}{2}(E_2 + E_1) \end{cases}$$

Source: Battin RH (1999) "An introduction to the mathematics and methods of astrodynamics" 1st Ed, AIAA

Proof of Lambert's Theorem (1788)

We express all parameters of Lambert's theorem $\sqrt{\mu}\Delta t = f(a, r_1 + r_2, c)$ in terms of ψ and ϕ:

- the sum of radial distances:

$$r_1 + r_2 = a(1 - e\cos E_1) + a(1 - e\cos E_2) = 2a(1 - \cos\psi\cos\phi)$$

- the chord:

$$c = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2\cos\Delta\theta} = \sqrt{(r_1 + r_2)^2 - 2r_1 r_2\cos\Delta\theta - 2r_1 r_2}$$

$$= \sqrt{(r_1 + r_2)^2 - 4r_1 r_2\cos^2\frac{\Delta\theta}{2}} = \sqrt{\left(2a(1 - \cos\psi\cos\phi)\right)^2 - 4a^2(\cos\psi - \cos\phi)}$$

$$\tan\left(\frac{E}{2}\right) = \sqrt{\frac{1-e}{1+e}}\tan\left(\frac{\theta}{2}\right) \begin{cases} \sin\frac{E}{2} = \sqrt{a(1-e)}\cdot\sqrt{r}\sin\frac{\theta}{2} \\ \cos\frac{E}{2} = \sqrt{a(1+e)}\cdot\sqrt{r}\cos\frac{\theta}{2} \end{cases}$$

$$\sqrt{r_1 r_2}\cos\frac{\Delta\theta}{2} = a(\cos\psi - \cos\phi)$$

$$= \sqrt{4a^2\sin^2\psi\cdot\sin^2\phi} = 2a\cdot\sin\psi\cdot\sin\phi$$

- the semi-perimeter:

$$s = \frac{r_1 + r_2 + c}{2} = a(1 - \cos(\phi + \psi))$$

Source: Battin RH (1999) "An introduction to the mathematics and methods of astrodynamics" 1st Ed, AIAA

Proof of Lambert's Theorem (1788)

We introduce the variables

$$\alpha = \phi + \psi$$
$$\beta = \phi - \psi$$

So that

$$2s = 2a(1 - \cos(\phi + \psi)) = 2a(1 - \cos\alpha)$$
$$2(s - c) = 2a(1 - \cos(\phi - \psi)) = 2a(1 - \cos\beta)$$

$$\begin{cases} \sin^2\frac{\alpha}{2} = \frac{s}{2a} \\ \cos^2\frac{\beta}{2} = \frac{s-c}{2a} \end{cases}$$

Finally, we can express the time equation of an **ellipse**

$$\sqrt{\mu}(t_2 - t_1) = 2\sqrt{a^3}[\psi - \sin\psi\cos\phi\ ]$$

In terms of α and β

$$\sqrt{\mu}(t_2 - t_1) = \sqrt{a^3}[(\alpha - \sin\alpha) - (\beta - \sin\beta)]$$

For **hyperbolic** orbits, following a similar procedure:

$$\psi = \frac{1}{2}(H_2 - H_1) \text{ and } \cosh\phi\ = e\cosh\frac{1}{2}(H_2 + H_1)$$
$$r_1 + r_2 = 2a(1 - \cosh\psi\cosh\phi)$$
$$c = -2a \cdot \sinh\psi \cdot \sinh\phi$$
$$\sinh^2\frac{\alpha}{2} = -\frac{s}{2a} \text{ and } \sinh^2\frac{\beta}{2} = -\frac{s-c}{2a}$$

$$\sqrt{\mu}(t_2 - t_1) = 2\sqrt{-a^3}[(\sinh\alpha - \alpha) - (\sinh\beta - \beta)]$$

$$\sqrt{\mu}\Delta t = f(a, r_1 + r_2, c)$$

Source: Battin RH (1999) "An introduction to the mathematics and methods of astrodynamics" 1st Ed, AIAA

Module 2 Requirements

Objectives:

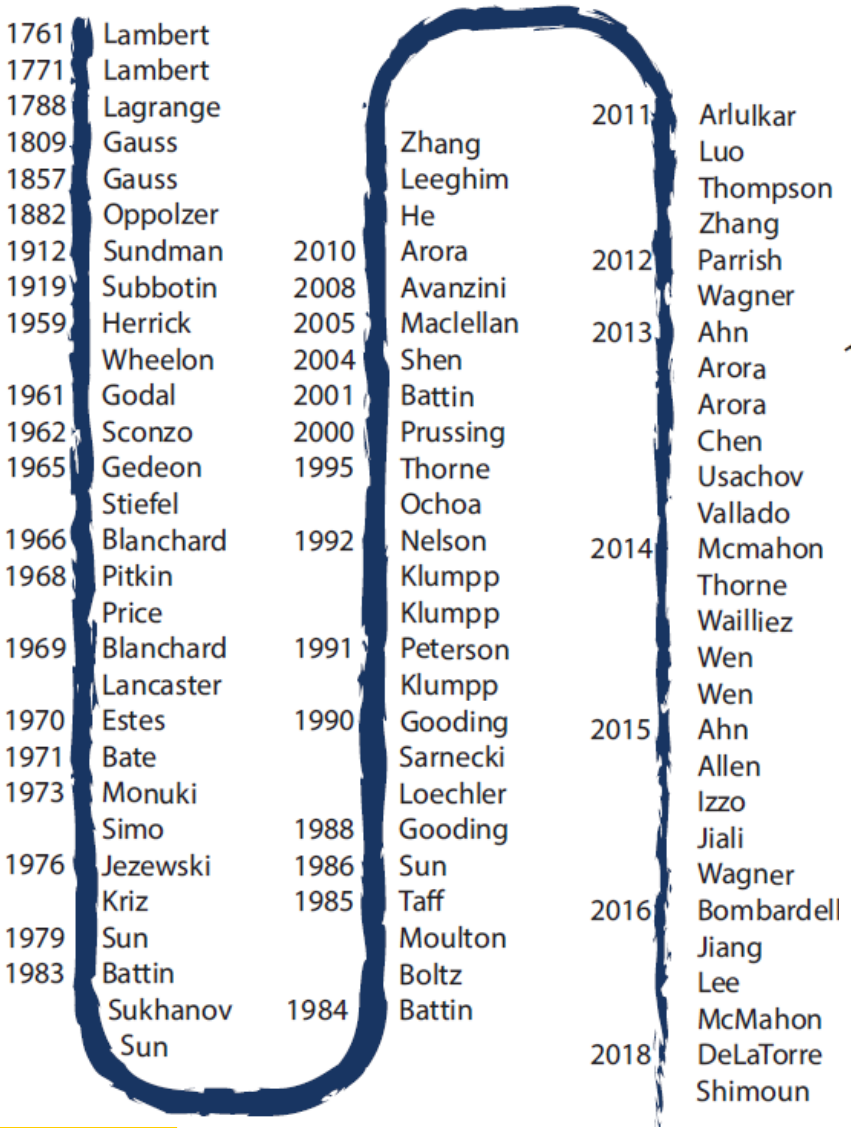
- **Problem definition, statement of the theorem of Lambert, sketch of proof**
- **Numerical aspects of the problem**
- **Classical methods of solution (Lagrange, Gauss)**
- **Introduction to modern methods**
- **Application to Optimization, Rendezvouz, Pork Chop Plots**
- **Application to the problem of interplanetary transfer**

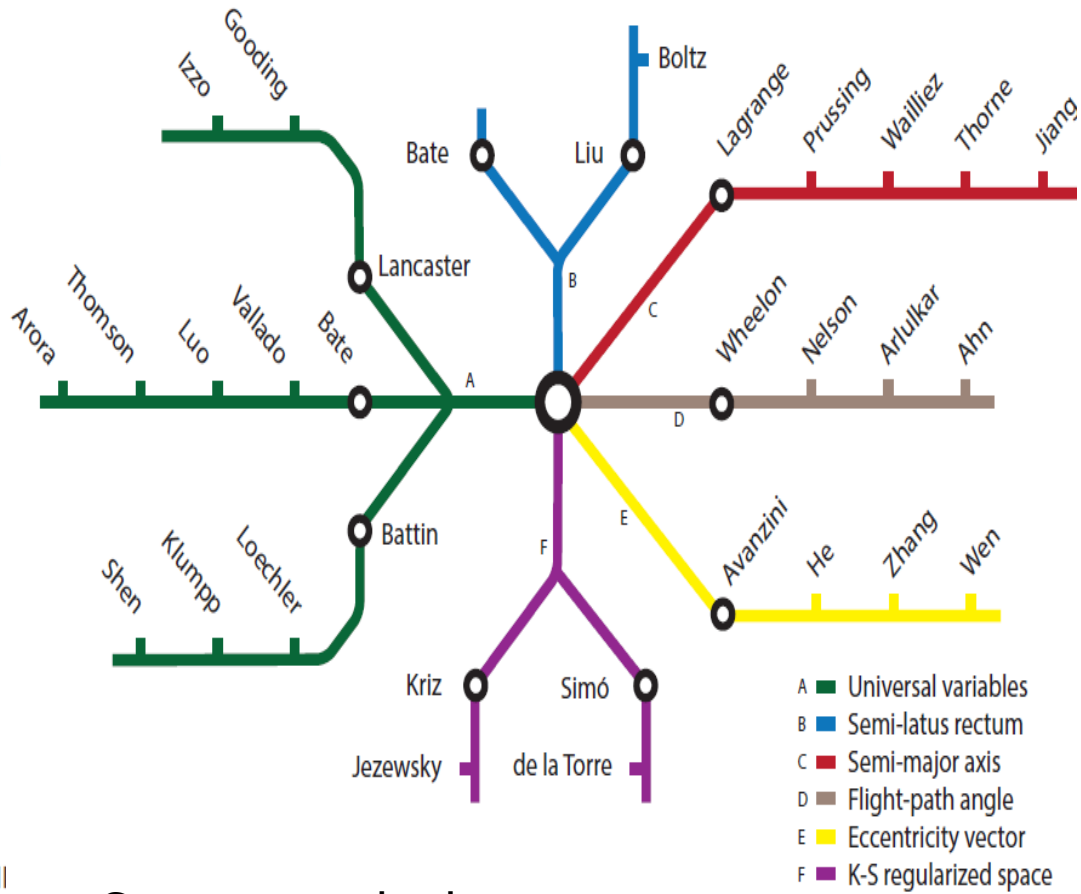
Time Allocation:

- **8 hours**

Methods to solve the Lambert Problem

>70 authors have proposed formulations and solutions to the problem





Current standards:

- NASA uses Gooding (1990)
- ESA uses Izzo (2015)

Source: Fig.4.1 and 4.2 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Pitfalls of the Lambert Problem

Issues arising while solving Lambert's problem:

- Differences handling the three types of conic sections:
  - Universal Methods (i.e. unique equation for ellipses, parabolas and hyperbolas → independent of eccentricity)
  - Case by case treatment (i.e. different equations for each orbit)
- Singularities, e.g.:
  - When $\Delta\theta = 180$ the plane of the transfer orbit is not determined
- Conic degeneracies, e.g.:
  - $\Delta\theta = 0$ or $\Delta\theta = 360$ (i.e. the rectilinear orbit)
- Choice of the initial guess
- Numerical divergence
- Loss of precision

Source:

Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications
De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Module 2 Requirements

Objectives:

- Problem definition, statement of the theorem of Lambert, sketch of proof
- Numerical aspects of the problem
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Time Allocation:

- 8 hours

The original Gauss method (1809)

Proposed by Gauss, while trying to locate Ceres asteroid (Dwarf planet since 2006) one year after its last observation. The method relies on the area swept by the satellite during the transfer, A_S.

The second law of Kepler states that the area is swept at a constant rate:

$$\frac{dA}{dt} = \frac{1}{2}h$$

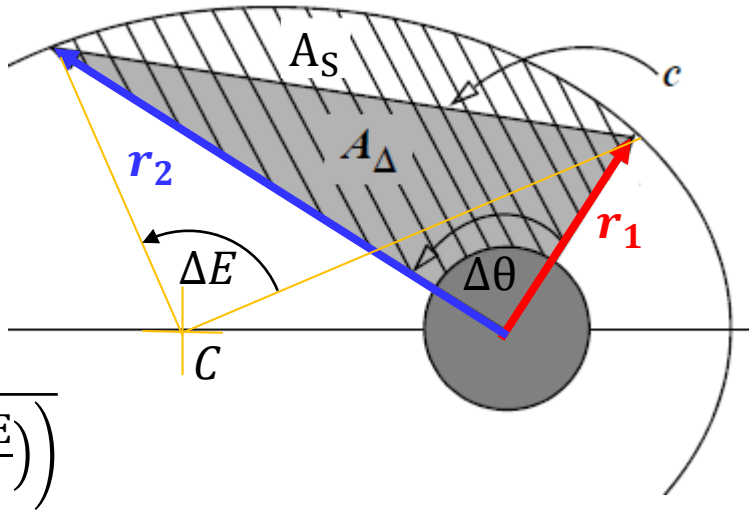
Hence the total area of the sector A_S during the transfer time Δt :

$$A_S = \frac{1}{2}h\Delta t = \frac{1}{2}\sqrt{\mu p}\Delta t$$

$$p = \frac{h^2}{\mu}$$

The area of the triangle formed by the two position vectors and the chord is:

$$A_\Delta = \frac{1}{2}r_1 \cdot r_2 \cdot \sin\Delta\theta$$



Its ratio:

$$y = \frac{A_S}{A_\Delta} = \frac{\sqrt{\mu p}\Delta t}{r_1 \cdot r_2 \cdot \sin\Delta\theta}$$

$$y^2 = \frac{\mu\Delta t^2 \sec^2\left(\frac{\Delta\theta}{2}\right)}{2r_1r_2 \cdot \left(r_1 + r_2 - 2\sqrt{r_1r_2}\cos\left(\frac{\Delta\theta}{2}\right)\cos\left(\frac{\Delta E}{2}\right)\right)}$$

Source: Fig 7.12 Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition, Space Technology Gauss CF (1809) "Theoria Motus Corporum Coelestium in Sectionibus Coicis Colem Ambientium (Theory of the Motion of the Heavenly Bodies Moving about the Sun in Conic Sections)" Hamburg: Friedrich Perthes and I.H. Besser

The original Gauss method (1809)

Introducing the temporary variables $\mathrm{l}, \mathrm{m}, x_1$:

$$l = \frac{\mathrm{r}_1 + \mathrm{r}_2}{4\sqrt{\mathrm{r}_1\mathrm{r}_2}\cos\left(\frac{\Delta\theta}{2}\right)} - \frac{1}{2} \qquad m = \frac{\mu\Delta\mathrm{t}^2}{\left(2\sqrt{\mathrm{r}_1\mathrm{r}_2}\cos\left(\frac{\Delta\theta}{2}\right)\right)^3} \qquad x_1 = \sin^2\left(\frac{\Delta\mathrm{E}}{4}\right)$$

Simplifies the notation to the 1st equation of Gauss:

$$\mathrm{y}^2 = \frac{m}{l + x_1}$$

A new temporary variable x_2 whose trigonometric functions are expanded with series as:

$$x_2 = \frac{\Delta\mathrm{E} - \sin\Delta\mathrm{E}}{\sin^3\left(\frac{\Delta\mathrm{E}}{4}\right)} \approx \frac{4}{3}\left(1 + \frac{6}{5}x_1 + \frac{6\cdot 8}{5\cdot 7}x_1^2 + \frac{6\cdot 8\cdot 10}{5\cdot 7\cdot 9}x_1^3 + \cdots\right)$$

Relates the change in eccentric anomaly $\Delta\mathrm{E}$ and y via the 2nd equation of Gauss:

$$\mathrm{y}^2(y-1) = mx_2$$

Joining the two equations of Gauss, we can obtain y :

$$y = (l + x_1)x_2 + 1$$

Once we find y, we obtain $\Delta\mathrm{E}$, the semi-major axis a and the semi-latus rectum (p parameter) of the orbit solution:

$$\cos\left(\frac{\Delta\mathrm{E}}{2}\right) = 1 - 2\mathrm{x}_1 \qquad \mathrm{a} = \frac{\mathrm{r}_1 + \mathrm{r}_2 - 2\sqrt{\mathrm{r}_1\mathrm{r}_2}\cos\left(\frac{\Delta\theta}{2}\right)\cos\left(\frac{\Delta\mathrm{E}}{2}\right)}{2\sin^2\left(\frac{\Delta\mathrm{E}}{2}\right)} \qquad \mathrm{p} = \frac{\mathrm{y}^2\cdot\mathrm{r}_1^2\cdot\mathrm{r}_2^2\cdot\sin^2(\Delta\theta)}{\mu\Delta\mathrm{t}^2}$$

An improved Gauss method (1971)

Bate, Mueller and White (1971) expanded Gauss solution to parabolic & hyperbolic orbits:

$$l = \frac{r_1 + r_2}{4\sqrt{r_1 r_2}\cos\left(\frac{\Delta\theta}{2}\right)} - \frac{1}{2} \qquad m = \frac{\mu \Delta t^2}{\left(2\sqrt{r_1 r_2}\cos\left(\frac{\Delta\theta}{2}\right)\right)^3}$$

Iterative algorithm with initial 1:1 triangle-to sector area ratio:

$y_0 = 1$

while $(y - y_0 < tolerance)$

$y_0 = y$

$$x_1 = \frac{m}{y^2} - l$$

$$x_2 = \frac{4}{3}\left(1 + \frac{6}{5}x_1 + \frac{6 \cdot 8}{5 \cdot 7}x_1^2 + \frac{6 \cdot 8 \cdot 10}{5 \cdot 7 \cdot 9}x_1^3 + \cdots\right)$$

$$y = (l + x_1)x_2 + 1$$

If $x_1 > 0$ eccentric solution orbit:

$$\cos\left(\frac{\Delta E}{2}\right) = 1 - 2x_1 \qquad p = \frac{r_1 \cdot r_2 \cdot (1 - \cos\Delta\theta)}{r_1 + r_2 - 2\sqrt{r_1 r_2}\cos\left(\frac{\Delta\theta}{2}\right)\cos\left(\frac{\Delta E}{2}\right)}$$

If $x_1 < 0$ hyperbolic solution orbit:

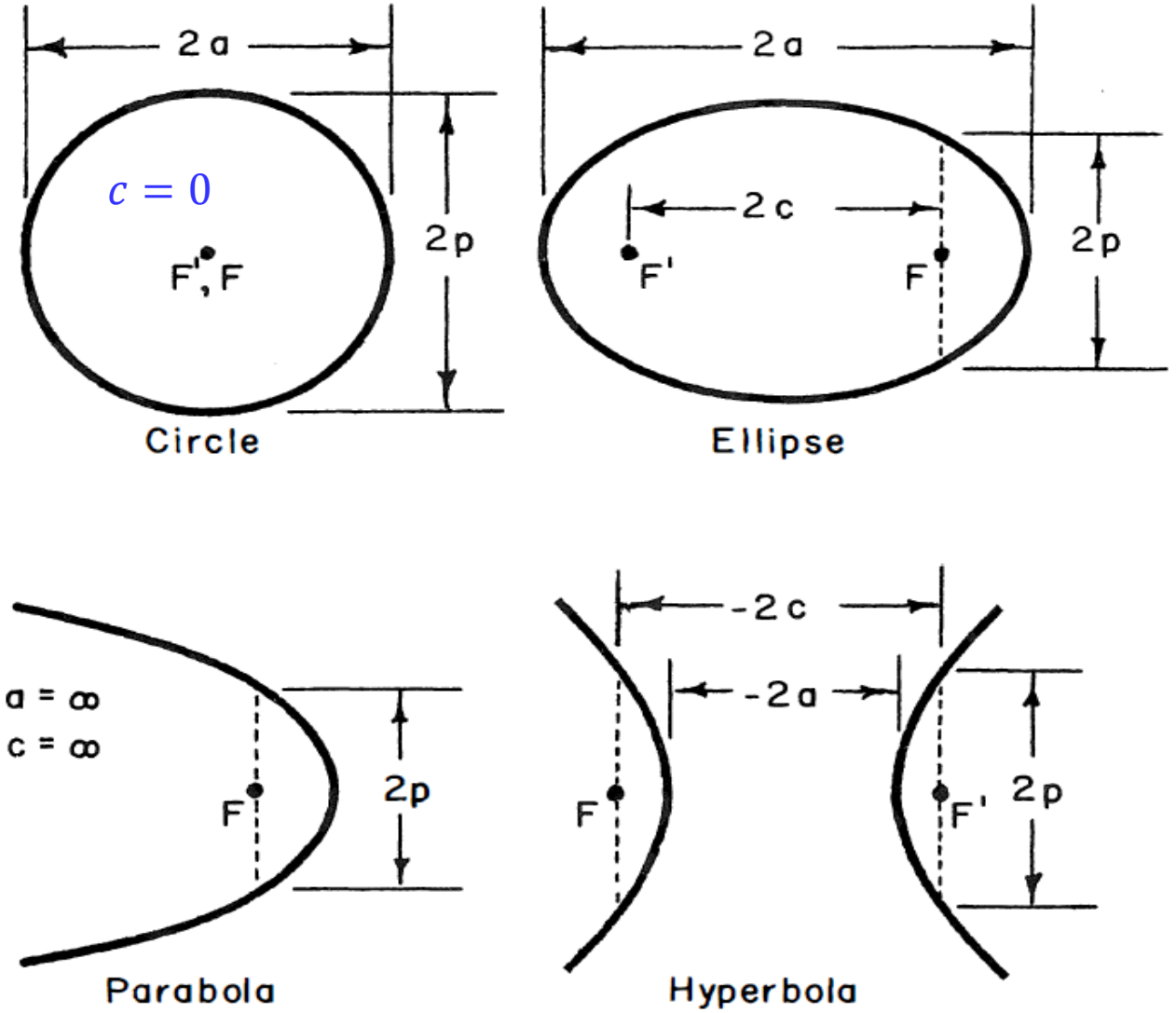
$$\cosh\left(\frac{\Delta H}{2}\right) = 1 - 2x_1 \qquad p = \frac{r_1 \cdot r_2 \cdot (1 - \cos\Delta\theta)}{r_1 + r_2 - 2\sqrt{r_1 r_2}\cos\left(\frac{\Delta\theta}{2}\right)\cosh\left(\frac{\Delta H}{2}\right)}$$

Source: Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition, Space Technology

Reminder!

$$p = \frac{h^2}{\mu}$$

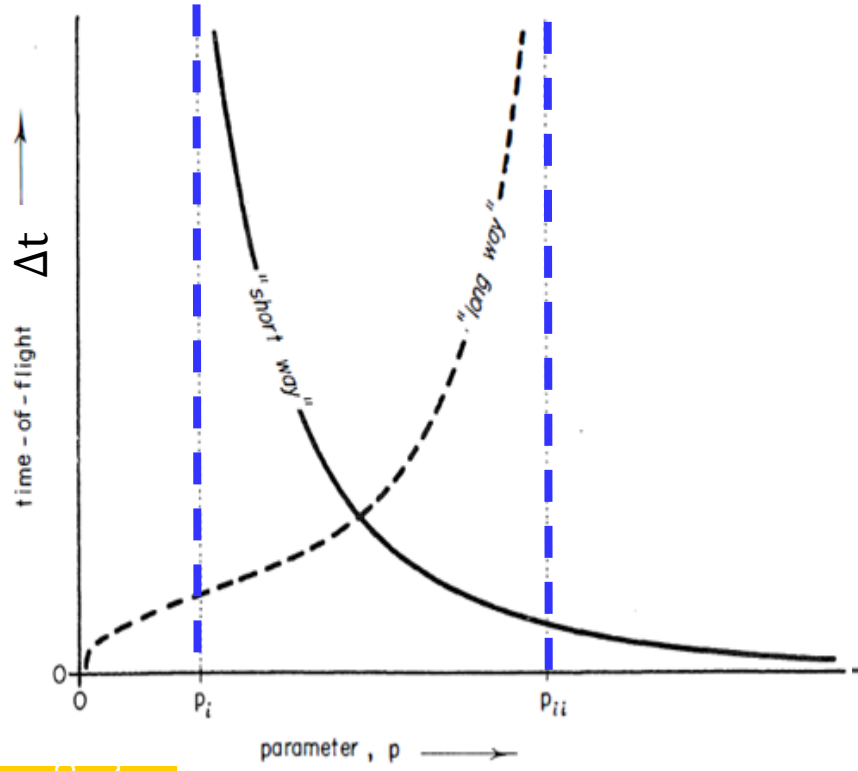
Geometrical dimensions common to all conic sections

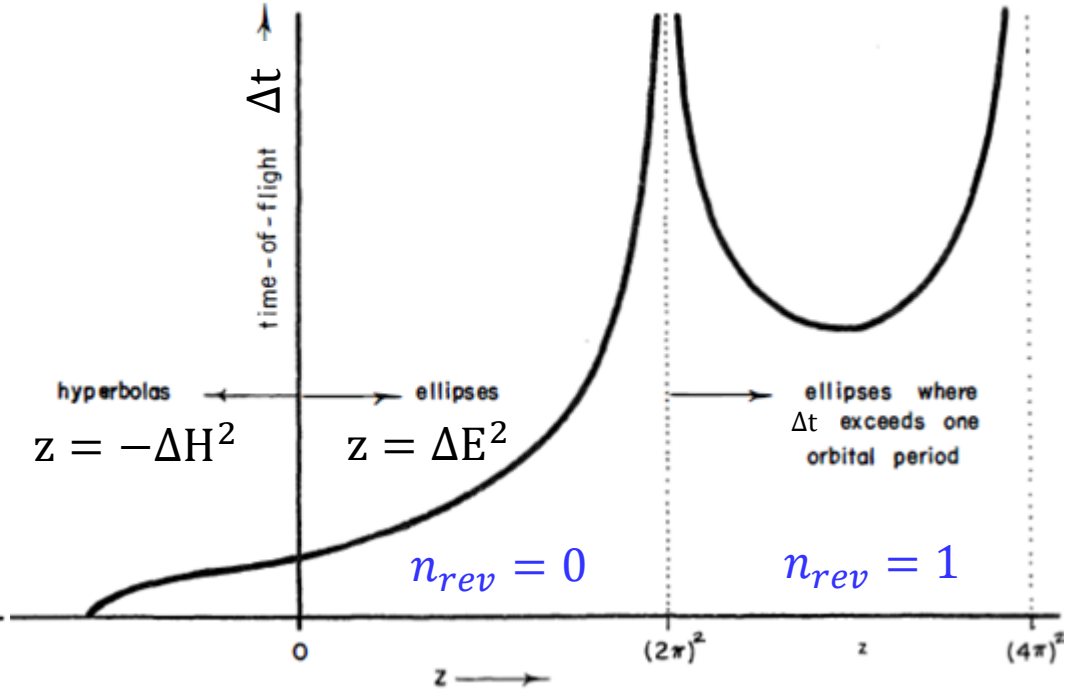


Source: Fig. 1.5-3 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

Typical flight time Δt plots

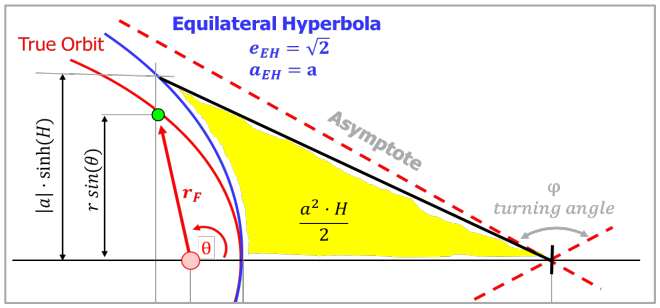
Different methods have different parameters to iterate and find the intersection with the given Δt, but we can recognize generic common shapes of $\sqrt{\mu}\Delta t = f(a, r_1 + r_2, c)$

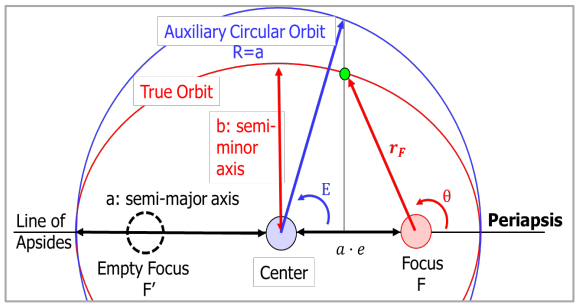




The limiting values of p correspond to the two parabolic orbits passing through r_1 and r_2

In parabolic orbits, $\Delta E = 0$

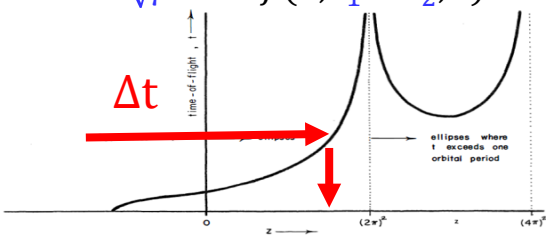




Source: Figs. 5.3-1 and 5.4-3 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

Iterative Methods

$\sqrt{\mu}\Delta t = f(a, r_1 + r_2, c)$



How do we solve for the given Δt ?

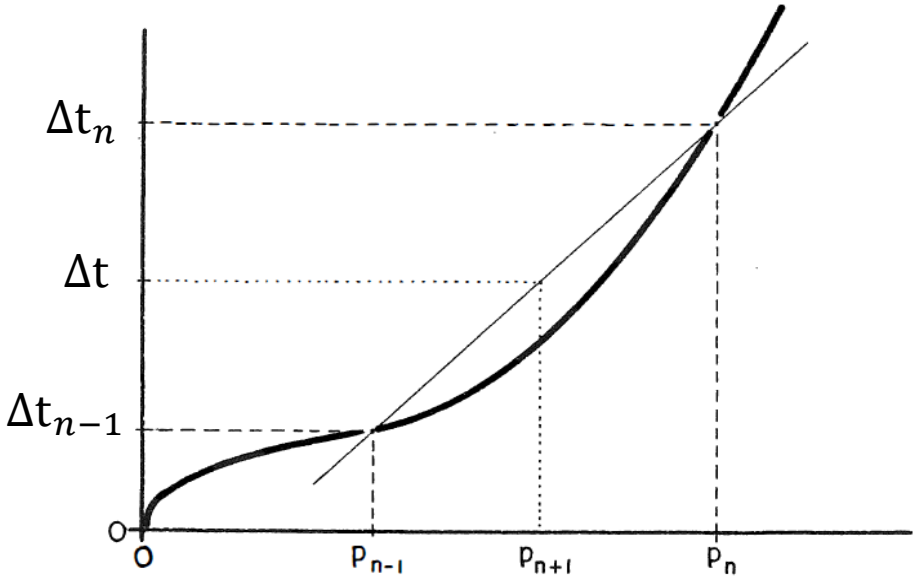
Bisection method we must find two trial values of p; one that gives too small a value for Δt, and one that gives too large a value.

The solution is then bracketed $[p_{low}, p_{\text{high}}]$ and, by choosing our next trial value half way between the first two, we can keep it bracketed while reducing the interval of uncertainty to some arbitrarily small value.

Linear interpolation method we choose two trial values of p: p_{n-1} and p_n with corresponding trial values times-of-flight Δt_{n-1} and Δt_n. Then we select a new value:

$$p_{n+1} = p_n + \frac{\Delta t - \Delta t_n}{\Delta t_n - \Delta t_{n-1}}(p_n - p_{n-1})$$

This scheme can be repeated, always retaining the latest two trial values of p (i.e. p_{n+1} and p_n) and their corresponding times-of-flight Δt_{n+1} and Δt_n for use in computing a still better trial value Δt_{n+2}.
It is not necessary that the initial two trial values bracket the answer.



Source: Fig. 5.4-4 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

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Astro-dynamics

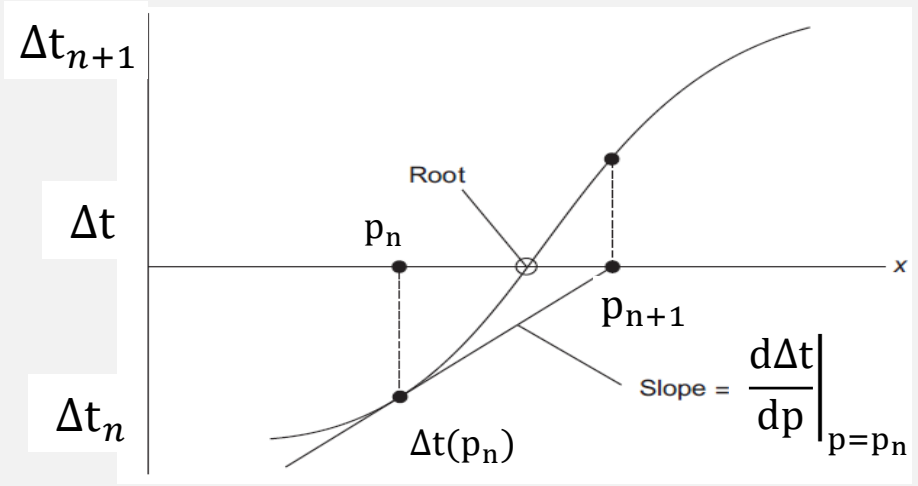
Adrià Rovira-Garcia

Iterative Householder Methods: we need to either obtain an expression for the derivative or compute it numerically

1st Order: Newton method, faster but requires that we compute the slope of the $\Delta t = \Delta t(p)$ at the last trial value of p.

If this last trial value is called p_n and Δt_n is the time-of-flight that results from it, then a better estimate of p may be obtained from

$$p_{n+1} = p_n + \frac{\Delta t - \Delta t_n}{\left.\frac{d\Delta t}{dp}\right|_{p=p_n}}$$



2nd Order: Halley method

$$p_{n+1} = p_n + \frac{(\Delta t - \Delta t_n)\left.\frac{d\Delta t}{dp}\right|_{p=p_n}}{\left(\left.\frac{d\Delta t}{dp}\right|_{p=p_n}\right)^2 + \frac{1}{2}\left.\frac{d^2\Delta t}{dp^2}\right|_{p=p_n}}$$

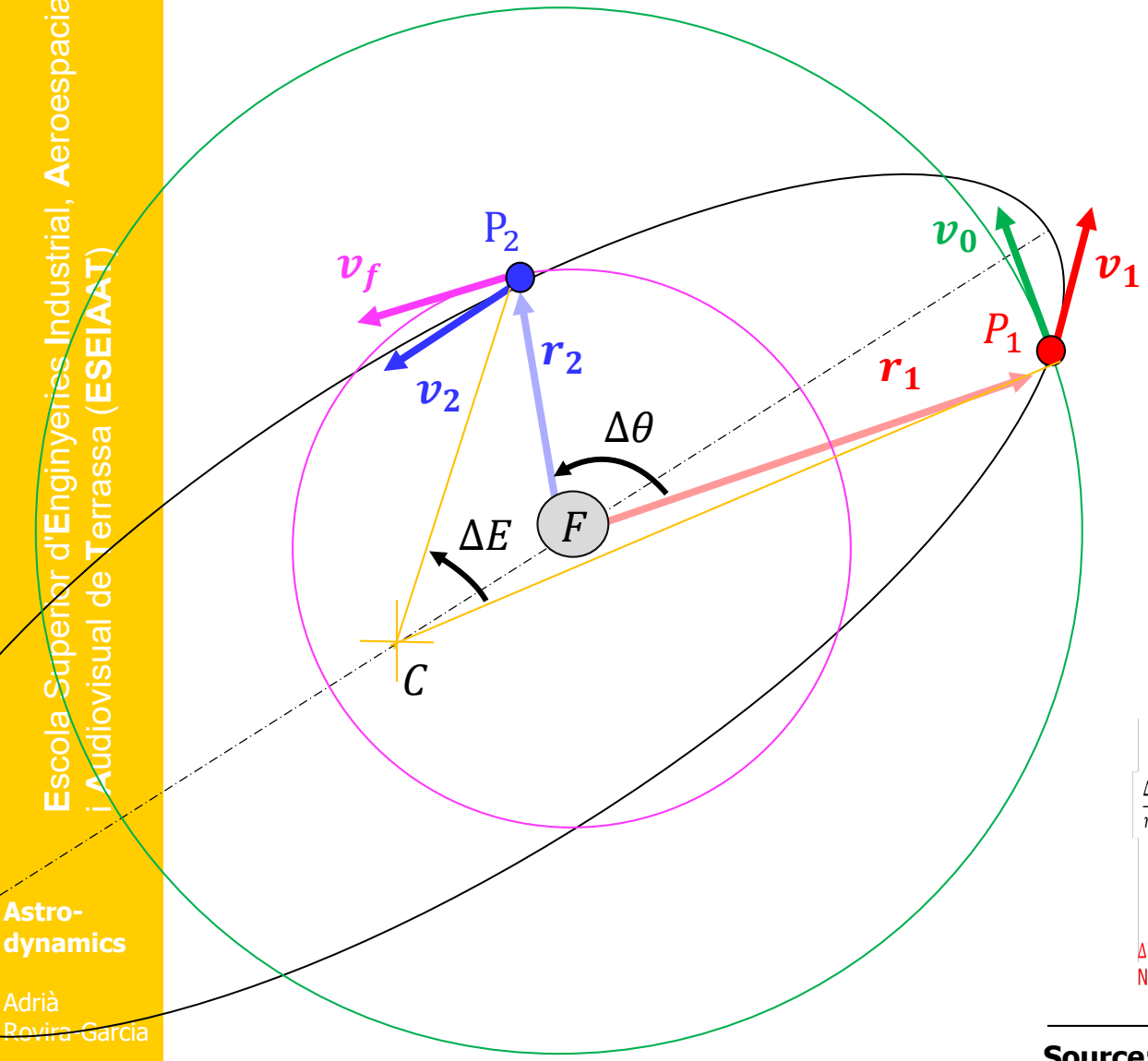
3rd Order

$$p_{n+1} = p_n \cdot (\Delta t - \Delta t_n)\frac{\left(\left.\frac{d\Delta t}{dp}\right|_{p=p_n}\right)^2 - \frac{1}{2}\left.\frac{d^2\Delta t}{dp^2}\right|_{p=p_n}}{\left.\frac{d\Delta t}{dp}\right|_{p=p_n} \cdot \left(\left(\left.\frac{d\Delta t}{dp}\right|_{p=p_n}\right)^2 - (\Delta t - \Delta t_n)\left.\frac{d^2\Delta t}{dp^2}\right|_{p=p_n}\right) + \frac{1}{6}\left.\frac{d^3\Delta t}{dp^3}\right|_{p=p_n} \cdot \frac{1}{6}\left.\frac{d^2\Delta t}{dp^2}\right|_{p=p_n}}$$

Source: Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications
Figure 3.7 of Curtis HD (2010) "Orbital Mechanics for Engineering Students" (3rd Edition) Elsevier
Householder AS (1970) "The Numerical Treatment of a Single Nonlinear Equation" McGraw-Hill

Terminal Velocities

Once solved for the given transfer time Δt, we need to compute the terminal velocities $\mathbf{v_1}$ and $\mathbf{v_2}$ at the boundaries of the orbital transfer from $\mathbf{P_1}$ to $\mathbf{P_2}$



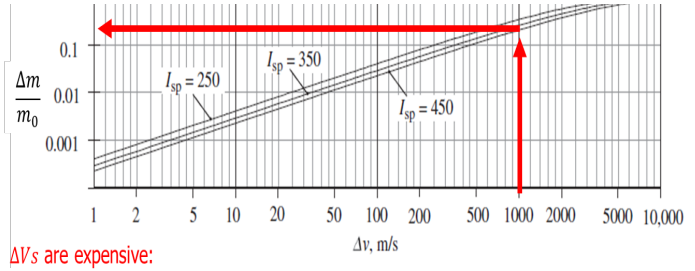
Velocity budget:

$$\Delta V = |\Delta V_1| + |\Delta V_2|$$

- $\Delta V_1 = \boldsymbol{v_1} - \boldsymbol{v_0}$ to embark the transfer orbit
- $\Delta V_2 = \boldsymbol{v_f} - \boldsymbol{v_2}$ to remind in the final orbit

Mass budget:

$$\frac{\Delta m}{m_0} = 1 - e^{-\frac{\Delta V}{I_{sp} \cdot g_{SL}}}$$

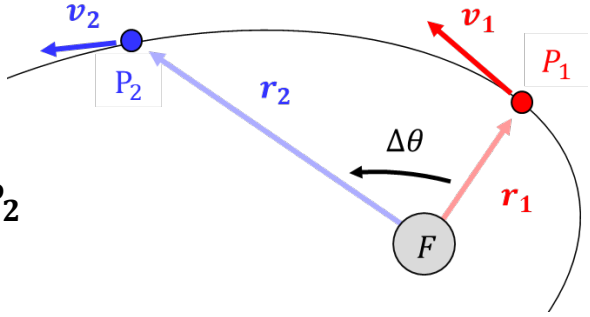


ΔVs are expensive:
Note that ΔV of ~1 km/s require propellant ~25% of the spacecraft mass before the burn

CCW ($t_m = +1$) as the Solar System

Source: Figure 6-1 Curtis HD (2014) "Orbital Mechanics for Engineering Students" (3rd Edition).

Terminal Velocities



Once solved the transfer time, compute terminal velocities $\mathbf{v_1}$ and $\mathbf{v_2}$ at the boundaries of the orbital transfer $\mathbf{P_1}$ and $\mathbf{P_2}$

Method 1 – Lagrange Coefficients

We have solved the a, e, p

$$\mathbf{v_1} = \frac{\mathbf{r_2} - f\mathbf{r_1}}{g} = \frac{\sqrt{\mu p}}{r_1 r_2 \sin\Delta\theta}\left[\mathbf{r_2} - \mathbf{r_1} + \frac{r_2}{p}\mathbf{r_1}(1-\cos\Delta\theta)\right]$$

$$\mathbf{v_2} = \frac{\dot{g}\mathbf{r_2} - \mathbf{r_1}}{g} = \frac{\sqrt{\mu p}}{r_1 r_2 \sin\Delta\theta}\left[\mathbf{r_2} - \mathbf{r_1} - \frac{r_1}{p}\mathbf{r_2}(1-\cos\Delta\theta)\right]$$

$$f = 1 - \frac{r_2}{p}(1-\cos\Delta\theta)$$

$$g = \frac{r_1 r_2}{\sqrt{\mu p}}\sin\Delta\theta$$

$$\dot{g} = 1 - \frac{r_1}{p}(1-\cos\Delta\theta)$$

Method 2 – Orbital Elements a, e, Ω, i, ω, θ

We have solved the a, e, p

$$\mathbf{v_1} = \sqrt{\frac{\mu}{p}}\begin{pmatrix} e\cdot\sin\theta_1 \\ 1 + e\cdot\cos\theta_1 \\ 0 \end{pmatrix}_{\mathbf{RSW}} \qquad \mathbf{v_2} = \sqrt{\frac{\mu}{p}}\begin{pmatrix} e\cdot\sin\theta_2 \\ 1 + e\cdot\cos\theta_2 \\ 0 \end{pmatrix}_{\mathbf{RSW}}$$

$$r_1 = \frac{p}{1 + e\cdot\cos\theta_1}$$

$$r_2 = \frac{p}{1 + e\cdot\cos\theta_2}$$

Method 3 – Radial and Along-track velocity components

We have solved the x and y

$$v_{r_1} = \gamma\frac{(\lambda y - x) - \rho(\lambda y + x)}{r_1} \qquad v_{\perp_1} = \gamma\,\sigma\frac{y + \lambda x}{r_1}$$

$$v_{r_2} = \gamma\frac{(\lambda y - x) - \rho(\lambda y + x)}{r_2} \qquad v_{\perp_2} = \gamma\,\sigma\frac{y + \lambda x}{r_2}$$

$$\mathbf{v_1} = v_{r_1}\frac{\mathbf{r_1}}{r_1} + v_{\perp_1}\frac{\mathbf{h}\times\mathbf{r_1}}{|\mathbf{h}\times\mathbf{r_1}|} \qquad \mathbf{v_2} = v_{r_2}\frac{\mathbf{r_2}}{r_2} + v_{\perp_2}\frac{\mathbf{h}\times\mathbf{r_2}}{|\mathbf{h}\times\mathbf{r_2}|}$$

$$\gamma = \sqrt{\frac{\mu s}{2}}$$

$$\rho = \frac{r_1 - r_2}{c}$$

$$\sigma = \sqrt{1-\rho^2}$$

$$\lambda^2 = 1 - \frac{c}{s}$$

$$h = |\mathbf{r}\times\mathbf{v}| \approx |\mathbf{r_1}\times\mathbf{r_2}|$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

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Time Allocation:

- 8 hours

Simó Method (1973)

Newton gravitational equation of motion of the two body problem is regularized (i.e. elimination of singularities) by adopting properly selected variables:

$$\ddot{\mathbf{r}} + \frac{\mu}{r^3}\mathbf{r} = 0$$

Introduce fictitious time "s" [s/m] (Sundman transformation) to eliminate $r \to 0$ singularity

$$dt = \frac{r}{\tau} ds$$

whose derivatives:

$$\frac{d}{dt} = \frac{1}{r}\frac{d}{ds}$$

r: distance [m] between the primary body and the orbiting mass

τ: constant [unitless] for scaling purposes, that can be set to $\tau = 1$

$$\frac{d^2}{dt^2} = -\frac{1}{r^2}\frac{dr}{dt}\frac{d}{ds} + \frac{1}{r}\frac{d}{dt}\left(\frac{d}{ds}\right) = -\frac{1}{r^2}\left(\frac{1}{r}\frac{dr}{ds}\right)\frac{d}{ds} + \frac{1}{r}\left(\frac{1}{r}\frac{d}{ds}\frac{d}{ds}\right) = \frac{1}{r^3}\left(-r'\frac{d}{ds} + r\frac{d^2}{ds^2}\right)$$

Conformal squaring: Transform between the physical $\mathbf{r}$ plane and complex $\mathbf{u}$ plane:

$$\mathbf{r} = \mathbf{u}^2$$

whose derivatives:

$$\mathbf{r}' = 2\,\mathbf{u}\cdot\mathbf{u}'$$
$$\mathbf{r}'' = 2\,(\mathbf{u}'\cdot\mathbf{u}' + \mathbf{u}\cdot\mathbf{u}'')$$

$\mathbf{u}'$: differentiation of $\mathbf{u}$ with respect to s

$\mathbf{u}''$: differentiation of $\mathbf{u}'$ with respect to s

with magnitude:

$$r = |\mathbf{u}|^2 = \mathbf{u}\cdot\bar{\mathbf{u}}$$
$$r' = \mathbf{u}'\cdot\bar{\mathbf{u}} + \mathbf{u}\cdot\bar{\mathbf{u}}'$$

$\bar{\mathbf{u}}$: the complex conjugate of $\mathbf{u}$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"
Simó C (1973) "Solución al problema de Lambert mediante regularización" Collectanea Mathematica 24(3):231-248

Simó Method (1973)

Newton's gravitational equation of motion:

$$\frac{d^2}{dt^2}(\boldsymbol{r}) + \frac{\mu}{\mathrm{r}^3}\mathbf{r} = 0$$

Is then transformed as

$$\frac{1}{\mathrm{r}^3}\left(-r'\frac{d}{ds}(\boldsymbol{r}) + r\frac{d^2}{ds^2}(\boldsymbol{r})\right) + \frac{\mu}{\mathrm{r}^3}\mathbf{r} = 0$$

$$-r'\frac{d}{ds}(\boldsymbol{r}) + r\frac{d^2}{ds^2}(\boldsymbol{r}) + \mu\mathbf{r} = 0$$

$$-r'\boldsymbol{r}' + r\boldsymbol{r}'' + \mu\mathbf{r} = 0$$

$$-(\boldsymbol{u}' \cdot \bar{\boldsymbol{u}} + \boldsymbol{u} \cdot \bar{\boldsymbol{u}}')(2\,\boldsymbol{u} \cdot \boldsymbol{u}') + (\boldsymbol{u} \cdot \bar{\boldsymbol{u}}) \cdot 2\,(\boldsymbol{u}' \cdot \boldsymbol{u}' + \boldsymbol{u} \cdot \boldsymbol{u}'') + \mu\boldsymbol{u}^2 = 0$$

$$2(\boldsymbol{u} \cdot \bar{\boldsymbol{u}} \cdot \boldsymbol{u}'') - 2(\boldsymbol{u} \cdot \boldsymbol{u}' \cdot \bar{\boldsymbol{u}}') + \mu\boldsymbol{u} = 0$$

$$2(\bar{\boldsymbol{u}} \cdot \boldsymbol{u}'') - 2(\boldsymbol{u}' \cdot \bar{\boldsymbol{u}}') + \mu = 0$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó Method (1973)

Introduce the specific mechanical energy -2ρ as a second independent equation:

$$\frac{1}{2}\mathrm{v}^2 - \frac{\mu}{r} = -2\rho$$

Then,

$$\frac{1}{2}\left|\frac{d\boldsymbol{r}}{dt}\right|^2 - \frac{\mu}{r} = -2\rho$$

$$\frac{1}{2}\left|\frac{1}{r}\frac{d\boldsymbol{r}}{ds}\right|^2 - \frac{\mu}{r} = -2\rho$$

$$\frac{1}{2\mathrm{r}^2}|\boldsymbol{r}'|^2 - \frac{\mu}{r} = -2\rho$$

Recalling that $|\boldsymbol{u}|^2 = \boldsymbol{u}\cdot\bar{\boldsymbol{u}}$

$$\frac{1}{2\mathrm{r}^2}(\boldsymbol{r}'\cdot\bar{\boldsymbol{r}}') - \frac{\mu}{r} = -2\rho$$

$$\frac{1}{2r}(\boldsymbol{r}'\cdot\bar{\boldsymbol{r}}') = \mu - 2\rho r$$

$$\frac{1}{2}\frac{(2\boldsymbol{u}\cdot\boldsymbol{u}')(2\bar{\boldsymbol{u}}\cdot\bar{\boldsymbol{u}}')}{\boldsymbol{u}\cdot\bar{\boldsymbol{u}}} = \mu - 2\rho\boldsymbol{u}\cdot\bar{\boldsymbol{u}}$$

$$2\boldsymbol{u}'\cdot\bar{\boldsymbol{u}}' = \mu - 2\rho\boldsymbol{u}\cdot\bar{\boldsymbol{u}}$$

$$2(\boldsymbol{u}'\cdot\bar{\boldsymbol{u}}') + 2\rho(\boldsymbol{u}\cdot\bar{\boldsymbol{u}}) - \mu = 0$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó Method (1973)

Merge the two previous equations to obtain a compact differential equation of motion:

- Newton gravitational equation of motion
- The specific mechanical energy

The transformed differential equation is linear.

$$2(\bar{\boldsymbol{u}} \cdot \boldsymbol{u}'') - 2(\boldsymbol{u}' \cdot \bar{\boldsymbol{u}}') + \mu = 0$$

$$+ \quad 2(\boldsymbol{u}' \cdot \bar{\boldsymbol{u}}') + 2\rho(\boldsymbol{u} \cdot \bar{\boldsymbol{u}}) - \mu = 0$$

$$2(\bar{\boldsymbol{u}} \cdot \boldsymbol{u}'') + 2\rho(\boldsymbol{u} \cdot \bar{\boldsymbol{u}}) = 0 \quad \rightarrow \quad 2 \cdot \bar{\boldsymbol{u}} \cdot (\boldsymbol{u}'' + \rho \boldsymbol{u}) = 0 \quad \rightarrow \quad \boldsymbol{u}'' + \rho \boldsymbol{u} = 0$$

Given the initial conditions $\boldsymbol{u_0}$ and $\boldsymbol{u_0'}$ at $s = s_0$, the solution to the differential equation of motion depends on the specific mechanical energy ρ:

$$\boldsymbol{u}(s) \begin{cases} \rho > 0 \text{ (ellipse)} & \boldsymbol{u} = \boldsymbol{u_0} \cos[\sqrt{\rho}(s - s_0)] + \boldsymbol{u_0'} \dfrac{1}{\sqrt{\rho}} \sin[\sqrt{\rho}(s - s_0)] \\ \rho = 0 \text{ (parabola)} & \boldsymbol{u} = \boldsymbol{u_0} + \boldsymbol{u_0'}(s - s_0) \\ \rho < 0 \text{ (hyperbola)} & \boldsymbol{u} = \boldsymbol{u_0} \cosh[\sqrt{-\rho}(s - s_0)] + \boldsymbol{u_0'} \dfrac{1}{\sqrt{-\rho}} \sinh[\sqrt{-\rho}(s - s_0)] \end{cases}$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó Method (1973)

We can use a Taylor expansion

$$f(x) = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot \left. \frac{d^n f(x)}{dx^n} \right|_{x=x_0} \cdot (x - x_0)^n$$

of $\boldsymbol{u}(s)$ around s_0 valid for all three conics (i.e. universal):

$$\boldsymbol{u}(s) = \sum_{k=0}^{\infty} \frac{\boldsymbol{u}^{(k)}(s_0)}{k!}(s - s_0)^k = \sum_{j=0}^{\infty} \frac{\boldsymbol{u}^{(2j)}(s_0)}{(2j)!}(s - s_0)^{2j} + \sum_{j=0}^{\infty} \frac{\boldsymbol{u}^{(2j+1)}(s_0)}{(2j+1)!}(s - s_0)^{(2j+1)}$$

From the previous transformed **linear** differential equation of motion $\boldsymbol{u}'' + \rho\boldsymbol{u} = 0$

$$\boldsymbol{u}'' = -\rho\boldsymbol{u}$$

Then,

$$\boldsymbol{u}^{(2j)}(s_0) = (-\rho)^j \boldsymbol{u_0}$$

Its derivative,

$$\boldsymbol{u}^{(2j+1)}(s_0) = (-\rho)^j \boldsymbol{u_0'}$$

$$= \boldsymbol{u_0} \sum_{j=0}^{\infty} \frac{(-1)^j \rho^j (s - s_0)^{2j}}{(2j)!} + \boldsymbol{u_0'}(s - s_0) \sum_{j=0}^{\infty} \frac{(-1)^j \rho^j (s - s_0)^{2j}}{(2j+1)!}$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Introducing the Stumpff functions

$$c_k(x) = \frac{1}{k!} - \frac{x}{(k+2)!} + \frac{x^2}{(k+4)!} + \cdots = \sum_{j=0}^{\infty} \frac{(-1)^j x^j}{(k+2j)!} \qquad \text{with } k = 0,1,2,3,\ldots$$

$$c_0(x) = \sum_{j=0}^{\infty} \frac{(-1)^j x^j}{(0+2j)!} = \begin{cases} x > 0 & \cos\sqrt{x} \\ x = 0 & 1 \\ x < 0 & \cosh\sqrt{-x} \end{cases}$$

$$c_1(x) = \sum_{j=0}^{\infty} \frac{(-1)^j x^j}{(1+2j)!} = \begin{cases} x > 0 & \sin\sqrt{x}/\sqrt{x} \\ x = 0 & 1 \\ x < 0 & \sinh\sqrt{-x}/\sqrt{-x} \end{cases}$$

$$c_{k+2}(x) = \frac{\frac{1}{k!} - c_k(x)}{x}$$

We can identify c_0 and c_1 of argument $\mathrm{x} = \rho \cdot (s - s_0)^2$:

$$\boldsymbol{u}(s) = \boldsymbol{u_0} \sum_{j=0}^{\infty} \frac{(-1)^j \rho^j (s-s_0)^{2j}}{(2j)!} + \boldsymbol{u_0'}(s-s_0) \sum_{j=0}^{\infty} \frac{(-1)^j \rho^j (s-s_0)^{2j}}{(1+2j)!}$$

$$\boldsymbol{u}(s) = \boldsymbol{u_0} c_0(\rho(s-s_0)^2) + \boldsymbol{u_0'}\,\rho(s-s_0) c_1(\rho(s-s_0)^2)$$

In the fictitious transfer time $\Delta s = s - s_0$ which we can assume $s_0 = 0$:

$$\boldsymbol{u}(s) = \boldsymbol{u_0} c_0(\rho s^2) + \boldsymbol{u_0'}\,\rho s \cdot c_1(\rho s^2)$$

Its derivative

$$\boldsymbol{u}'(s) = \boldsymbol{u_0'} c_0(\rho s^2) + \boldsymbol{u_0''}\,\rho s \cdot c_1(\rho s^2)$$

$$\boldsymbol{u}'(s) = \boldsymbol{u_0'} c_0(\rho s^2) - \boldsymbol{u_0}\,\rho s \cdot c_1(\rho s^2)$$

Source: Stumpff K (1956) "Himmelsmechanik" Vol 1, Deutscher Verlag der Wissenschaften, Berlin, 1956.

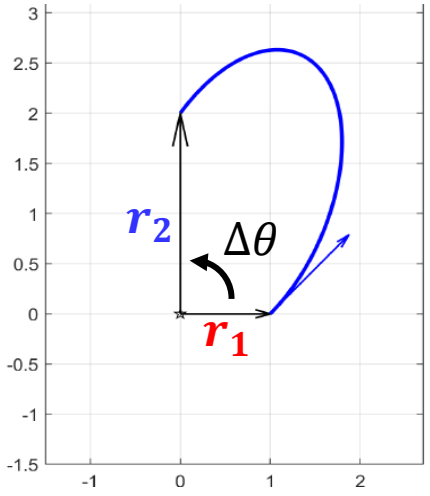
Given the time of transfer $\Delta t = t_2 - t_1$ from $\boldsymbol{r_1}$ to $\boldsymbol{r_2}$, and its associated vectors $\boldsymbol{u_1}$ and $\boldsymbol{u_2}$ in the complex plane:

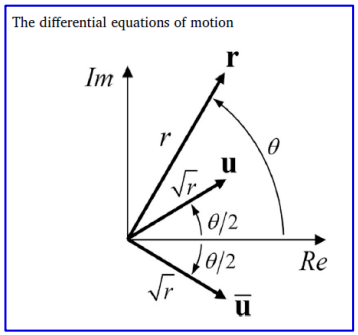
$$\boldsymbol{u_1^2} = \boldsymbol{r_1}$$
$$\boldsymbol{u_2^2} = \boldsymbol{r_2}$$

Built as

$$\boldsymbol{u_1} = \begin{pmatrix} A \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{r_1} \\ 0 \end{pmatrix}$$

$$\boldsymbol{u_2} = \begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} \sqrt{r_2}\cos(\Delta\theta/2) \\ \sqrt{r_2}\sin(\Delta\theta/2) \end{pmatrix}$$





Since the angle $\Delta\theta = \langle \boldsymbol{r_1}, \boldsymbol{r_2} \rangle = 2\langle \boldsymbol{u_1}, \boldsymbol{u_2} \rangle$ as $z^2 = \left(re^{i\theta}\right)^2 = r^2 e^{i2\theta}$

So that we can determine $\boldsymbol{u}(s)$ and $\boldsymbol{u}'(s)$ components $\boldsymbol{u_0}$ and $\boldsymbol{u_0'}$

$$\boldsymbol{u}(s) = \boldsymbol{u_0} c_0(\rho s^2) + \boldsymbol{u_0'}\, \rho s \cdot c_1(\rho s^2)$$
$$\boldsymbol{u}'(s) = \boldsymbol{u_0'} c_0(\rho s^2) - \boldsymbol{u_0}\, \rho s \cdot c_1(\rho s^2)$$

built as

$$\boldsymbol{u_0} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

$$\boldsymbol{u_0'} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

from

$$\begin{cases} \boldsymbol{u_1} = \boldsymbol{u}(s_1) = \boldsymbol{u_0} c_0(0) \\ \boldsymbol{u_2} = \boldsymbol{u}(s_2) = \boldsymbol{u_0} c_0(\rho s_2^2) + \boldsymbol{u_0'} s_2 c_1(\rho s_2^2) \end{cases} \quad \begin{cases} \begin{pmatrix} A \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} c_0(0) \\ \begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} c_0(\rho s_2^2) + \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} s_2 c_1(\rho s_2^2) \end{cases} \rightarrow \begin{cases} \alpha_1 = A \\ \alpha_2 = 0 \\ \beta_1 = \dfrac{B - A c_0(\rho s_2^2)}{s_2 c_1(\rho s_2^2)} \\ \beta_2 = \dfrac{C}{s_2 c_1(\rho s_2^2)} \end{cases}$$

Source: Fig 2. De La Torre, Flores R, Fantino E (2018) On the solution of Lambert's problem by regularization Acta Astronautica 163:26-38

Simó Method (1973)

This provides the expression for the distance $r(s)$:

$$r(s) = \boldsymbol{u} \cdot \boldsymbol{u} = \boldsymbol{u_0} \cdot \boldsymbol{u_0} \cdot c_0^2(\rho s^2) + \boldsymbol{u_0'} \cdot \boldsymbol{u_0'} \cdot s^2 \cdot c_1^2(\rho s^2) + 2 \cdot \boldsymbol{u_0} \cdot \boldsymbol{u_0'} \cdot c_0(\rho s^2) \cdot s \cdot c_1(\rho s^2)$$

Note that

$$\boldsymbol{u_0} \cdot \boldsymbol{u_0} = \alpha_1^2 + \alpha_2^2$$
$$\boldsymbol{u_0'} \cdot \boldsymbol{u_0'} = \beta_1^2 + \beta_2^2$$
$$\boldsymbol{u_0} \cdot \boldsymbol{u_0'} = \alpha_1 \cdot \beta_1 + \alpha_2 \cdot \beta_2$$

Then, the transfer time can be computed from the transformation $\mathrm{dt} = \frac{\mathrm{r}}{\tau}\mathrm{ds}$:

$$\Delta t = t_2 - t_1 = \int_0^{s_2} r\, ds$$
$$= (\alpha_1^2 + \alpha_2^2) \int_0^{s_2} c_0^2(\rho s^2)\, ds$$
$$+(\beta_1^2 + \beta_2^2) \int_0^{s_2} s^2 \cdot c_1^2(\rho s^2)\, ds$$
$$+2(\alpha_1 \cdot \beta_1 + \alpha_2 \cdot \beta_2) \int_0^{s_2} c_0(\rho s^2) \cdot s \cdot c_1(\rho s^2)\, ds$$

Integrating:

$$\Delta t = A^2 s_2 - 2 \cdot A^2 \cdot \rho \cdot s_2^3 \cdot c_3(4 \cdot \rho s_2^2)$$
$$+ \frac{[B^2 + C^2 + A^2 c_0^2(\rho s_2^2) - AB c_0(\rho s_2^2)] \cdot 2 s_2 c_3(4 \cdot \rho s_2^2)}{c_1^2(\rho s_2^2)}$$
$$+ \frac{2A[B - A c_0(\rho s_2^2)] \cdot s_2 c_2(4 \cdot \rho s_2^2)}{c_1(\rho s_2^2)}$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó Method (1973)

This provides the expression for the distance $r(s)$:

$$r(s) = \boldsymbol{u} \cdot \boldsymbol{u} = \boldsymbol{u_0} \cdot \boldsymbol{u_0} \cdot c_0^2(\rho s^2) + \boldsymbol{u_0'} \cdot \boldsymbol{u_0'} \cdot s^2 \cdot c_1^2(\rho s^2) + 2 \cdot \boldsymbol{u_0} \cdot \boldsymbol{u_0'} \cdot c_0(\rho s^2) \cdot s \cdot c_1(\rho s^2)$$

Note that

$$\boldsymbol{u_0} \cdot \boldsymbol{u_0} = \alpha_1^2 + \alpha_2^2$$
$$\boldsymbol{u_0'} \cdot \boldsymbol{u_0'} = \beta_1^2 + \beta_2^2$$
$$\boldsymbol{u_0} \cdot \boldsymbol{u_0'} = \alpha_1 \cdot \beta_1 + \alpha_2 \cdot \beta_2$$

Then, the transfer time can be computed from the transformation $\mathrm{dt} = \frac{\mathrm{r}}{\tau}\mathrm{ds}$:

$$\Delta t = t_2 - t_1 = \int_0^{s_2} r\, ds$$
$$= (\alpha_1^2 + \alpha_2^2) \int_0^{s_2} c_0^2(\rho s^2)\, ds$$
$$+(\beta_1^2 + \beta_2^2) \int_0^{s_2} s^2 \cdot c_1^2(\rho s^2)\, ds$$
$$+2(\alpha_1 \cdot \beta_1 + \alpha_2 \cdot \beta_2) \int_0^{s_2} c_0(\rho s^2) \cdot s \cdot c_1(\rho s^2)\, ds$$

Integrating:

$$\Delta t = A^2 s_2 - 2 \cdot A^2 \cdot \rho \cdot s_2^3 \cdot c_3(4 \cdot \rho s_2^2)$$
$$+ \frac{[B^2 + C^2 + A^2 c_0^2(\rho s_2^2) - AB c_0(\rho s_2^2)] \cdot 2 s_2 c_3(4 \cdot \rho s_2^2)}{c_1^2(\rho s_2^2)}$$
$$+ \frac{2A[B - A c_0(\rho s_2^2)] \cdot s_2 c_2(4 \cdot \rho s_2^2)}{c_1(\rho s_2^2)}$$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó Method (1973)

We multiply both sides by $\frac{c_1^2(\rho s_2^2)}{s_2}$

$$\frac{c_1^2(\rho s_2^2)}{s_2}\cdot\Delta t = A^2c_1^2(\rho s_2^2) - 2\cdot A^2\cdot\rho\cdot s_2^2\cdot c_3(4\cdot\rho s_2^2)c_1^2(\rho s_2^2)$$
$$+[B^2+C^2+A^2c_0^2(\rho s_2^2) - ABc_0(\rho s_2^2)]\cdot 2c_3(4\cdot\rho s_2^2)$$
$$+2A[B - Ac_0(\rho s_2^2)]\cdot c_2(4\cdot\rho s_2^2)\cdot c_1(\rho s_2^2)$$

Denoting $z = \rho s_2^2$

$$\frac{c_1^2(z)}{s_2}\cdot\Delta t = A^2c_1^2(z) - 2\cdot A^2\cdot z\cdot c_3(4\cdot z)c_1^2(z)$$
$$+[B^2+C^2+A^2c_0^2(z) - ABc_0(z)]\cdot 2c_3(4\cdot z)$$
$$+2A[B - Ac_0(z)]\cdot c_2(4\cdot z)\cdot c_1(z)$$

Noting that

$$c_1^2(z) = 2\cdot c_2\,(4\cdot z)$$
$$c_1(4\cdot z) = 1 - 4\cdot z\cdot c_3(4\cdot z)$$
$$c_3(4\cdot z)\cdot c_0^2(z) = c_3(4\cdot z)\cdot[1 - 2\cdot z\cdot c_2(4\cdot z)]$$
$$c_0(z)\cdot c_2(4\cdot z)\cdot c_1(z) = c_2(4\cdot z)\cdot c_1(4\cdot z)$$

The transfer time equation becomes:

$$\frac{c_1^2(z)}{s_2}\cdot\Delta t = 2\cdot P\cdot c_3(4\cdot z) + 2AB[c_1(z)\cdot c_2(4\cdot z) - 2\cdot c_0(z)\cdot c_3(4\cdot z)]$$

with $P = A^2 + B^2 + C^2 = r_1 + r_2$

Source: De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó (1973): transfer time equation

We eliminate the dependence from s_2 using again the specific mechanical energy

$$\mu = 2\rho\left[\frac{P - 2\cdot A\cdot B\cdot c_0(z)}{z\cdot[c_1(z)]^2}\right]$$

Using the identity $c_0^2(z) + z\cdot c_1^2(z) = 1$ and defining

$$Q = 2\cdot A\cdot B = 2\sqrt{r_1\cdot r_2}\cos\left(\frac{\Delta\theta}{2}\right)$$

We finally arrive to singularity-free and universal transfer time $\Delta t = \Delta t(z, r_1, r_2, \Delta\theta)$

$$\Delta t = \frac{2\cdot P\cdot c_3(4\cdot z) + Q\cdot[c_1(z)\cdot c_2(4\cdot z) - 2\cdot c_0(z)\cdot c_3(4\cdot z)]}{[c_1(z)]^3}\sqrt{2\frac{P - Q\cdot c_0(z)}{\mu}}$$

Recall:

$$A = \sqrt{r_1} \qquad B = \sqrt{r_2}\cos\left(\frac{\Delta\theta}{2}\right) \qquad C = \sqrt{r_2}\sin\left(\frac{\Delta\theta}{2}\right) \qquad P = A^2 + B^2 + C^2 = r_1 + r_2$$

$$z = \rho s_2^2 \qquad -2\rho = \frac{v^2}{2} - \frac{\mu}{r}$$

$$c_0(z) = \begin{cases} z > 0 & \cos\sqrt{z} \\ z = 0 & 1 \\ z < 0 & \cosh\sqrt{-z} \end{cases} \qquad c_1(z) = \begin{cases} z > 0 & \sin\sqrt{z}/\sqrt{z} \\ z = 0 & 1 \\ z < 0 & \sinh\sqrt{-z}/\sqrt{-z} \end{cases}$$

$$c_2(z) = \frac{1 - c_0(z)}{z} \qquad c_3(z) = \frac{1 - c_1(z)}{z}$$

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Simó (1973): transfer time equation

We solve Δt given a geometry and the orbital energy ρ

$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q \cdot [c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{[c_1(z)]^3} \sqrt{2 \frac{P - Q \cdot c_0(z)}{\mu}}$$

The semi-major axis a of the conic solution is determined by:

$$a = \frac{P - Q \cdot c_0(z)}{2 \cdot z \cdot [c_1(z)]^2}$$

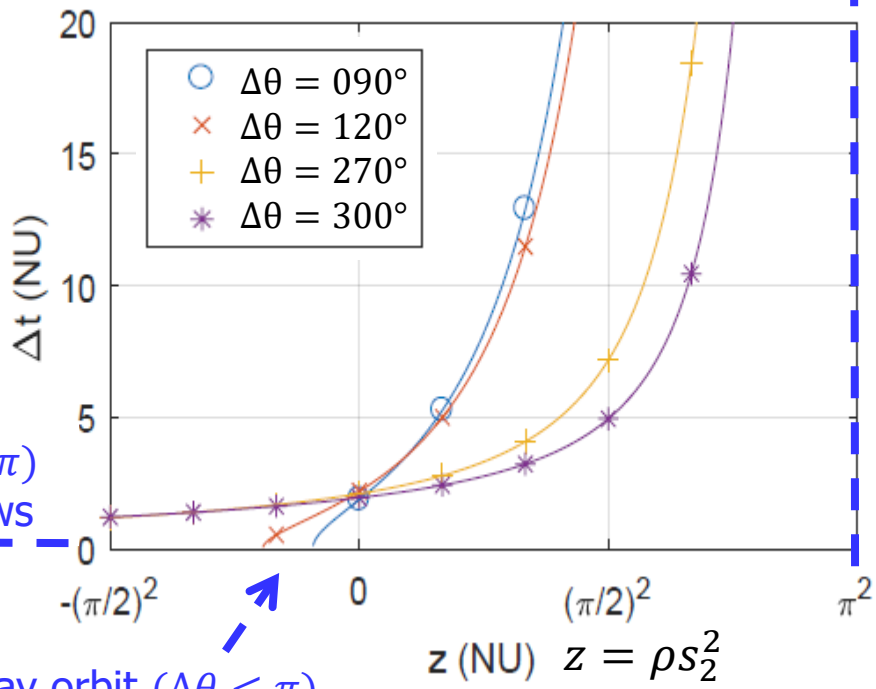
$a > 0$ for ellipses
$a = 0$ straight line ($\Delta t = 0$ and infinite velocity)
$a < 0$ for hyperbolas

The conic eccentricity:

$$e^2 = 1 - \frac{A^2 \cdot C^2}{a^2 \cdot z \cdot [c_1(z)]^2}$$

The conic p parameter (semilatus rectum):

$$p = \frac{h^2}{\mu} = \frac{2 \cdot A^2 \cdot C^2}{P - Q \cdot c_0(z)}$$



Assymptote parabolic orbit at $z = \pi^2$
$\Delta t = a = \infty$ and $e = 1$ since $c_1(\pi^2) = 0$

Assymptote long way orbit ($\pi < \Delta\theta < 2\pi$) at $\Delta t = 0$ → Hyperbola eccentricity grows

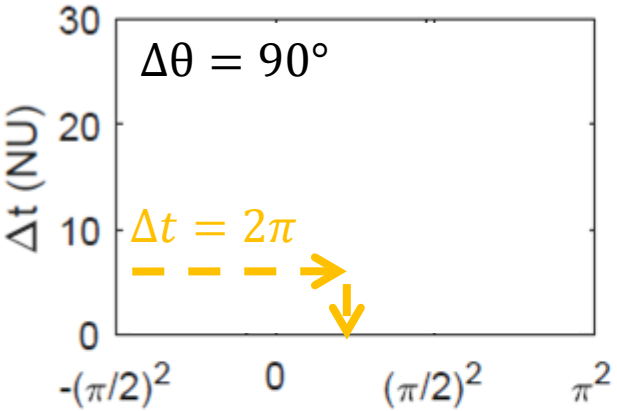
short way orbit ($\Delta\theta < \pi$) straight line (hyperbola $\Delta t = 0$ with $e = \infty$)

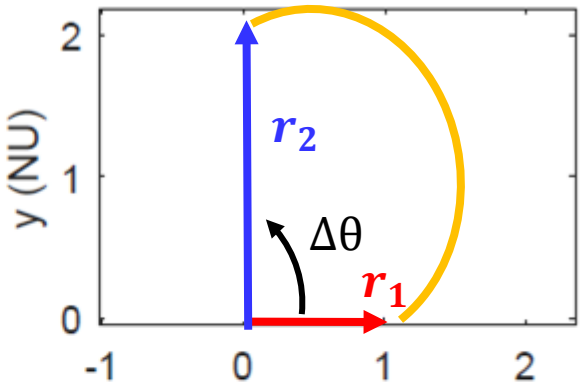
Source: Fig.3.2 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Simó (1973): transfer time equation

We solve Δt given a geometry and the orbital energy ρ

$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q[c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{[c_1(z)]^3} \sqrt{2 \frac{P - Q \cdot c_0(z)}{\mu}}$$

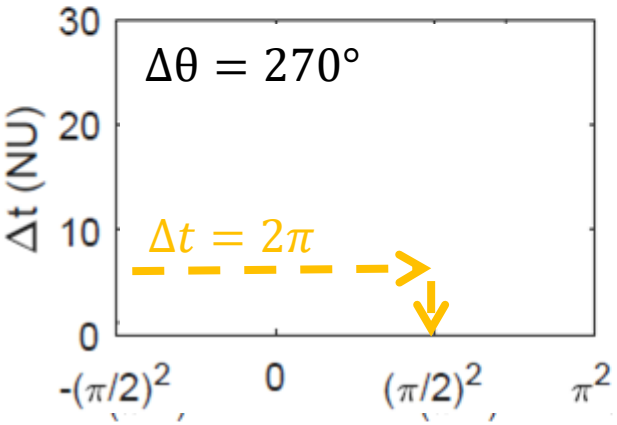


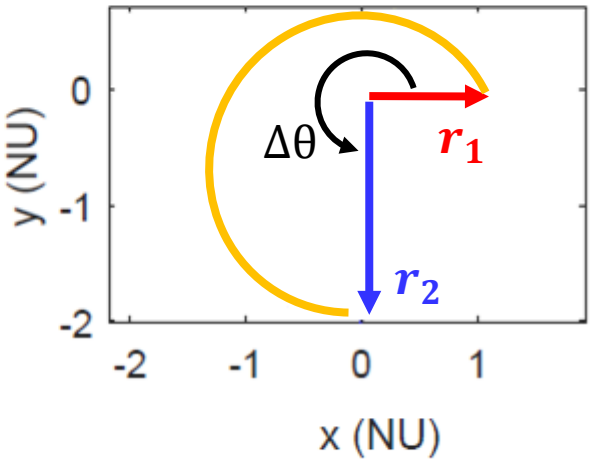


Counter Clock Wise

↓

short way $\Delta\theta < 180°$





Counter Clock Wise

↓

long way $\Delta\theta > 180°$

Source: Fig.3.4 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Geometrical meaning of $z = \rho s_2^2$

Elliptical Orbits

$$M = E - e \cdot sin\, E$$

$$\dot{M} = \dot{E} - e \cdot cos\, E \cdot \dot{E} = (1 - e \cdot \cos E)\frac{\mathrm{dE}}{dt} = \sqrt{\frac{\mu}{a^3}}$$

Since $r = a \cdot (1 - e \cdot \cos E)$

$$\frac{r}{a}\frac{\mathrm{dE}}{dt} = \sqrt{\frac{\mu}{a^3}}$$

$$r \cdot \mathrm{dE} = \sqrt{\frac{\mu}{a}} \cdot dt$$

Since $2\rho = \frac{v^2}{2} - \frac{\mu}{r} = \frac{\mu}{2a}$

$$r \cdot \mathrm{dE} = \sqrt{4\rho} \cdot dt$$

We again use the fictitious time "s" $dt = \frac{r}{\tau} ds$ with $\tau = 1$

$$r \cdot \mathrm{dE} = 2\sqrt{\rho} \cdot r \cdot ds$$

Integrating

$$\int_{E_1}^{E_2} \mathrm{dE} = 2 \int_{s_1=0}^{s_2} \sqrt{\rho} \cdot ds$$

$$\frac{\Delta E}{2} = \sqrt{\rho} \cdot s_2 = \sqrt{z}$$

$\sqrt{z}$ is equal to the half of the variation in eccentric anomaly between P_1 and P_2 on the solution orbit

Hyperbolic Orbits

Same rationale starting from $M = e \cdot sinh\, H - H$ → $\frac{\Delta H}{2} = \sqrt{-\rho} \cdot s_2 = \sqrt{-z}$

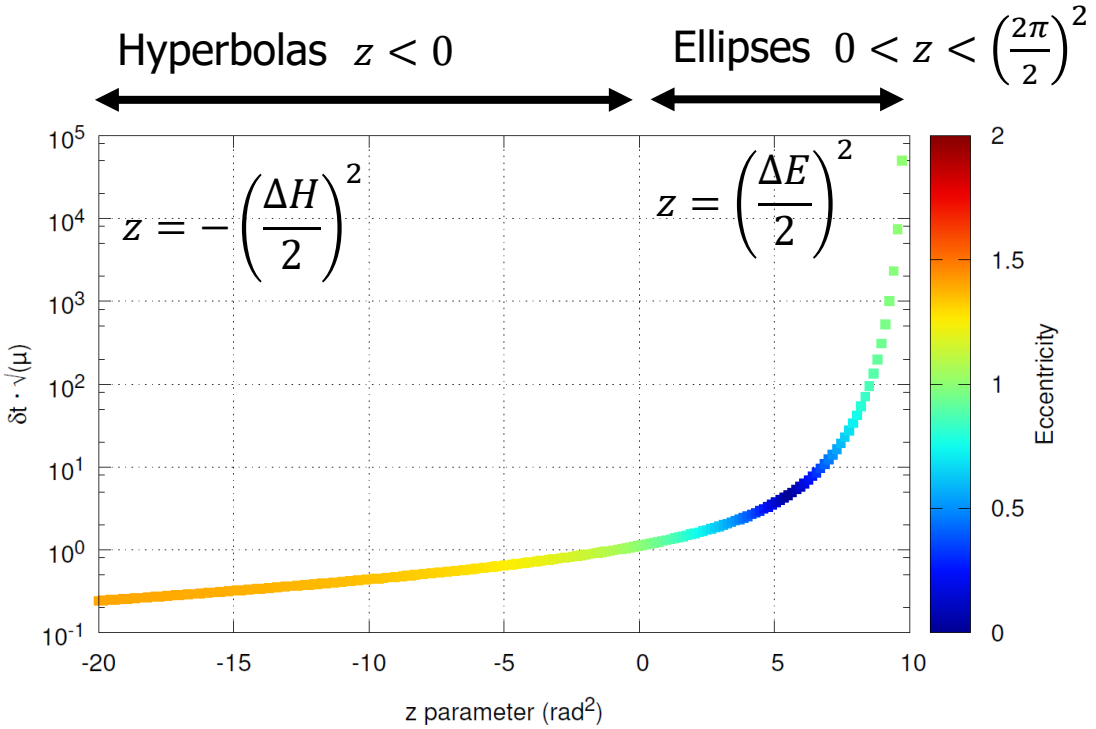
Parabolic Orbits

Limit case: $z = 0$ because $\rho = 0$

Geometrical meaning of $z = \rho s_2^2$

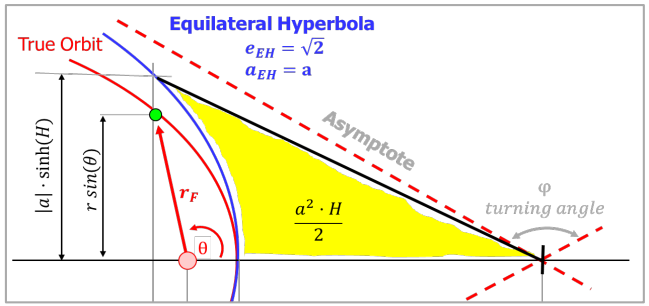
We solve Δt given a geometry and the orbital energy ρ

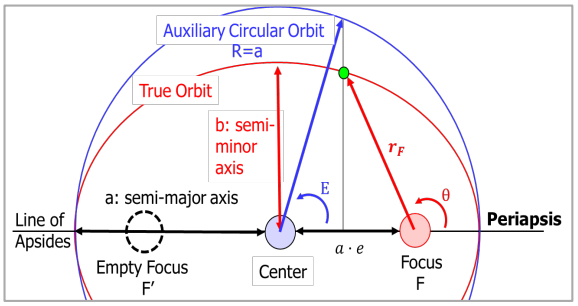
$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q \cdot [c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{[c_1(z)]^3} \sqrt{2 \frac{P - Q \cdot c_0(z)}{\mu}}$$



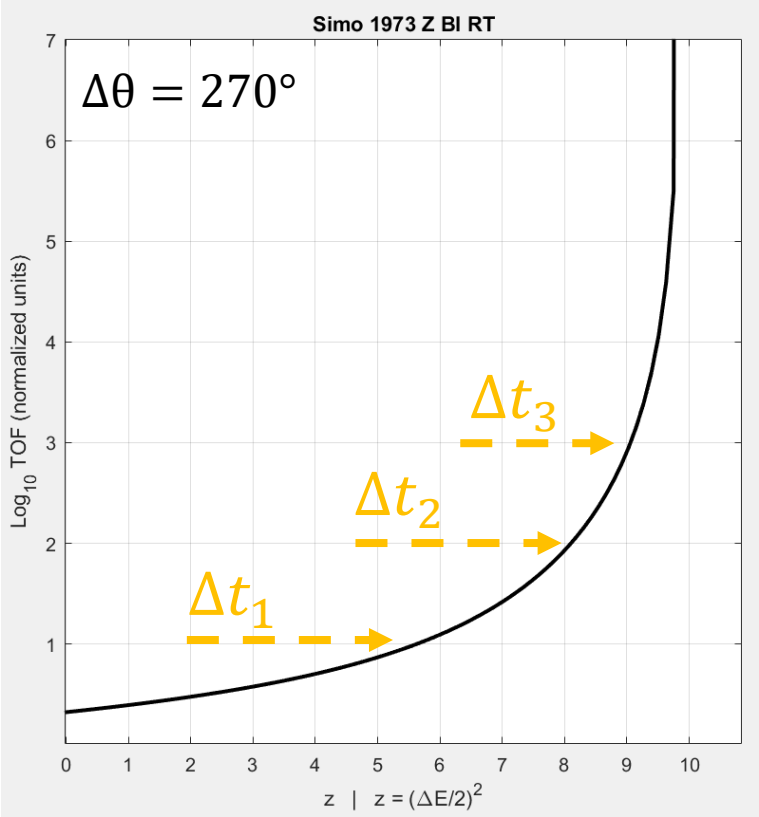
The conic eccentricity:

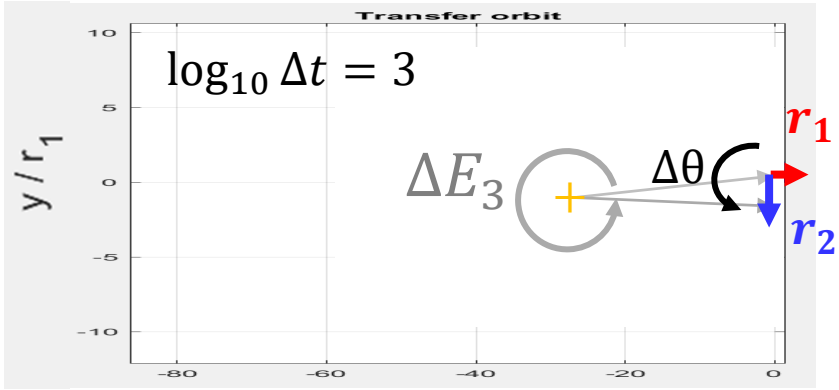
$$e^2 = 1 - \frac{A^2 \cdot C^2}{a^2 \cdot z \cdot c_1^2(z)}$$

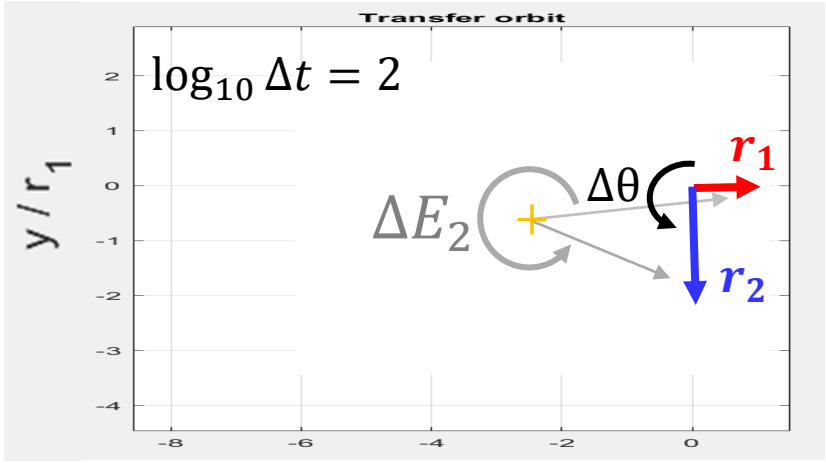


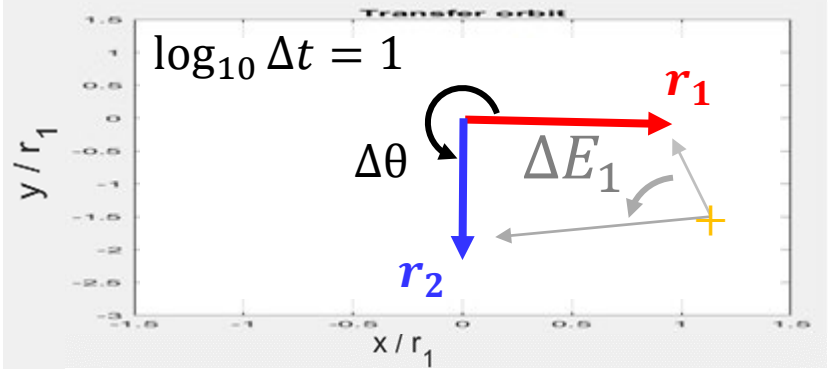


Geometrical meaning of $z = \rho s_2^2$









$\sqrt{z}$ is equal to the half of the variation in eccentric anomaly between P_1 and P_2 on the solution orbit

$$z = \rho s_2^2 = \left(\frac{\Delta E}{2}\right)^2$$

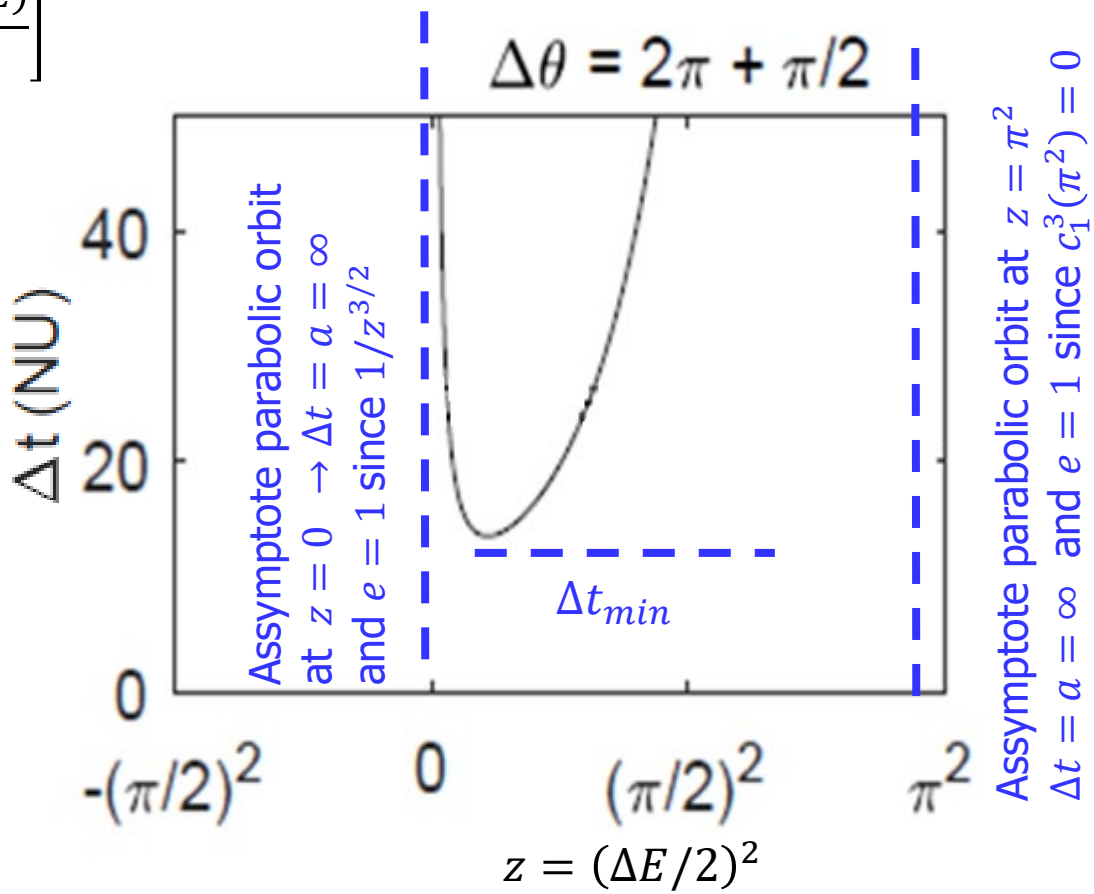
The transfer time equation: multi rev

In **elliptical** transfer orbits, we can account for $\Delta t > 2\pi$ which correspond to the secondary mass completing an integer number $\boldsymbol{n_{rev}}$ of revolutions (i.e. $\Delta\theta > 2\pi$) around the primary mass before arriving to P_2

$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q[c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{[c_1(z)]^3} \sqrt{2\frac{P - Q \cdot c_0(z)}{\mu}}$$

$$+ \frac{\pi \cdot \boldsymbol{n_{rev}}}{[c_1(z)]^3} \sqrt{\frac{1}{2\mu}\left[\frac{P - Q \cdot c_0(z)}{z}\right]^3}$$

- Δt includes the time to complete $\boldsymbol{n_{rev}}$ revolutions (Δt_{min}) plus the time to travel the final arc
- For each $\Delta t > \Delta t_{min}$, we have two elliptical orbits with
  - short period: $T_1 = 2\pi\sqrt{\frac{a(z_1)^3}{\mu}}$
  - long period: $T_2 = 2\pi\sqrt{\frac{a(z_2)^3}{\mu}}$



Source: Fig.3.4 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

The transfer time equation: multi rev

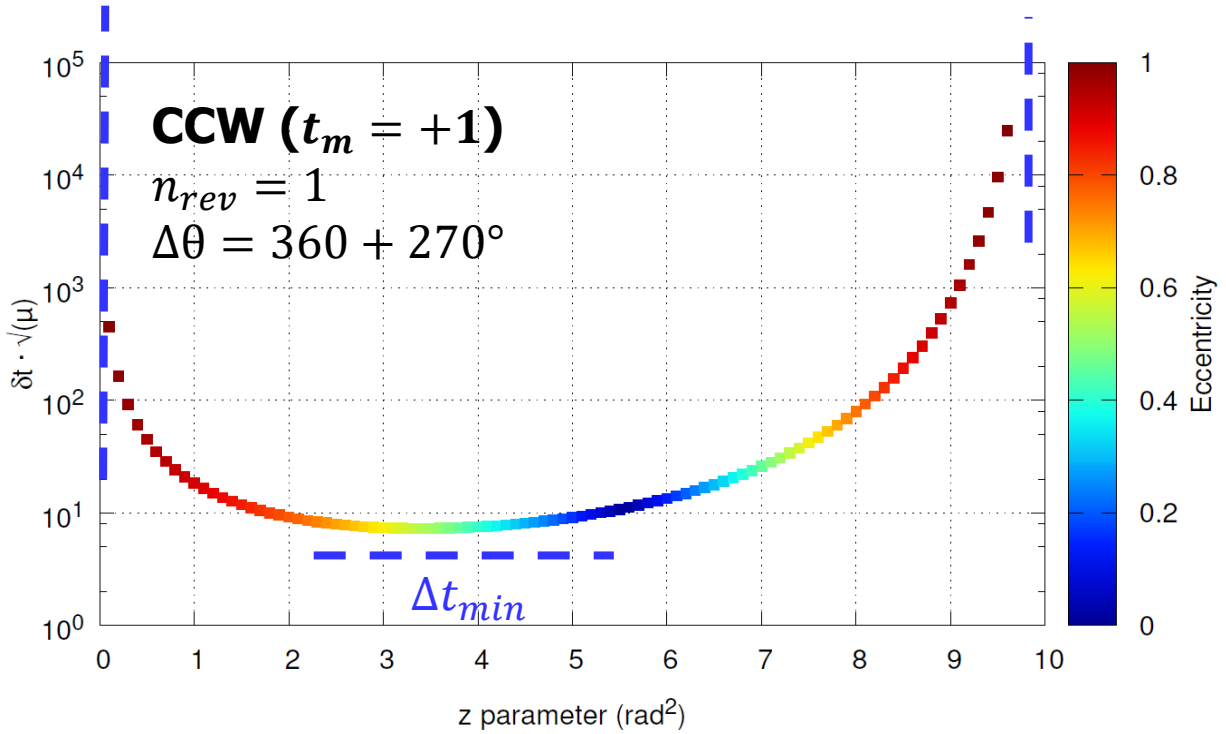
$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q \cdot [c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{[c_1(z)]^3} \sqrt{2 \frac{P - Q \cdot c_0(z)}{\mu}}$$

$$+ \frac{\pi \cdot n_{rev}}{[c_1(z)]^3} \sqrt{\frac{1}{2\mu} \left[\frac{P - Q \cdot c_0(z)}{z}\right]^3}$$

The conic eccentricity:

$$e^2 = 1 - \frac{A^2 \cdot C^2}{a^2 \cdot z \cdot [c_1(z)]^2}$$

Asymptote parabolic orbit at $z = 0 \rightarrow \Delta t = a = \infty$ and $e = 1$ since $1/z^{3/2}$



Asymptote parabolic orbit at $z = \pi^2$ $\Delta t = a = \infty$ and $e = 1$ since $c_1(\pi^2) = 0$

The transfer time equation: multi rev

In **elliptical** transfer orbits, we can account for $\Delta\theta > 2\pi$ which correspond to the secondary mass completing an integer number n_{rev} of revolutions around the primary mass before arriving to P_2

$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q \cdot [c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{[c_1(z)]^3} \sqrt{2\frac{P - Q \cdot c_0(z)}{\mu}}$$

$$+ \frac{\pi \cdot n_{rev}}{[c_1(z)]^3} \sqrt{\frac{1}{2\mu}\left[\frac{P - Q \cdot c_0(z)}{z}\right]^3}$$

For each transfer time Δt, two elliptical orbits with periods given by

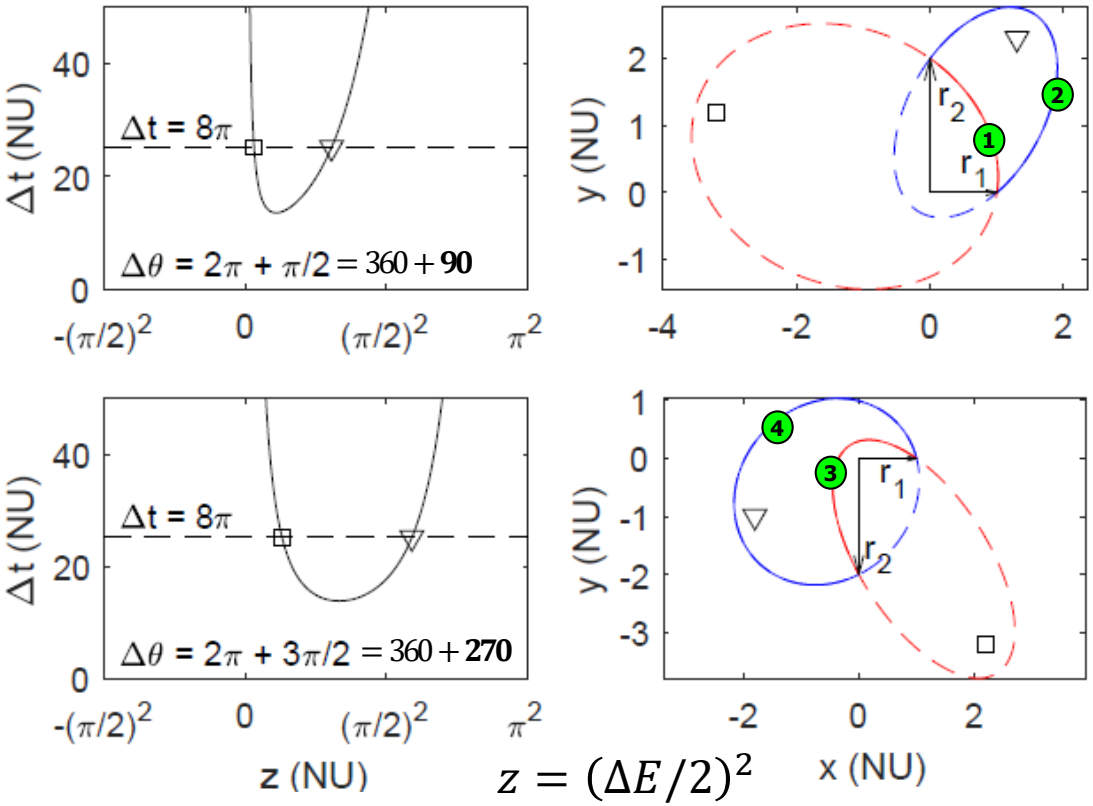
$$T = 2\pi\sqrt{\frac{[a(z)]^3}{\mu}} = 2\pi\sqrt{\frac{1}{\mu}\left[\frac{P - Q \cdot c_0(z)}{2 \cdot z \cdot [c_1(z)]^2}\right]^3}$$

Combinations of periods and ways as follows:

Counter clockwise:

1. short way $\Delta\theta < 180$ long period
2. short way $\Delta\theta < 180$ short period

3. long way $\Delta\theta > 180$ long period
4. long way $\Delta\theta > 180$ short period



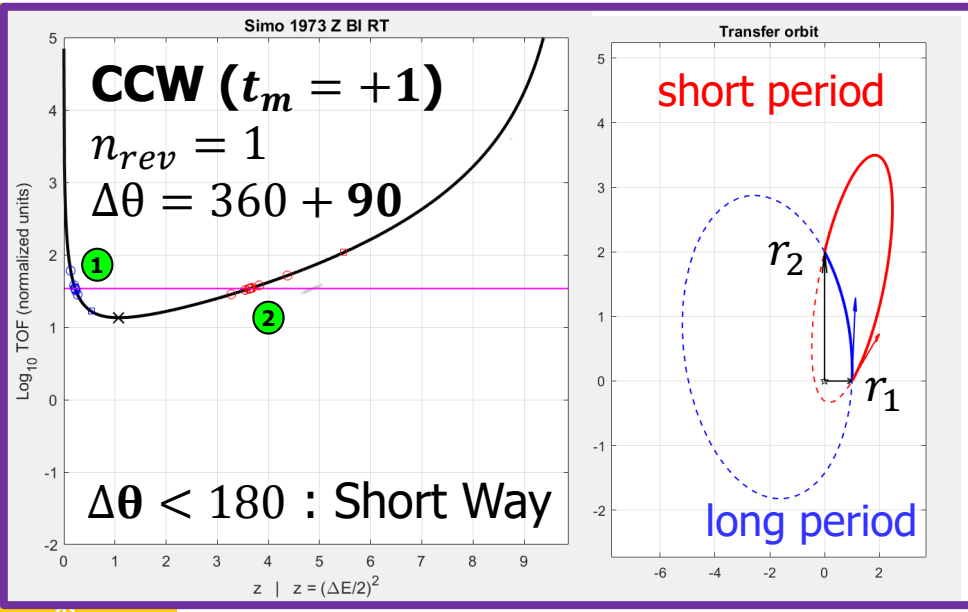
source: Fig. 3.4 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

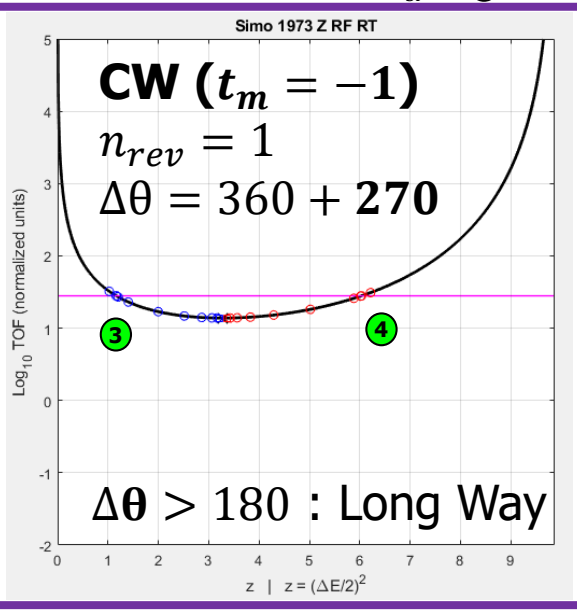
Parameter $Q = 2\sqrt{r_1 \cdot r_2}\cos\left(\frac{\Delta\theta}{2}\right)$ accounts the rotation direction CCW (as in the Solar System) or CW

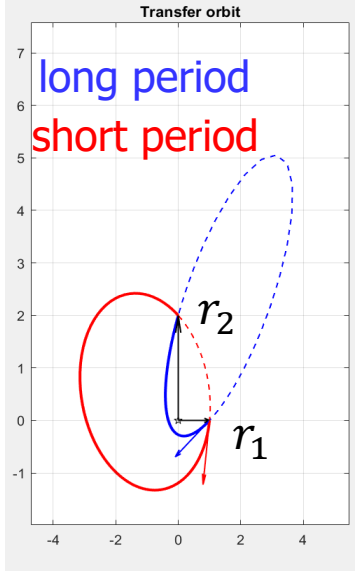
Every r_1 and r_2, with $P = r_1 + r_2$, has 2 possible transfer times Δt_{CW} and Δt_{CCW} and 4 transfer orbits with periods T as follows:

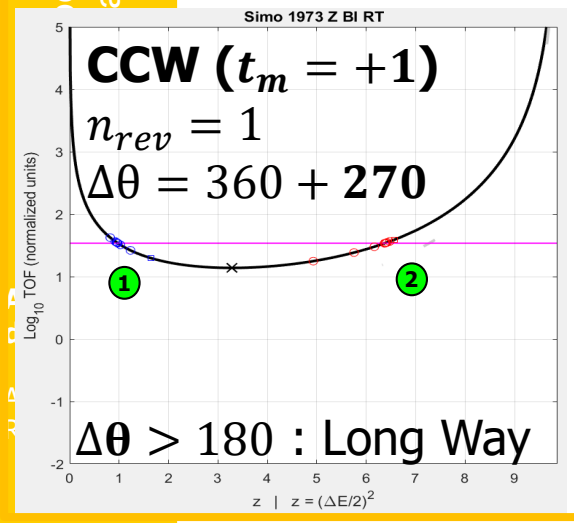
$$T = 2\pi\sqrt{\frac{[a(z)]^3}{\mu}} = 2\pi\sqrt{\frac{1}{\mu}\left[\frac{P - Q\cdot c_0(z)}{2\cdot z\cdot [c_1(z)]^2}\right]^3}$$

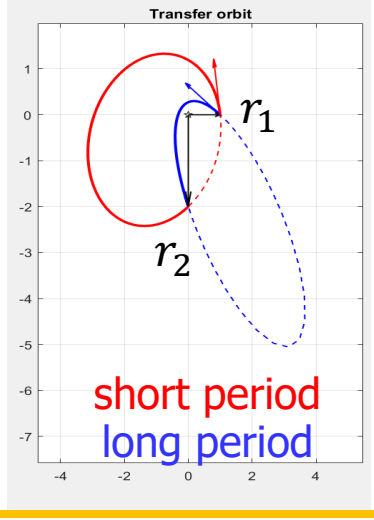
- Δt_{CCW}
  - $T_{CCW_{LP}}$ (1)
  - $T_{CCW_{SP}}$ (2)
- Δt_{CW}
  - $T_{CW_{LP}}$ (3)
  - $T_{CW_{SP}}$ (4)

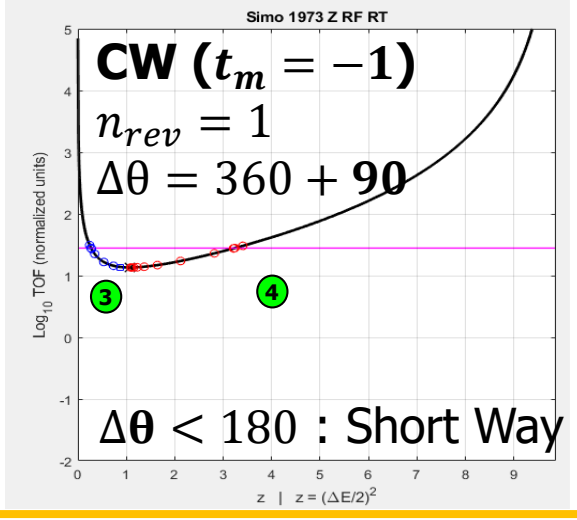


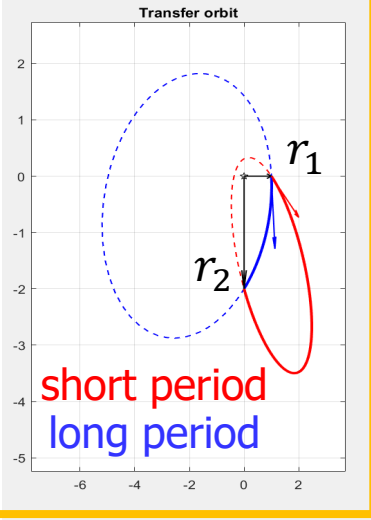












Generic Lambert procedure (e.g. Simó)

The transfer-time equation $\Delta t = f(z, \boldsymbol{r_1}, \boldsymbol{r_2}, \Delta\theta)$ does not have a closed- analytical-solution. Thus, z must be approximated with numerical methods

Given $\mu, \Delta t, \boldsymbol{r_1}, \boldsymbol{r_2}, \Delta\theta$ and a choice between long- and short- period if $\boldsymbol{n_{rev}} > 0$, we can proceed as follows:

1. Computation of geometrical constants P and Q

2. Choose a value of n_{rev}. If $n_{rev} > 0$
   a) determine the Δt_{min} and the corresponding z_{min}
   b) If $\Delta t < \Delta t_{min}$ the problem does not admit solution

3. Choice of an initial value of z_0. For instance:
   Elliptical orbits: $z_0 = \left(\frac{\Delta\theta}{2}\right)^2$
   Hyperbolic orbits: $z_0 = 0$

4. Approximate z by application of a numerical method to equation $\Delta t = f(z, r_1, r_2, \Delta\theta)$ and iterate on the transfer time equation until convergence.

5. Computation of the terminal velocity vectors
   a) v_1 at r_1 from the orbital elements and the true anomaly θ_1
   b) v_2 at r_2 from the orbital elements and the true anomaly $\theta_2 = \theta_1 + \Delta\theta$

Source: Fig.3.4 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

Module 2 Requirements

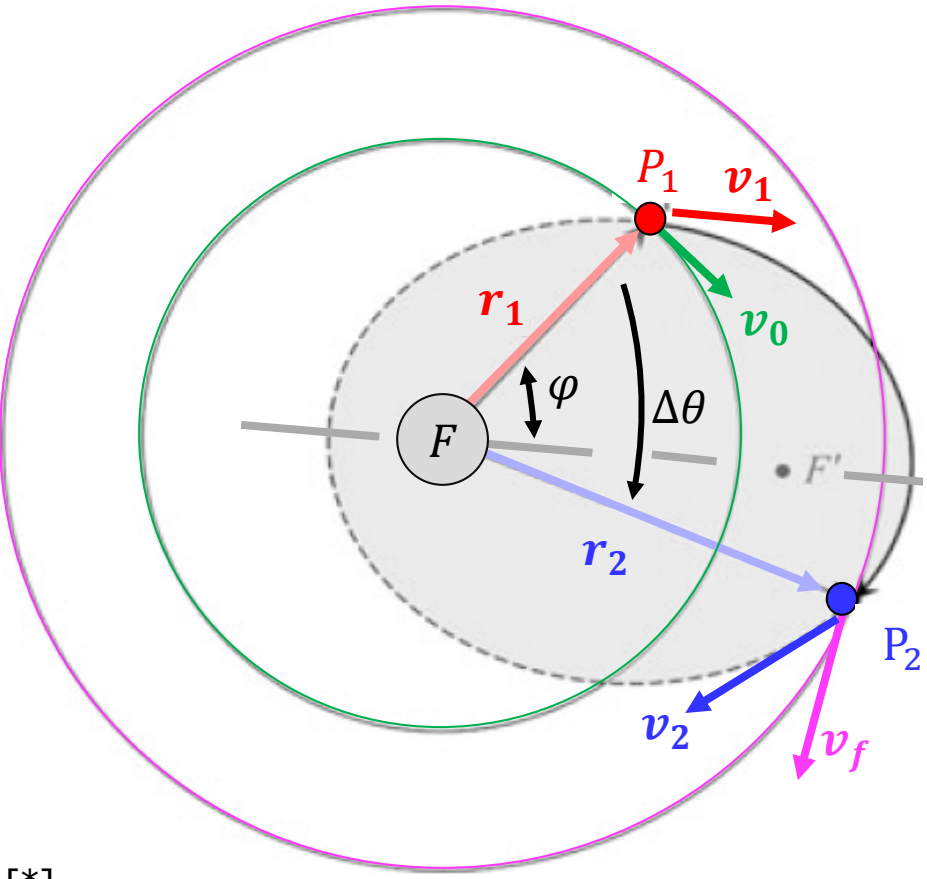
Objectives:

- Problem definition, statement of the theorem of Lambert, sketch of proof
- Numerical aspects of the problem
- Classical methods of solution (Lagrange, Gauss)
- Introduction to modern methods
- Application to Optimization, Rendezvouz, Pork Chop Plots
- Application to the problem of interplanetary transfer

Time Allocation:

- 8 hours

Application: Optimization (ΔV)



- Given two transfer points P_1 and P_2 with velocities $\mathbf{v_0}$ and $\mathbf{v_f}$
- What is the Lambert transfer orbit that minimizes the total $\Delta V_T = \Delta V_1 + \Delta V_2$:
  - $\Delta V_1 = \boldsymbol{v_1} - \boldsymbol{v_0}$ to embark the transfer orbit
  - $\Delta V_2 = \boldsymbol{v_f} - \boldsymbol{v_2}$ to remind in the final orbit
- The two orbital equations [*] at P_1 and P_2:

$$p = a(1-e^2) = r_1 + \boldsymbol{e}\cdot\boldsymbol{r_1}$$
$$p = a(1-e^2) = r_2 + \boldsymbol{e}\cdot\boldsymbol{r_2}$$

- We choose as the free element the orientation angle φ of the transfer orbit with respect $\boldsymbol{r_1}$

$$\boldsymbol{e}\cdot\boldsymbol{r_1} = e\cdot r_1\cos\varphi$$
$$\boldsymbol{e}\cdot\boldsymbol{r_2} = e\cdot r_2\cos(\Delta\theta-\varphi)$$

- Then, substituting in the two orbit equations

$$p = r_1(1+e\cdot\cos\varphi)$$
$$p = r_2(1+e\cdot\cos(\Delta\theta-\varphi))$$

$$e = \frac{r_2 - r_1}{r_1\cos\varphi - r_2\cos(\Delta\theta-\varphi)} \qquad p = \frac{r_1}{r_1(1+e\cdot\cos\varphi)}$$

- Function to optimize:

$$\Delta V_T = \Delta V_T(\varphi) = [\boldsymbol{v_1}(\varphi) - \boldsymbol{v_0}] + [\boldsymbol{v_f} - \boldsymbol{v_2}(\varphi)]$$

[*]

$$r = \frac{p}{1+e\cdot\cos\theta} \qquad p = r\cdot(1+e\cdot\cos\theta)$$

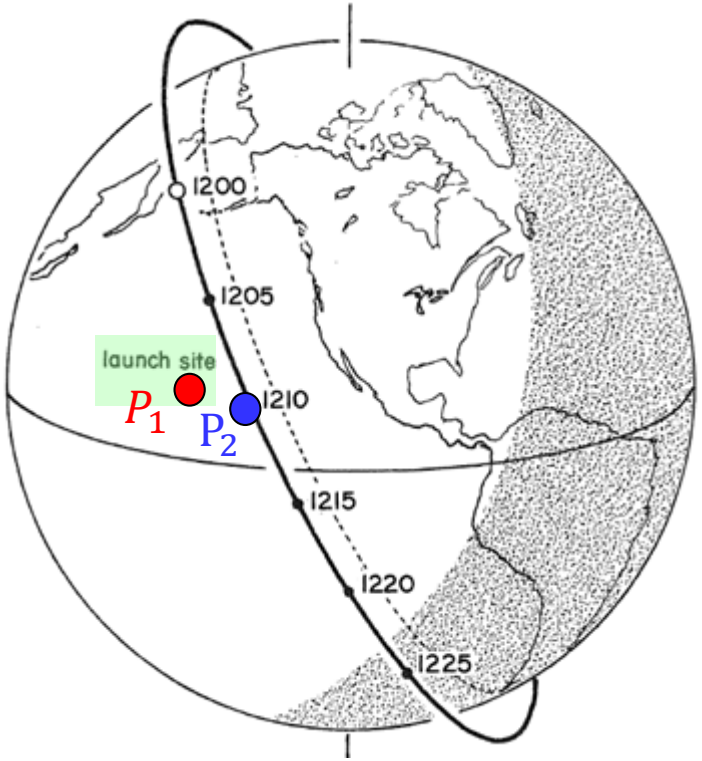
$$p = \frac{h^2}{\mu} \qquad h^2 = \mu\cdot a\cdot(1-e^2) \qquad p = a\cdot(1-e^2)$$

Source: Fig 8.14 Walter U (2019) "Astronautics. Th
Physics of Space Flight" 3rd Edition Springer

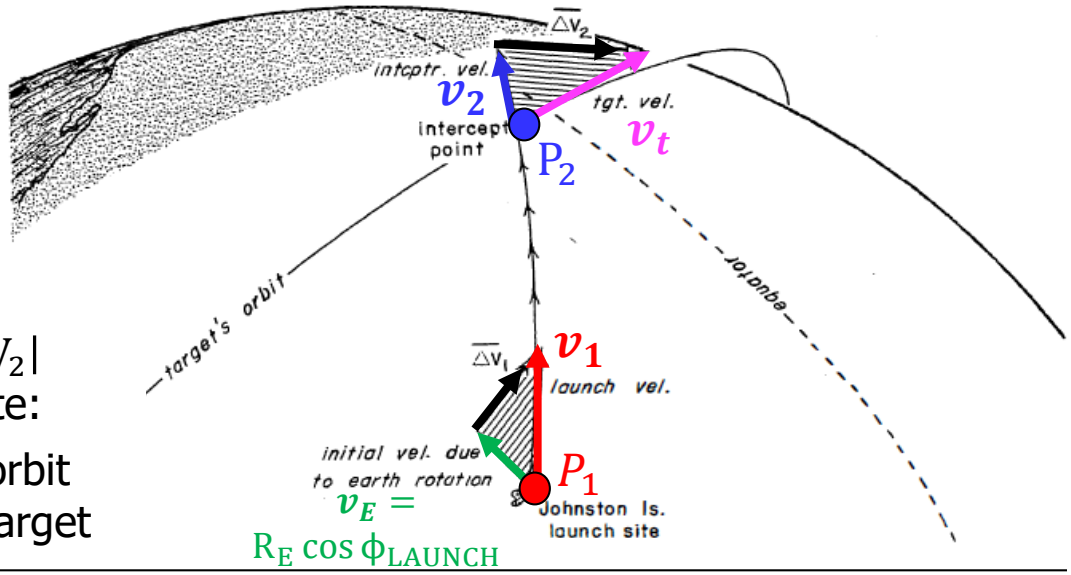
Application: Intercept & Rendezvous

Problem statement:

- We know the position and velocity of a target satellite at time $t_0 = 12h00$
- We wish to intercept this satellite from a ground launch site (λ_{LAUNCH},ϕ_{LAUNCH}).
- What is the launch time and the Δt for the interceptor that optimizes ΔV?



1. Determine target orbit:
   From $\mathbf{r}_{target}, \boldsymbol{v}_{target}(12h00)$ → $a, e, \Omega, i, \omega, \theta_{12h00}$
2. Determine interception:
   Assuming launch time $12h05$ and flight time $\Delta t = 5$ min
   From $a, e, \Omega, i, \omega, \theta_{12h10}$ → $\boldsymbol{r}_{target}(12h10)$, $\boldsymbol{v}_{target}(12h10)$
3. Solve Lambert Problem:
   $\boldsymbol{P_1} = \boldsymbol{r}_{launch}(12h05)$, $\boldsymbol{P_2} = \boldsymbol{r}_{target}(12h10)$, $\Delta t = 5$ min → $\boldsymbol{v_1}, \boldsymbol{v_2}$

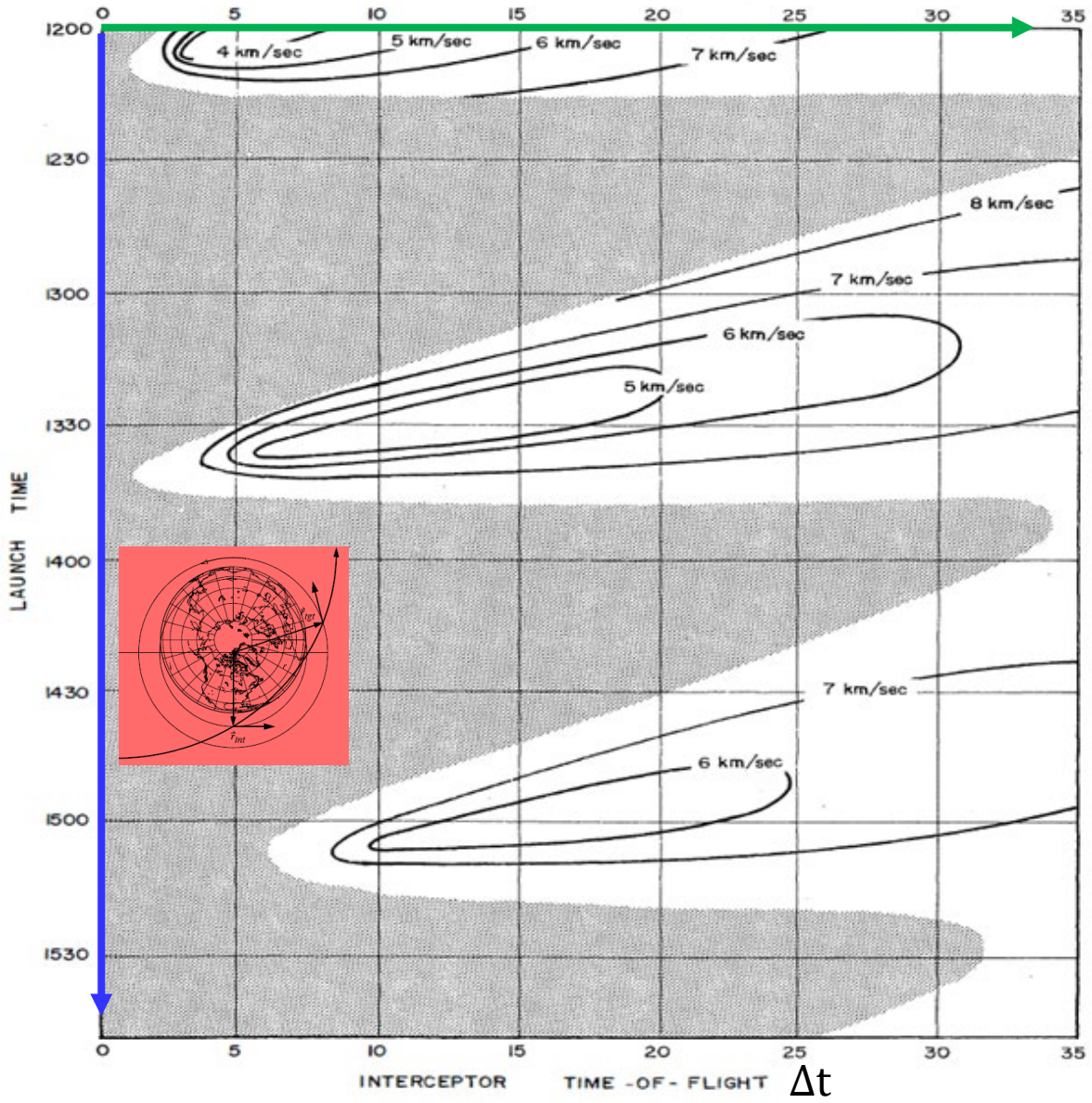


4. Determine the total $\Delta V_T = |\Delta V_1| + |\Delta V_2|$ budget to intercept the target satellite:

- $\Delta \mathbf{V_1} = \boldsymbol{v_1} - \boldsymbol{v_E}$ to embark the transfer orbit
- $\Delta \mathbf{V_2} = \boldsymbol{v_t} - \boldsymbol{v_2}$ to rendezvous with the target

Source: Figs. 5.7-2 and 5.7-3 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

Pork Chop Plot (PCP)



- results for various combinations of launch time and time-of-flight Δt
- **lines of constant ΔV_T indicate the regions of interest**
- shaded areas: interceptor strikes Earth en-route to the target: far interception point and short Δt
- Earth rotates! Although the target satellite returns to the same XYZ in inertial space after one orbital period $(\mathrm{e.g.}\, T = 90\ \mathrm{min})$, the launch site moves eastward by 15º/hour
- After $12\ \mathrm{h}$ the launch site will again pass through the orbit plane
- Where is the Optimum ?
  - ΔV_{min} maximizes the payload mass → when the transfer orbit is **coplanar** with the target orbit
  - Δt_{min} if target is to be intercepted as quick as possible

Source: Fig. 5.7-4 Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications
Fig 7.16 Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition

Pork Chop Plot (PCP)

PCPs are **contours** of trajectory performance and geometry data as a function of launch and arrival between two bodies orbiting a common mass.

PCPs are employed for many mission planning purposes, including:

- To determine the launch date (or successive launch windows).
- To determine the arrival date at the target body.

- To provide a preliminary estimate of **the amount of propellant** to be carried onboard the spacecraft $\frac{\Delta m}{m_0} = 1 - e^{-\frac{\Delta V}{I_{sp} \cdot g_{SL}}}$:
  - This estimate dictates the launch vehicle to be used.

- To provide information on the **capture orbit** when the spacecraft arrives at the target body.
  - Arrival date that minimizes the amount of propellant used to change the spacecraft speed to allow it to be captured into orbit.
  - It is also important to the scientists for meeting data-taking requirements, for instance, if aerobraking is used with a target atmosphere.

- To verify that **launch constraints** such as the Launch Azimuth is within the allowable safety range.

Source: Hintz GR (2015) "Orbital Mechanics and Astrodynamics" 1st Edition, Springer

Module 2 Requirements

Objectives:

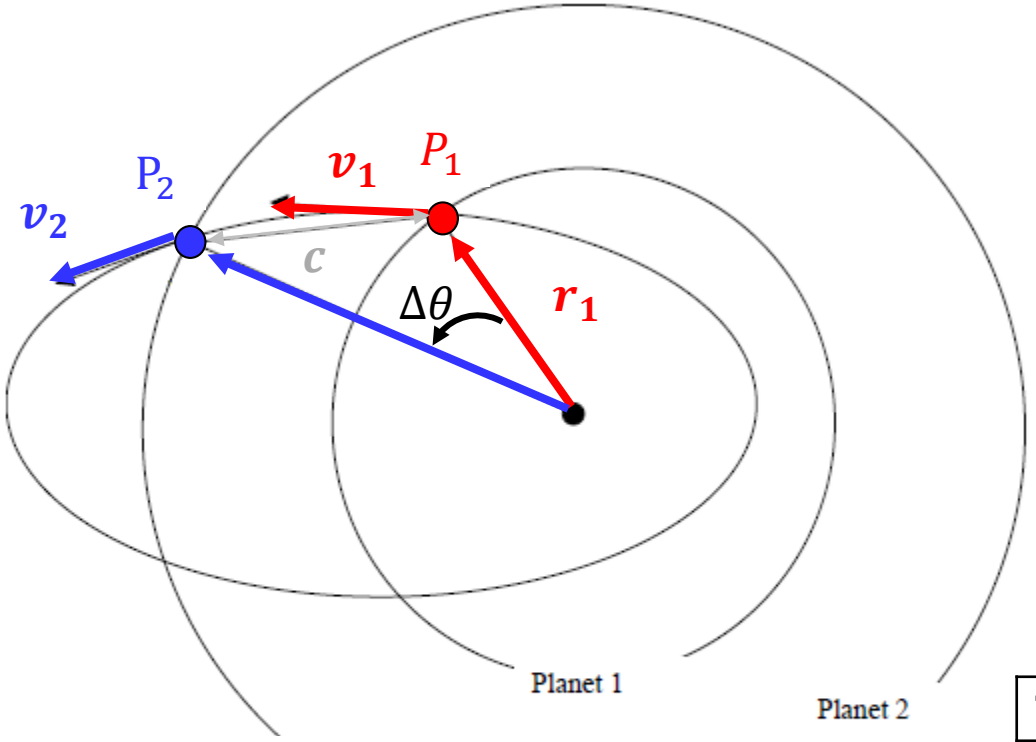
- Problem definition, statement of the theorem of Lambert, sketch of proof
- Numerical aspects of the problem
- Classical methods of solution (Lagrange, Gauss)
- Introduction to modern methods
- Application to Optimization, Rendezvouz, Pork Chop Plots
- Application to the problem of interplanetary transfer

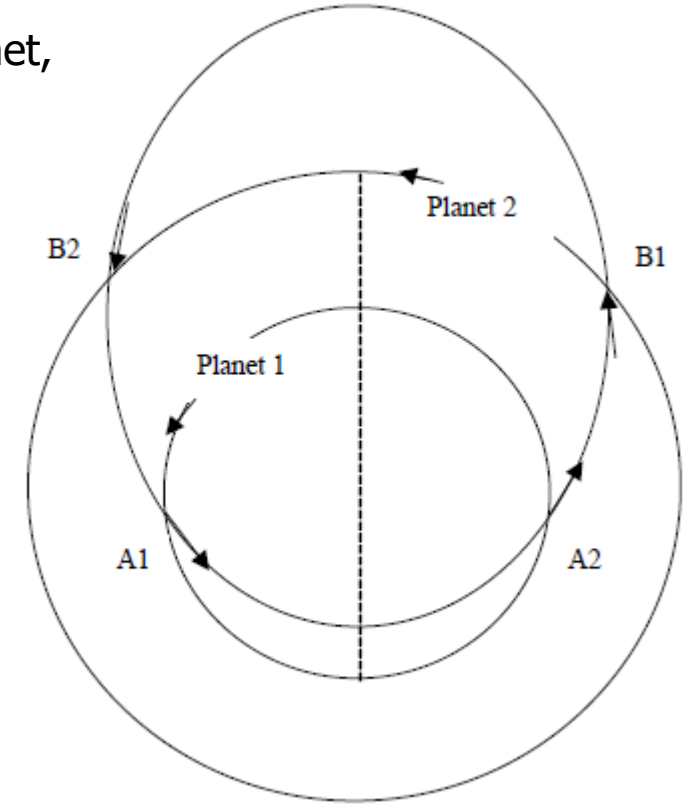
Time Allocation:

- 8 hours

Application: Interplanetary Flight

We can apply Lambert problem to find heliocentric orbits between two given points (e.g. planet to planet, or planet to asteroid/comet/moon)





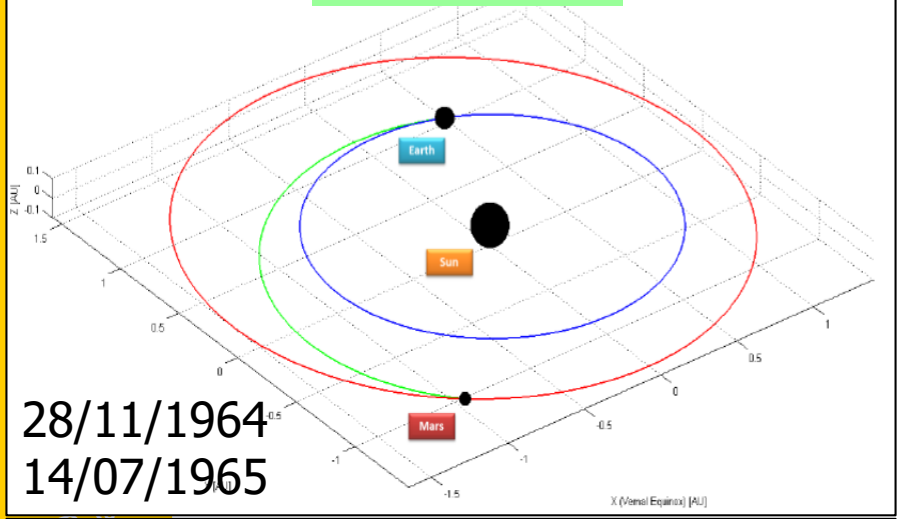
- the more tangent the velocity vectors are at departure/arrival, the lesser ΔV_1 and ΔV_2.
- minimum ΔV_T when $\Delta\theta = 180$ (i.e. a Hohmann transfer) with perihelion at planet 1 and aphelion at planet 2

Transfer	Departure	Arrival
A2 to B1	Radially outwards	Radially outwards
A2 to B2	Radially outwards	Radially inwards
A1 to B1	Radially inwards	Radially outwards
A1 to B2	Radially inwards	Radially inwards

Source: Figs. 1.1.5 and 1.1.12 Kemble S (2006) "Interplanetary Mission Analysis and Design" 1st Edition, Springer

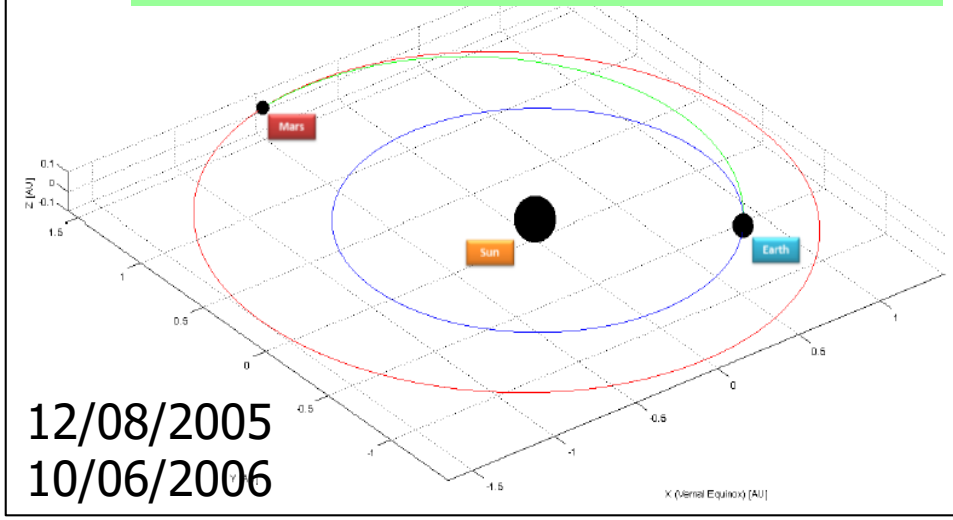
Application: Interplanetary Flight

Mariner 4



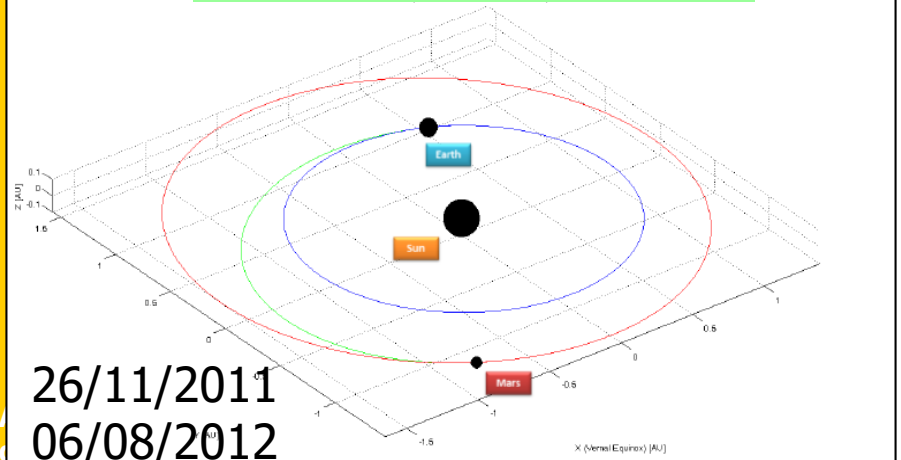
28/11/1964
14/07/1965

Mars Reconnaissance Orbiter (MRO)



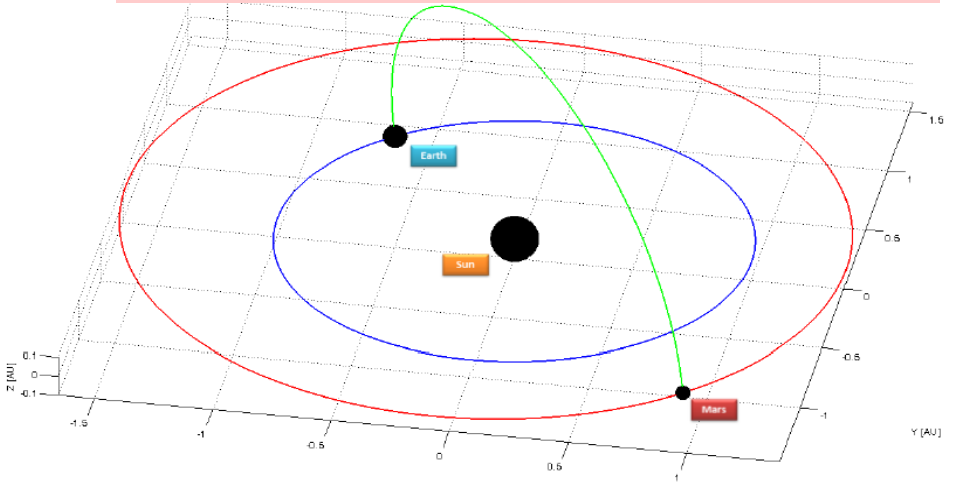
12/08/2005
10/06/2006

Mars Science Laboratory



26/11/2011
06/08/2012

Theoretical highly-inclined transfer orbit



$$\Delta V_{change\ of\ plane} = 2V_0 \sin \Delta i/2$$

Source: Figs. 16-19 Conte D (2013) "Survey of Earth-Mars Trajectories using Lambert's Problem and Applications" PhD Thesis, Embry-Riddle Aeronautical University

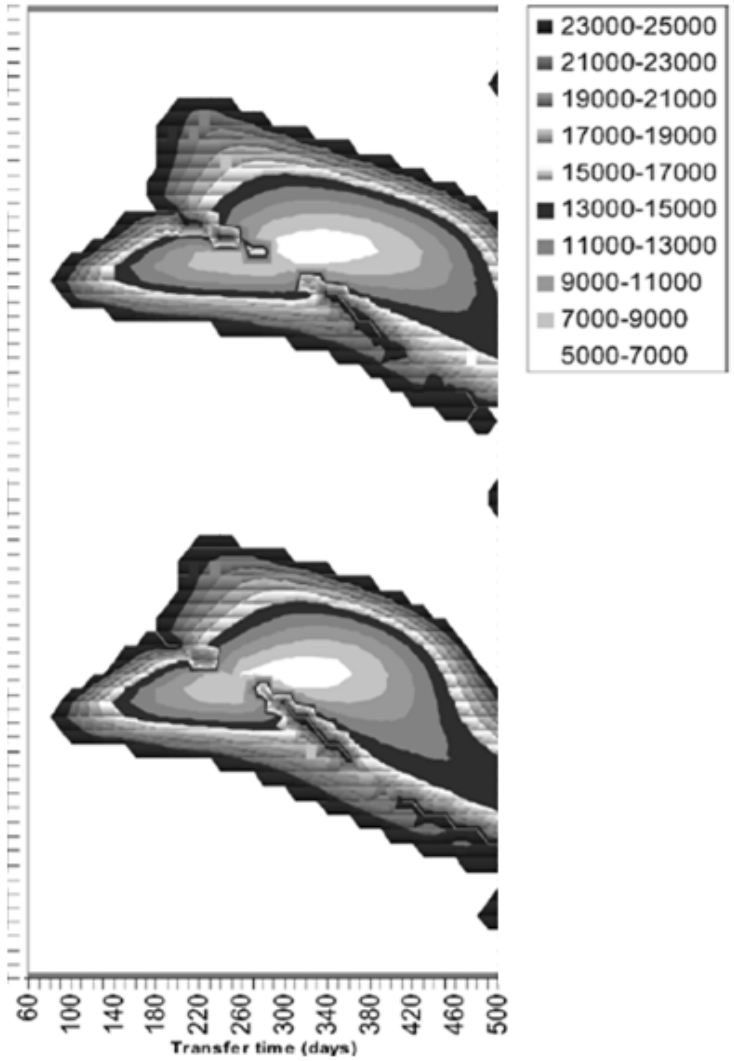
Application: Interplanetary Flight

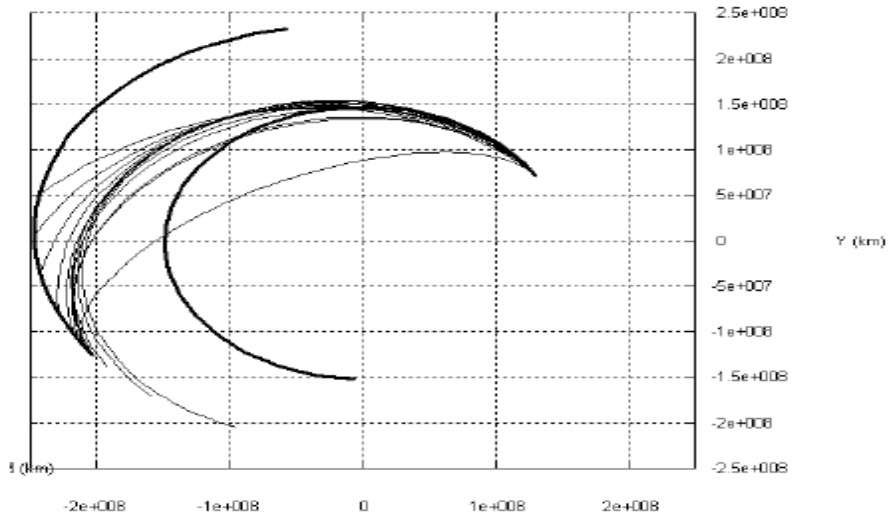
PCPs are derived from multiple Lambert transfers (e.g. Earth to Mars)

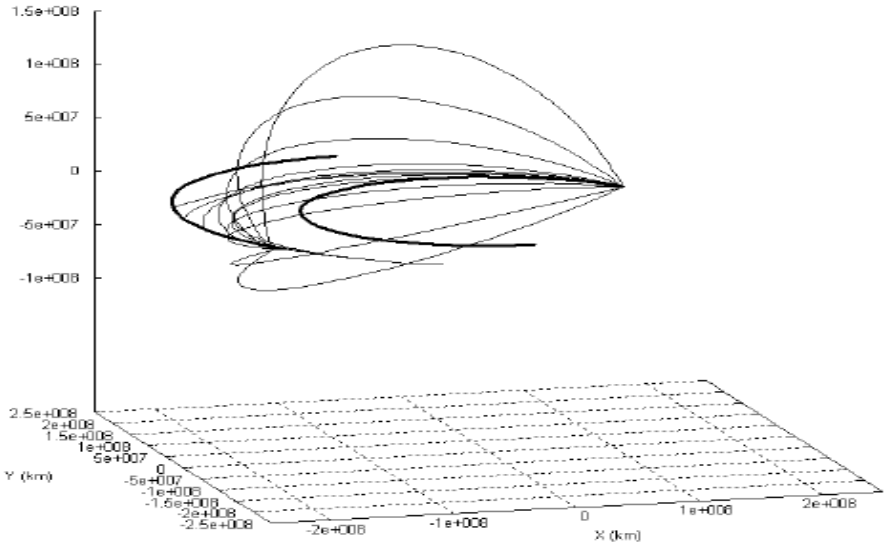
$\Delta V_T = |\Delta V_1| + |\Delta V_2|$

$\Delta V_1 = \boldsymbol{v_1} - \boldsymbol{v_0}$

$\Delta V_2 = \boldsymbol{v_f} - \boldsymbol{v_2}$



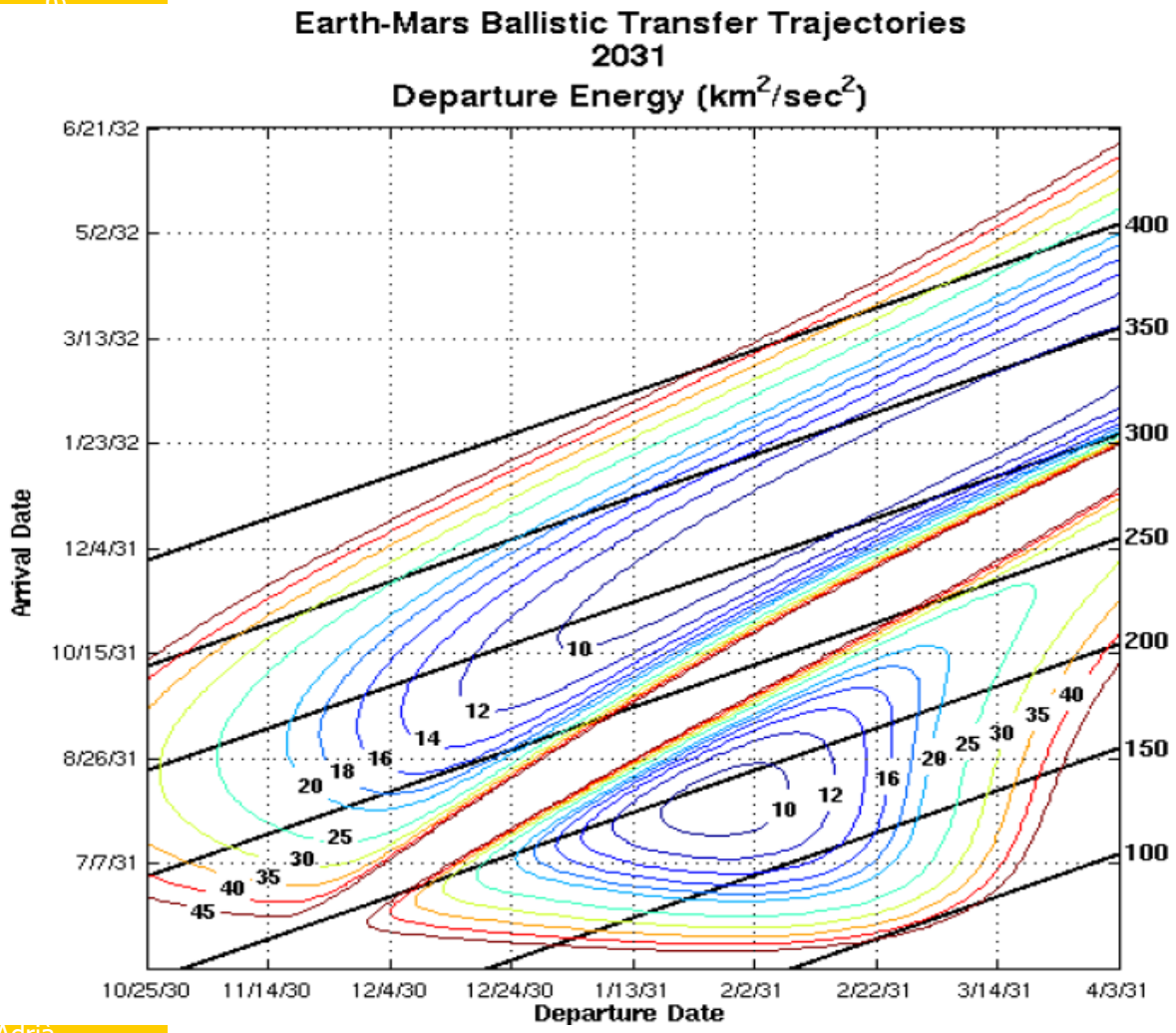


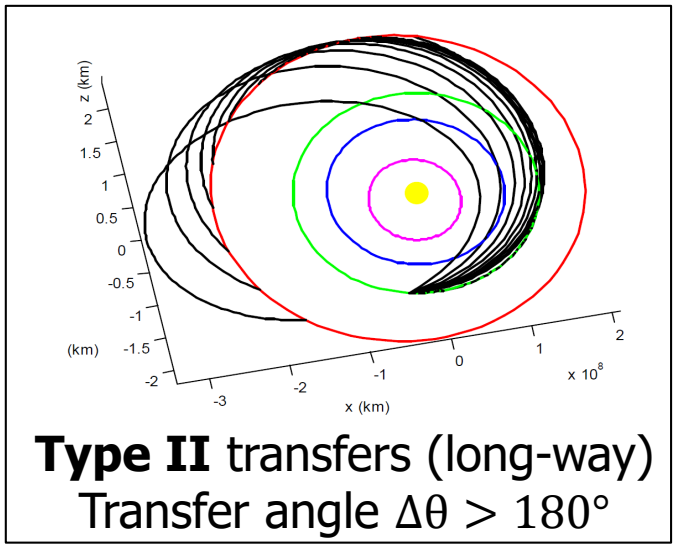


Source: Figs. 1.1.5 and 1.1.12 Kemble S (2006) "Interplanetary Mission Analysis and Design" 1st Edition, Springer

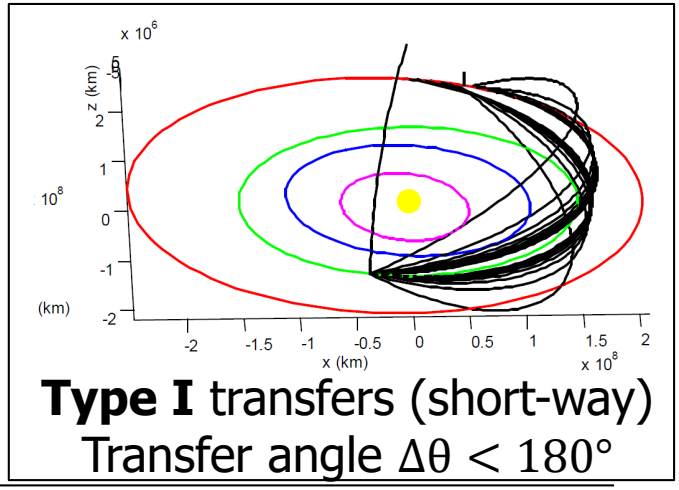
Pork Chop Plots (PCPs)

The straight lines depict the transfer duration in days.
The solid curved contours are the departure characteristic energy $C_3 = v_\infty^2$





Type II transfers (long-way)
Transfer angle $\Delta\theta > 180°$



Type I transfers (short-way)
Transfer angle $\Delta\theta < 180°$

Source: Fig.3 of Burke LM, Falck RD, McGuire ML (2010) "Interplanetary Mission Design Handbook: Earth-to-Mars Mission Opportunities 2026 to 2045" NASA Glenn Research Center
De La Torre (2017) "Astrodynamics Lectures Notes"

Trajectory Correction Maneuvers (TCMs)

Spacecrafts can perform ΔVs to correct its course after the launch, during the deep-space cruise and its final approach.

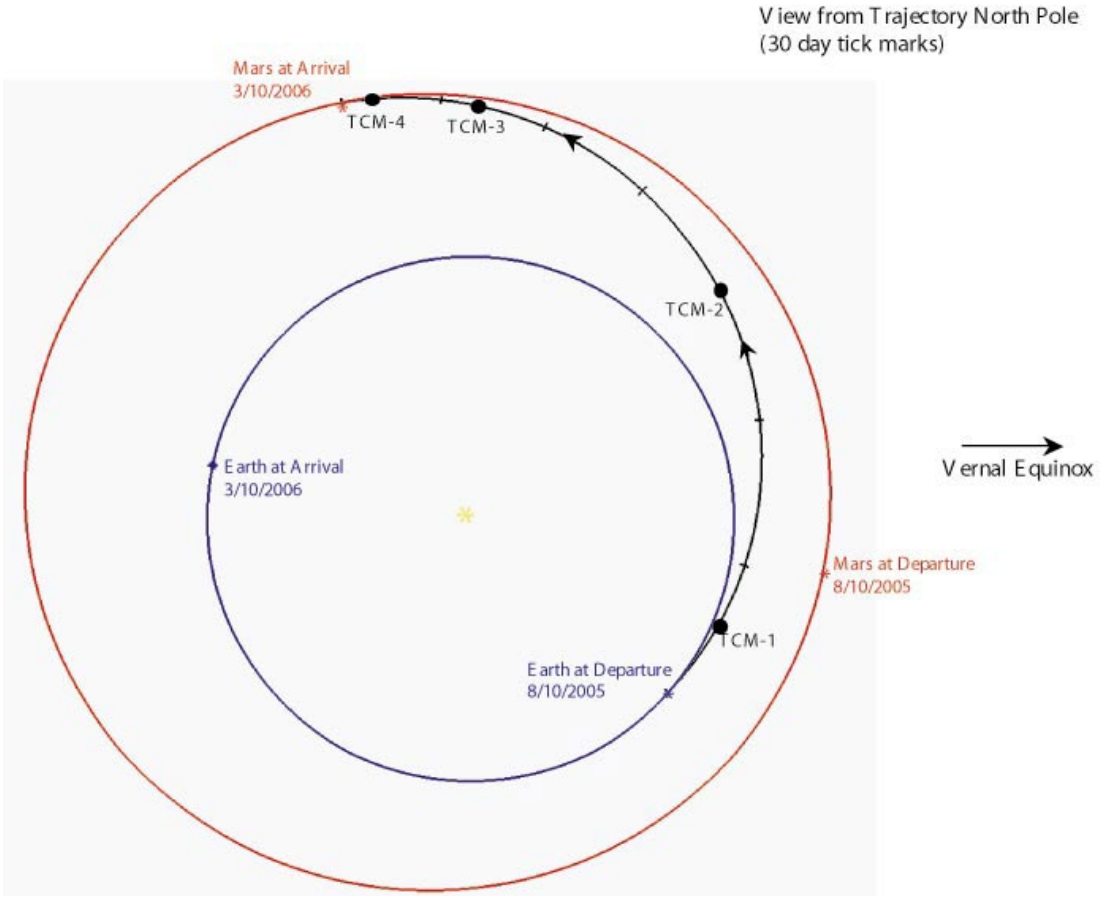
Example: TCMs for NASA mission Mars Reconnaissance Orbiter (MRO)

TCM-1 to remove
- launch vehicle injection **errors**
- the injection aim point **bias**:
  an intentional pointing error to ensure that the upper stage of the launch vehicle does not hit Mars after separating from MRO

TCM-2 to remove
- TCM-1 execution errors
- other errors accumulated in flight

TCM-3 planned but not used

TCM-4 to correct errors on the final aim point for Mars orbit insertion on a southern approach trajectory



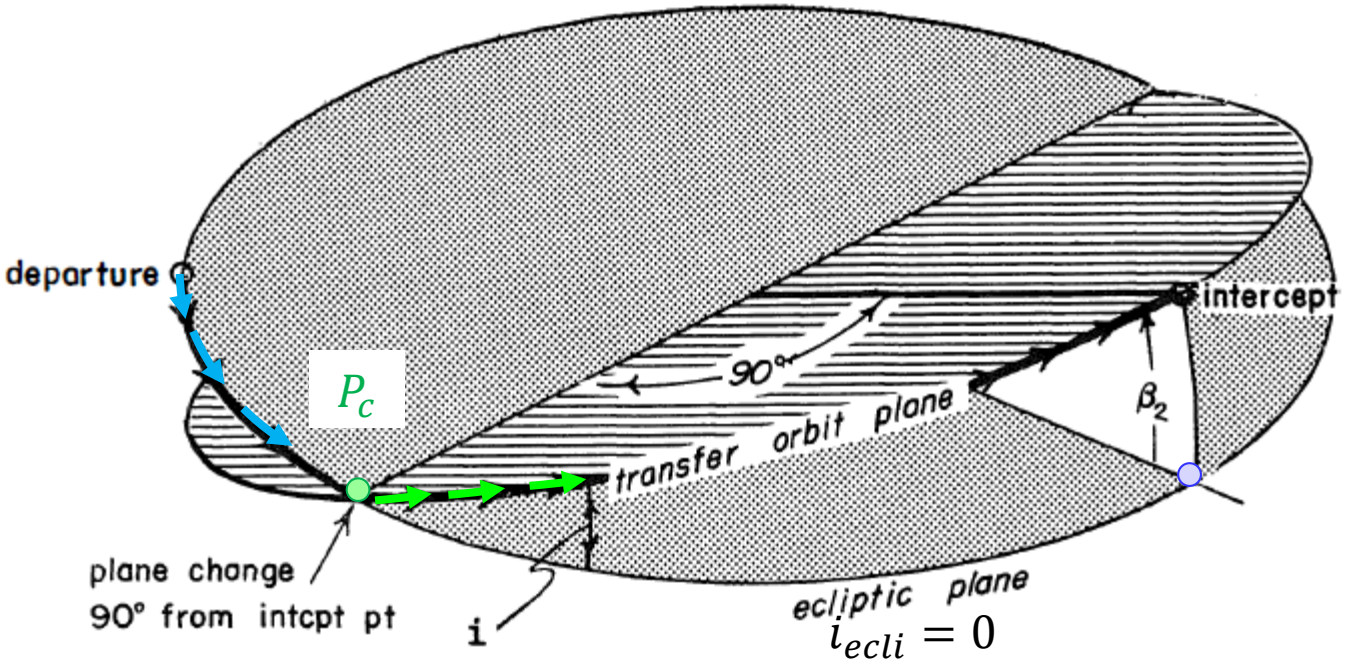
Source: https://mars.nasa.gov/mro/mission/timeline/mtapproach/approachtcms/

Noncoplanar Interplanetary Trajectories

A good TCM procedure when the target (e.g. a planet) lies above or below the ecliptic plane at intercept is the ***broken plane maneuver***:

- to launch the spacecraft into a transfer orbit which lies in the ecliptic plane targeting the projection of the interceptor on the ecliptic
- to make a simple Plane Change (PC) during mid-course when the true anomaly change remaining to intercept is $\Delta\theta = 90^{o}$
- This minimizes the magnitude of the required plane change ΔV to

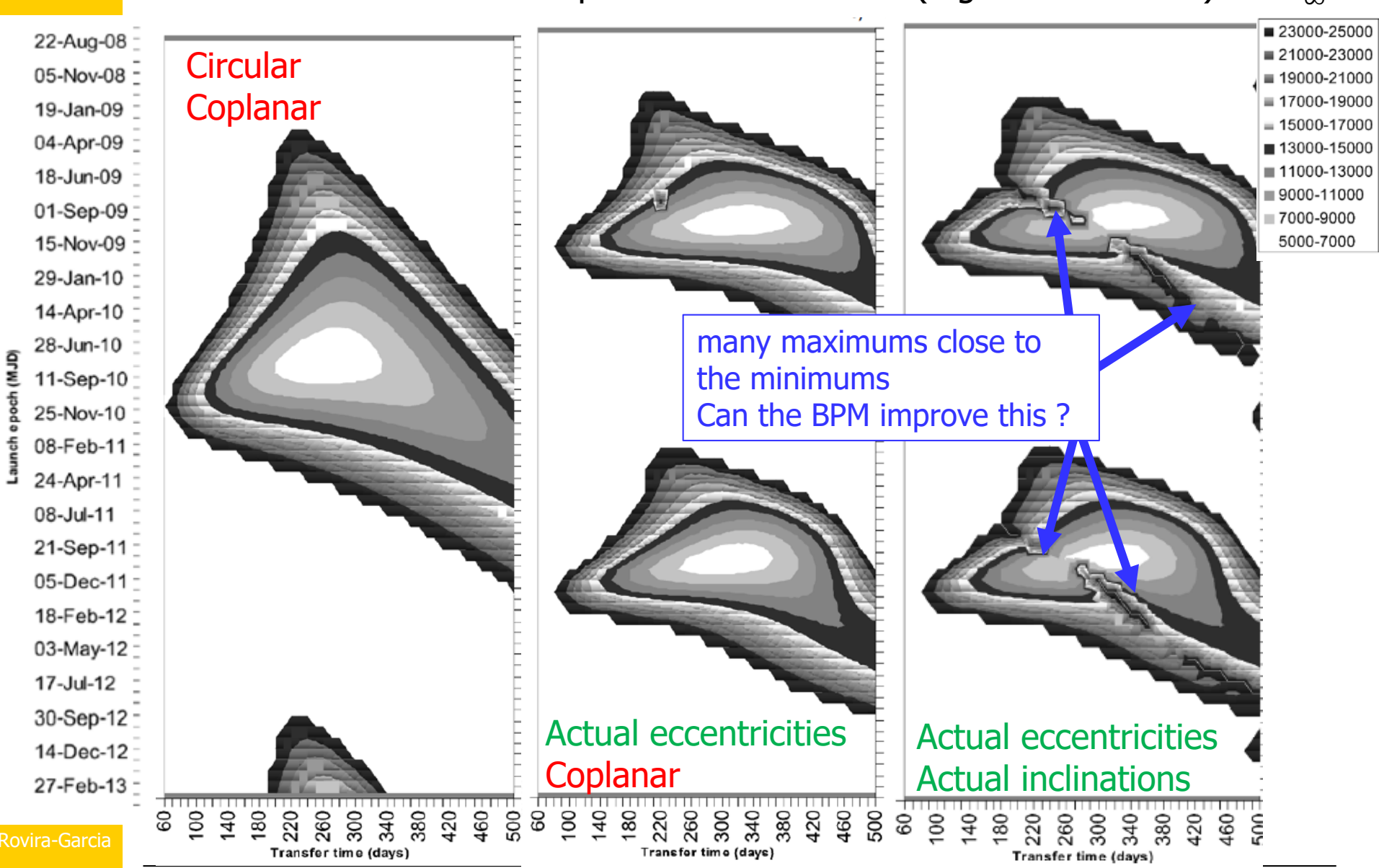
$$\Delta V = 2V_{P_c} \sin \frac{\Delta i}{2}$$



Source: Fig. 8.4-1. Bate RR, Mueller DD, White JE (1976) "Fundamentals of Astrodynamics" Dover Publications

Application: Interplanetary Flight

PCPs are derived from multiple Lambert transfers (e.g. Earth to Mars)



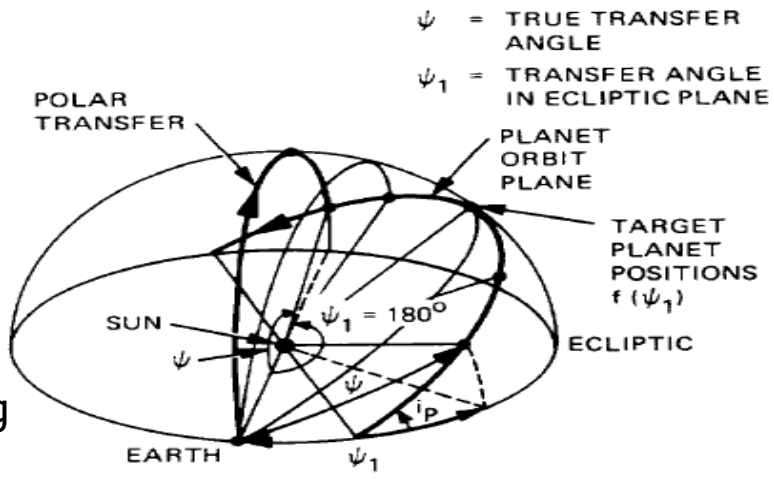
Source: Figs. 1.1.7 to 1.1.9 Kemble S (2006) "Interplanetary Mission Analysis and Design" 1st Edition, Springer

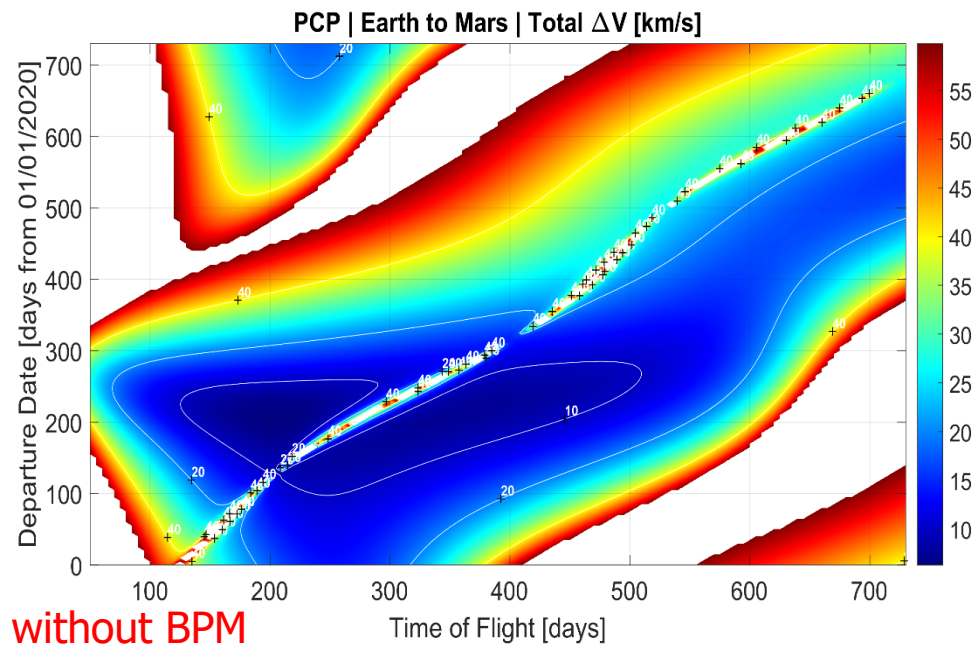
PCPs with Broken Plane Maneuvers

In 3D space, planetary orbits are not strictly coplanar.

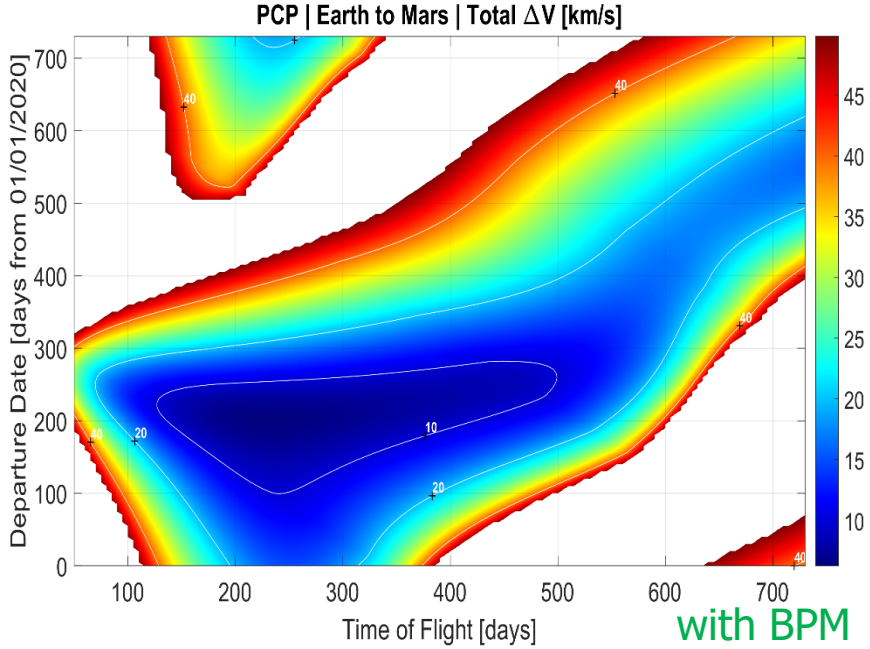
Diametric transfer arcs require high ecliptic inclinations → **worst-case:** polar flight path for an exact $\Delta\theta = 180°$ (hence these transfers are close to the PCP minimums)

The Broken Plane Maneuver avoids performing large (i.e. costly) out of plane $\Delta V = 2V_0 \sin \frac{\Delta i}{2}$





without BPM



with BPM

Source: De La Torre (2017) "Astrodynamics Lectures Notes"
Andrey B. Sergeyevsky AB, Snyder GC, Cunniff RA (1984) "Interplanetary Mission Design Handbook, Volume I, Part 2 Earth to Mars Ballistic Mission Opportunities, 1990-2005"

Course Project

Build your Interplanetary Mission Analysis Design tool.

- **Objective:** To compute the ΔV budget trading off different departure and arrival dates.

Design, implement and test a Pork Chop Plot tool by the building blocks:

- Planetary positions and velocities at a given departure and arrival dates.
- Solve the Lambert problem (any algorithm is OK) given the $\boldsymbol{r_1}$, $\boldsymbol{r_2}$ and specified Δt with the following steps:
  - Verify elemental cases against examples present in the GUI and/or in the theory slides
  - Implement a planar PCP ignoring the inclination and eccentricity of the departure/arrival planets
  - Implement a planar PCP assuming the actual planets eccentricities
  - Implement the actual PCP with actual inclinations and eccentricities
- Compute the required $\Delta V_T = |\Delta V_1| + |\Delta V_2|$ to embark the transfer orbital segment and to reach the target planet
- Generate the PCP and plot the transfer orbit

Deadline: *One week before the Final Exam*
Maximum Pages: *50*
Groups of three

Minimum Report Content

Introduction with the aim and theory behind the problem you are solving

Methodology of the developed solution

- Explain you main software functions → All code must be yours! No proprietary built-in functions are allowed

Results

- Validation of the code with elementary cases
  - Assess one example with respect to a Hohmann transfer
  - Bring your code to the limit: rectilinear transfers $\Delta\theta = 0°$ or $180°$
  - Numerically quantify the discrepancies: replace "little", "big", "good" by a number!!
- Full PCP and ΔV computations
  - For every planet, in a given time window 2020-2025, minimum ΔV and launch opportunity
  - For an example, quantify the increment of ΔV due to changing the transfer's direction of motion from a CCW to CW
  - Make a table for the minimum ΔV and orbital elements to all solar system planets assuming the project deadline as launch date
  - Execution time of one PCP: mention assumptions (e.g. grid, CPU)

Conclusion

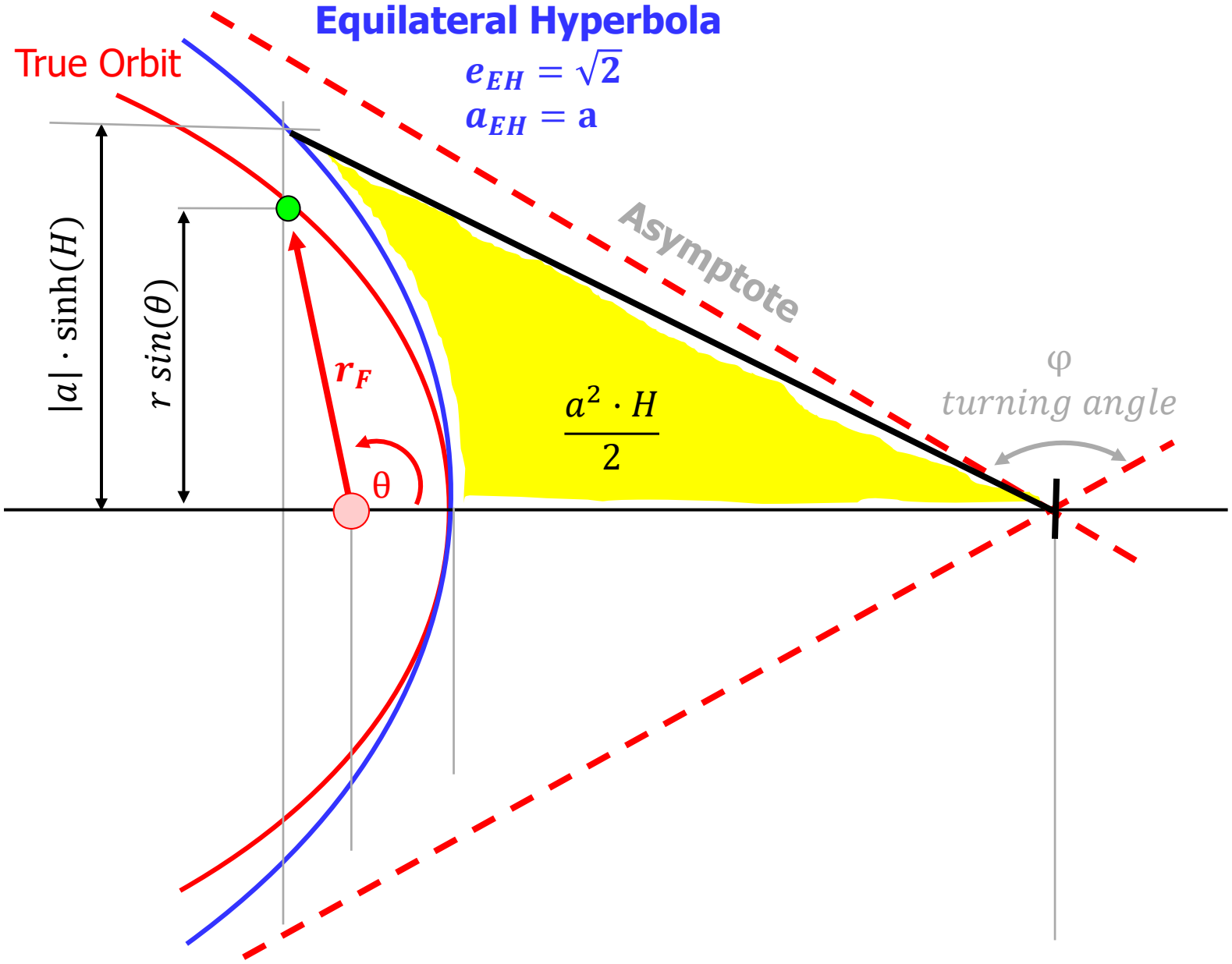
- Summary
- What your learnt, difficulties faced
- Future improvements

Back Up Slides

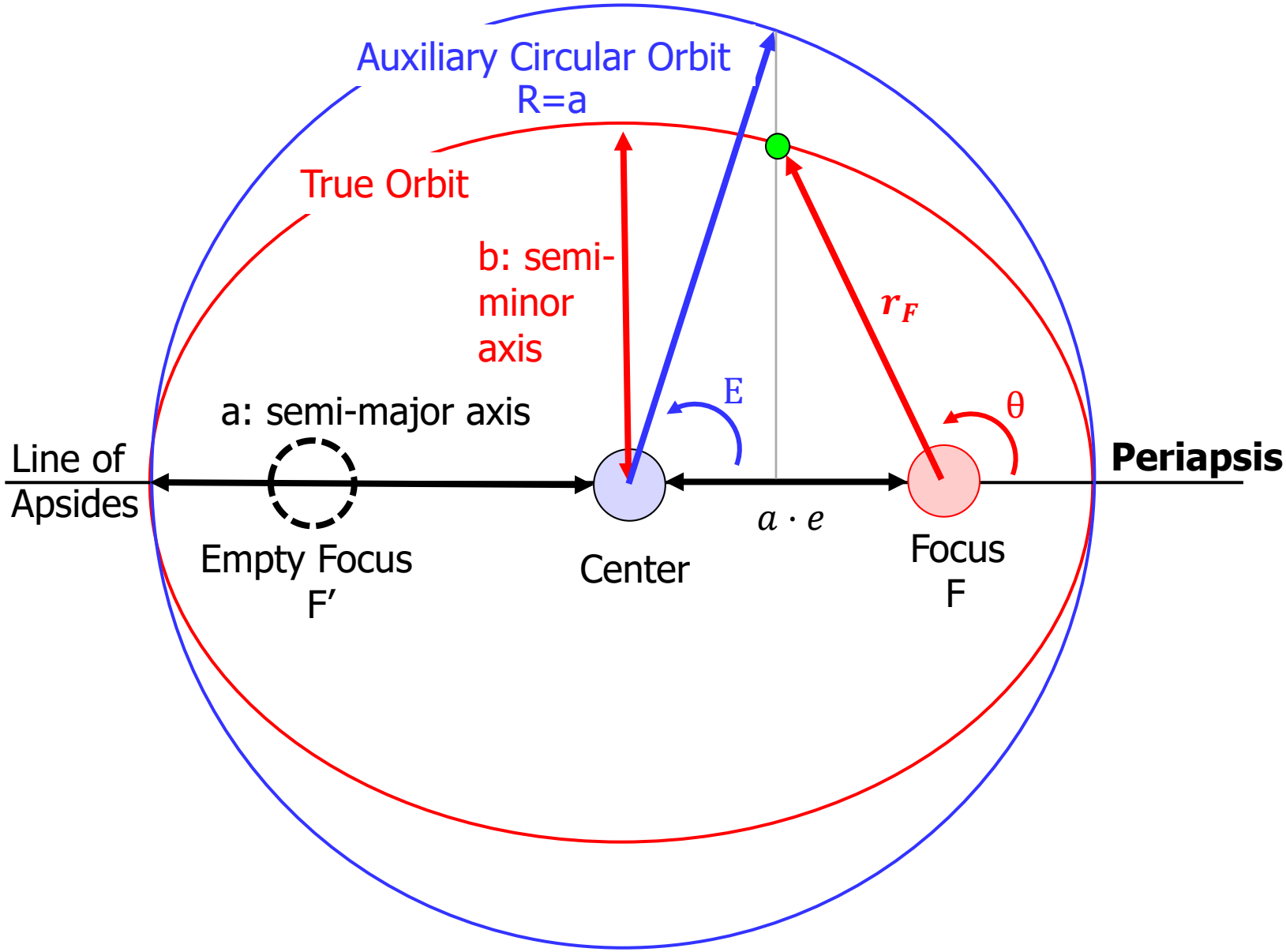
Escola Superior d'Enginyeries Industrial, Aeroespacial i Audiovisual de Terrassa (ESEIAAT)

Astro-dynamics

Adrià Rovira-Garcia



Elliptical Orbit (0<e<1)



Source: Curtis HD (2010) "Orbital Mechanics for Engineering Students" (3rd Edition) Elsevier.

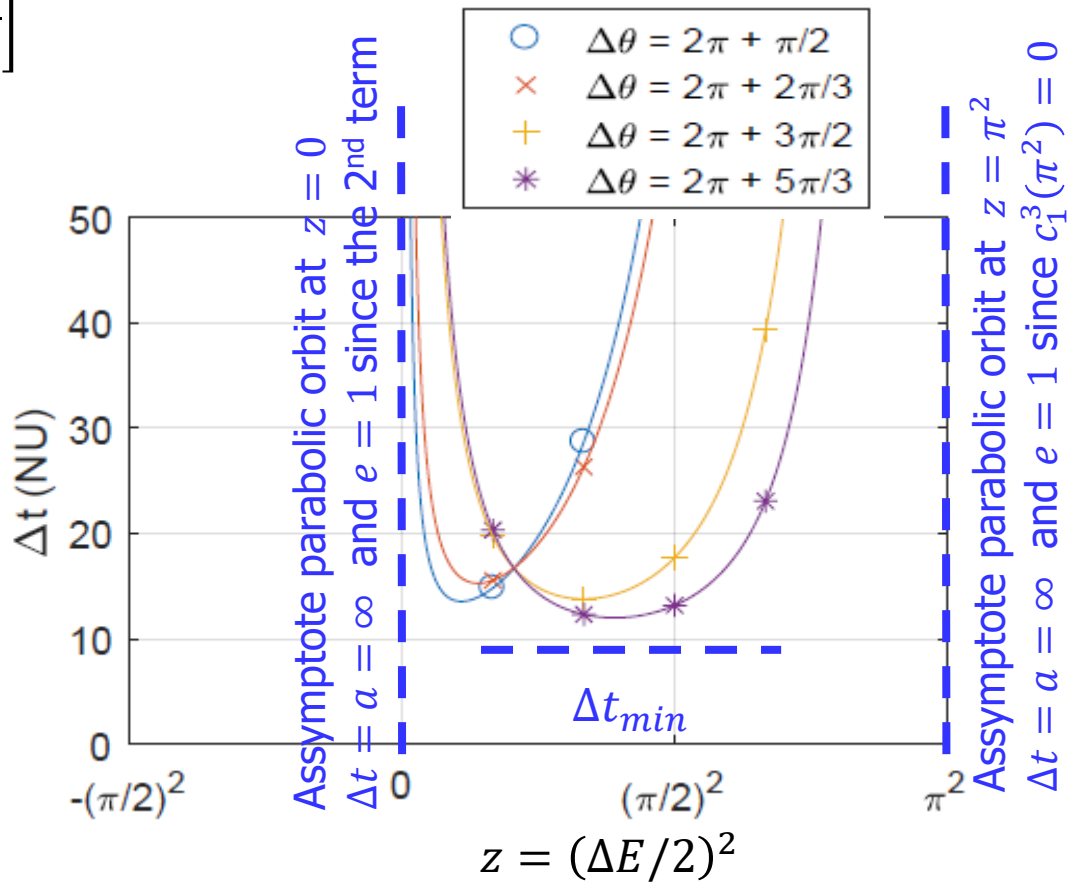
The transfer time equation: multi rev

In **elliptical** transfer orbits, we can account for $\Delta t > 2\pi$ which correspond to the secondary mass completing an integer number $\boldsymbol{n_{rev}}$ of revolutions around the primary mass before arriving to P_2

$$\Delta t = \frac{2 \cdot P \cdot c_3(4 \cdot z) + Q[c_1(z) \cdot c_2(4 \cdot z) - 2 \cdot c_0(z) \cdot c_3(4 \cdot z)]}{c_1^3(z)} \sqrt{2\frac{P - Q \cdot c_0(z)}{\mu}}$$

$$+ \frac{\pi \cdot \boldsymbol{n_{rev}}}{c_1^3(z)} \sqrt{\frac{1}{2\mu}\left[\frac{P - Q \cdot c_0(z)}{z}\right]^3}$$

- Δt includes the time to complete $\boldsymbol{n_{rev}}$ revolutions (Δt_{min}) plus the time to travel the final arc
- For each Δt, we have two elliptical orbits with period $T = 2\pi\sqrt{\frac{a^3}{\mu}}$: short period and long period.



Source: Fig.3.4 De La Torre (2020) "Optimization of Interplanetary Trajectories with Gravity Assist"

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Astro-dynamics

Adrià Rovira-Garcia

Back Up Slides Launch

Launch Geometry

The Injection orbit is determined by:

- Launch Azimuth A (Angle measured from the North)
- Launch site coordinates (λ_{LAUNCH},ϕ_{LAUNCH})
- Launch Time θ_{GMST}

The RAAN:

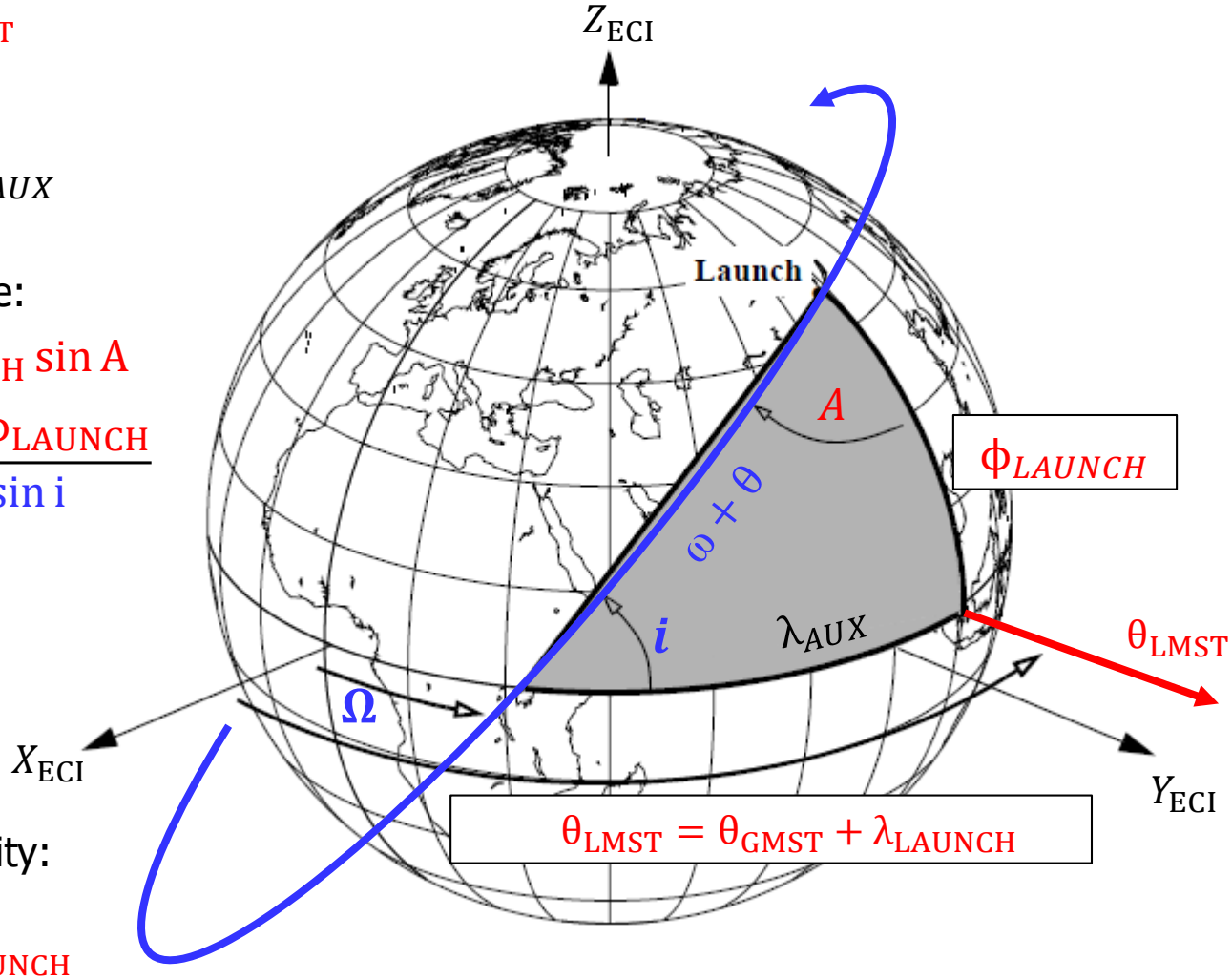
$$\Omega = \theta_{LMST} - \lambda_{AUX}$$

Spherical right triangle:

$$\cos i = \cos \phi_{LAUNCH} \sin A$$

$$\sin(\omega + \theta) = \frac{\sin \phi_{LAUNCH}}{\sin i}$$

$$\cos \lambda_{AUX} = \frac{\cos A}{\sin i}$$



Initial Eastward Velocity:

$$v_0 = \dot{\Omega}_E \times R_{LAUNCH}$$
$$= \dot{\Omega}_E R_E \cos \phi_{LAUNCH}$$
$$= 0.465 \cos \phi_{LAUNCH} \text{ km/s}$$

Source: Figure 6-8 Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition

Launch Geometry

Consequence of

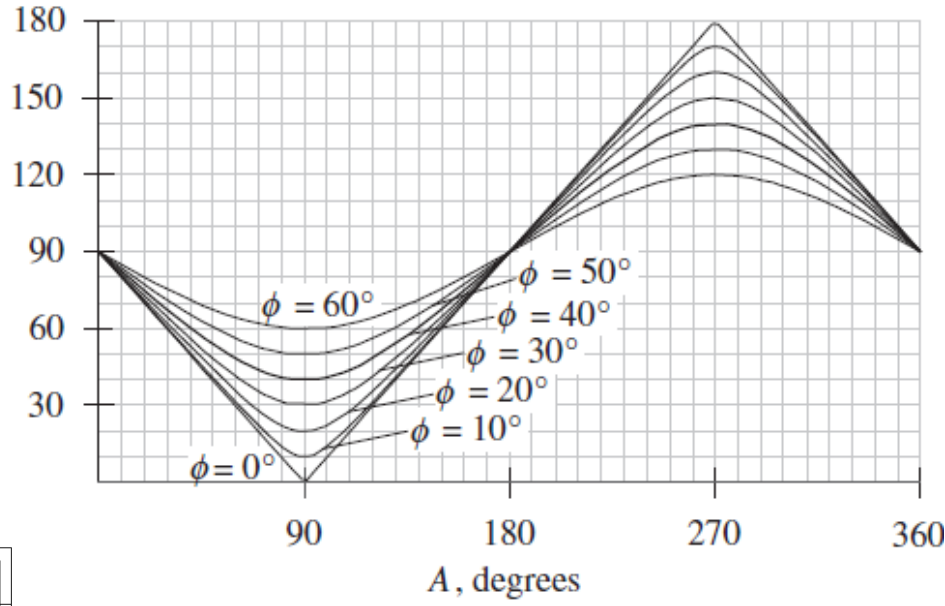
$$\cos i = \cos \phi_{LAUNCH} \sin A$$

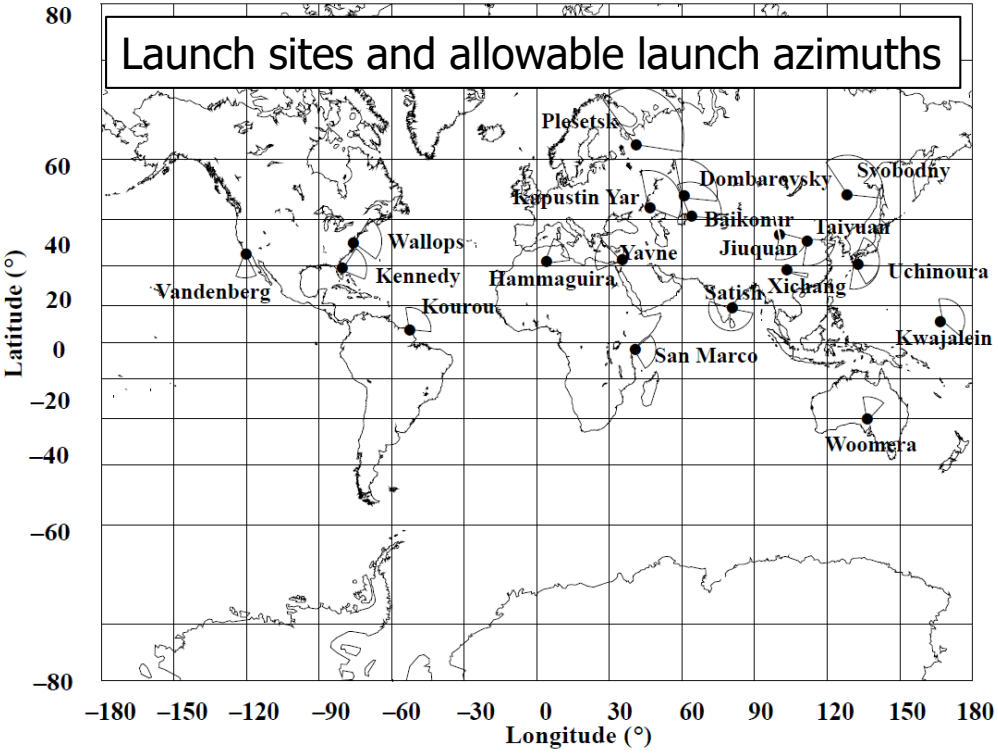
Because

$$|\sin A| \leq 1$$

Then, for a direct launch (i.e. without maneuvers), the orbit inclination:

$$i \geq \phi_{LAUNCH}$$

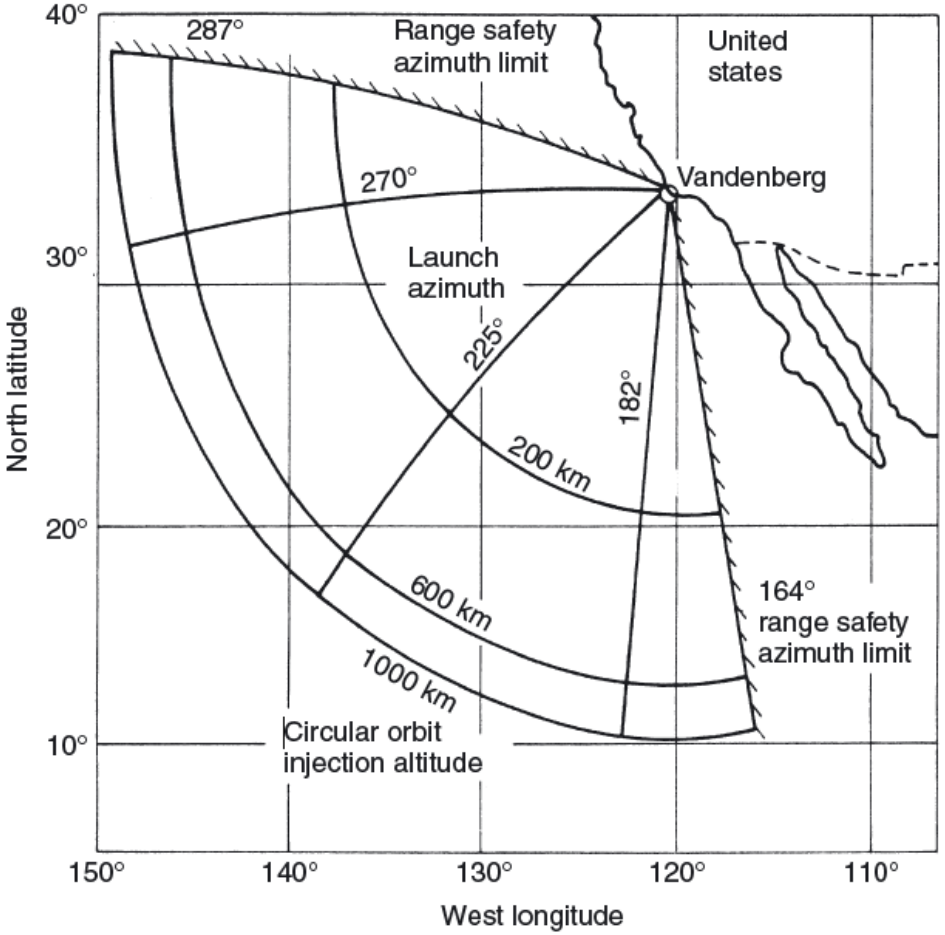


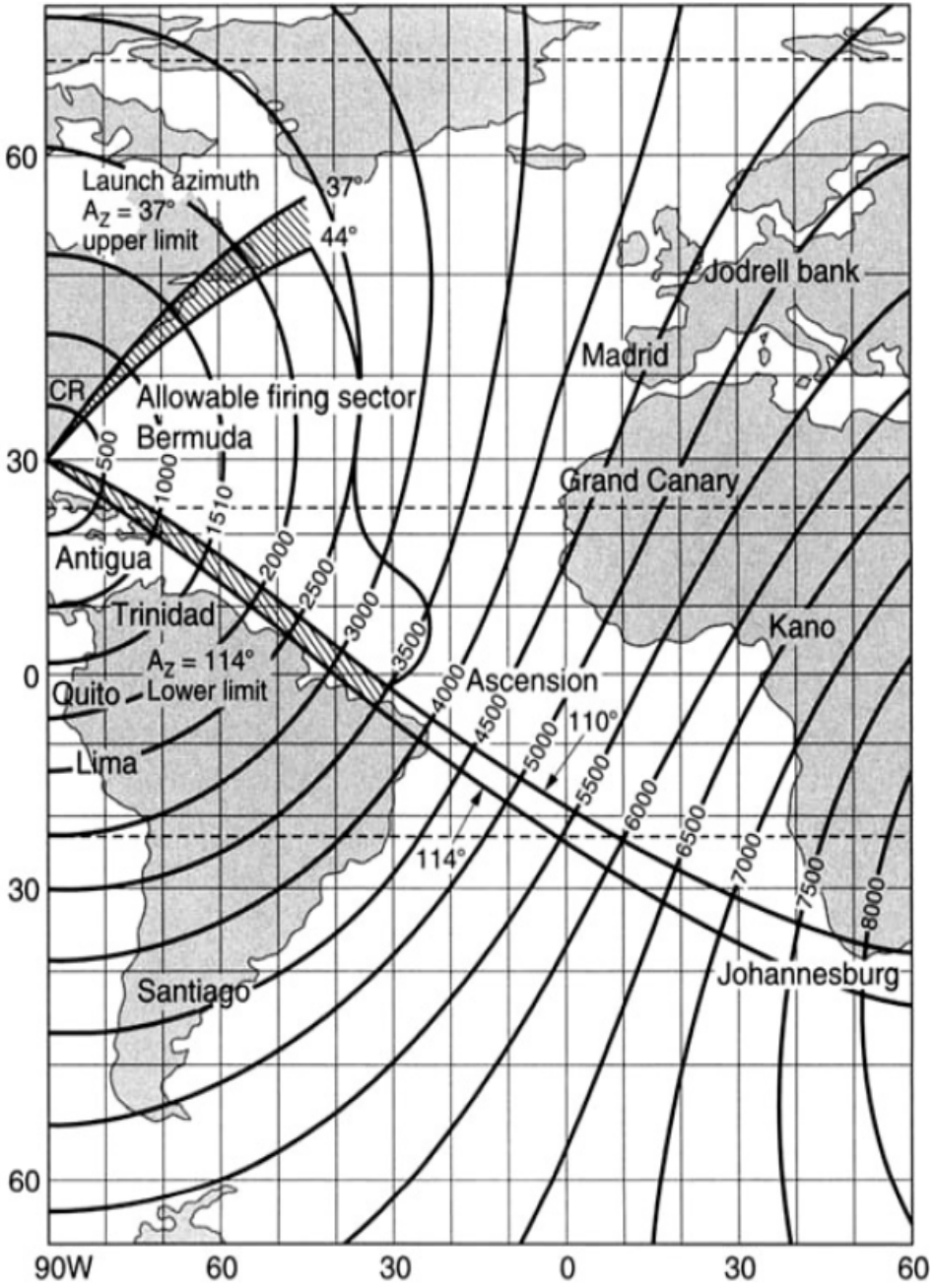


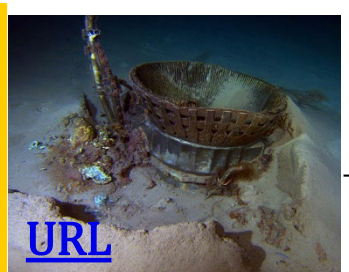
Name	Location	Lat	Lon	Min Az	Max Az	Min *i*	Max *i*
Alcantara Space Center	Alcantara, Brazil	2.283	-44.38	343	90	2.3	107.0
Baikonur Cosmodrome, aka Tyuratam Missile and Space Center	Kazakhstan	45.997	63.301	340	90	46.0	103.7
Dombarovsky Launch Site, Russia	Dombarovsky, Russia	50.803	59.512	also Yasny nearby			
Guiana Space Center	Kourou, French Guiana	5.235	-52.772	351	95	5.2	99.0
Hammaguira Space Track Range	Algeria	30.876	3.066	closed			
Jiuquan Space Center	China	40.966	100.285	101	190	42.2	97.5
Kapustin Yar Missile & Space Complex	Russia	48.648	46.017	350	107	48.6	96.6
Kodiak Launch Complex	Kodiak, Alaska	57.436	-152.338	110	220	59.6	110.2
Plesetsk Missile and Space Complex	Russia	62.922	40.661	330	90	62.9	103.2
Reagan Test Site	Kwajalein Atoll	8.720	167.732	-15	145	8.7	104.8
San Marco Launch Platform	Indian Ocean (Kenya)	-2.941	40.213	50	150	2.9	40.1
Satish Dhawan Space Center, formerly Sriharikota Launching Range	India	13.727	80.232	100	290	16.9	155.9
Svobodny Launch Complex	Russia	51.880	128.374	closed, Vostochny replacement			
Taiyuan Space Center	China	38.849	111.608	90	190	38.8	97.8
Tanegashima Space Center	Yoshinobu, Japan	30.400	130.974	90	190	30.4	98.6
Uchinoura Space Center, formerly Kagoshima Space Center	Japan	31.251	131.081	20	150	31.3	73.0
USAF Eastern Test Range	Cape Canaveral, FL	28.533	-80.575	32	111	28.5	62.3
USAF Western Test Range	Vandenburg AFB, CA	34.669	-120.61	152	202	67.3	107.9
Wallops Island	Wallops Island, VA	37.843	-75.479	90	160	37.8	74.3
Woomera, Australia	Woomera, Austrailia	-30.943	136.521	350	45	52.7	98.6
Xichang	China	28.246	102.027	94	105	31.7	28.5
Yavne Launch Facility	Israel	31.898	34.701				

Source: Figure 6-28 Curtis HD (2014) "Orbital Mechanics for Engineering Students" (3rd Edition) Elsevier.
Figure 6-9 and Table 6-2 of Vallado DA (2013) "Fundamentals of Astrodynamics and Applications" 4th Edition

Launch Azimuths

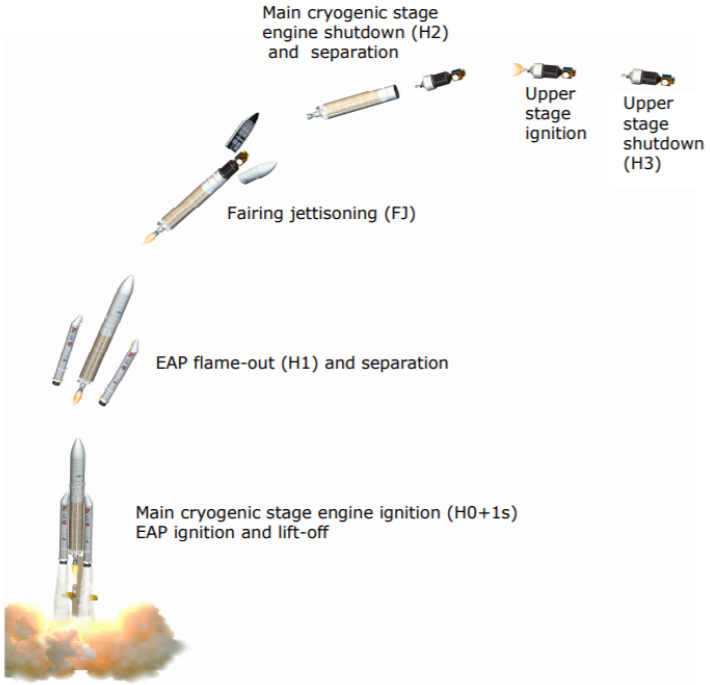


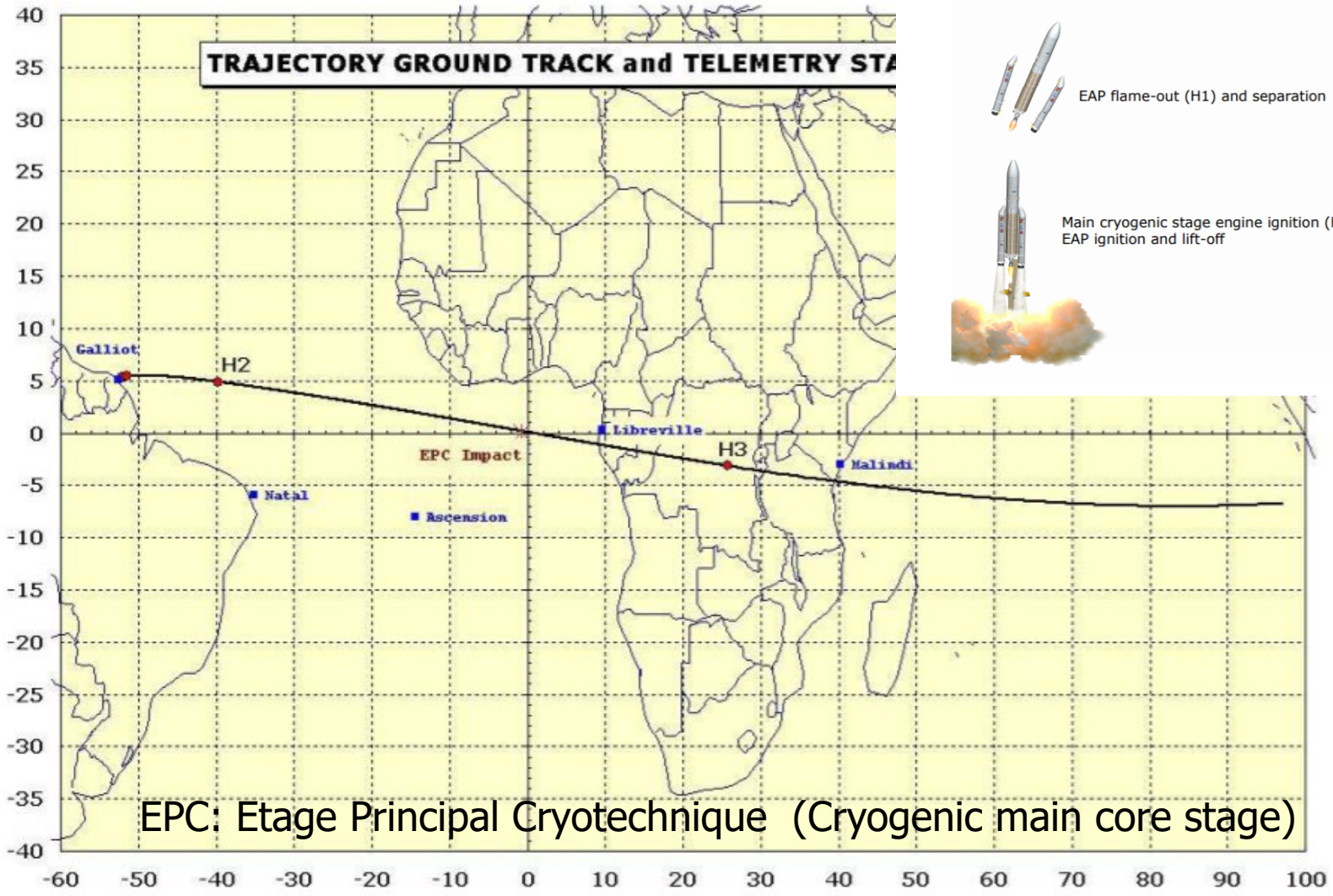




Source: Fig. 7-11 Fortescue PW, Swinerd G, Stark JPW (2011). Spacecraft systems engineering
Fig. 4-12 Pisacane VL (2005) "Fundamentals of space systems". 2nd ed. Oxford University Press

Launch Trajectory



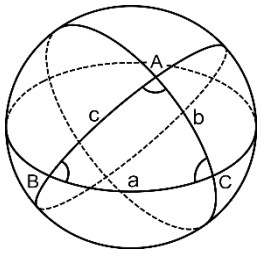


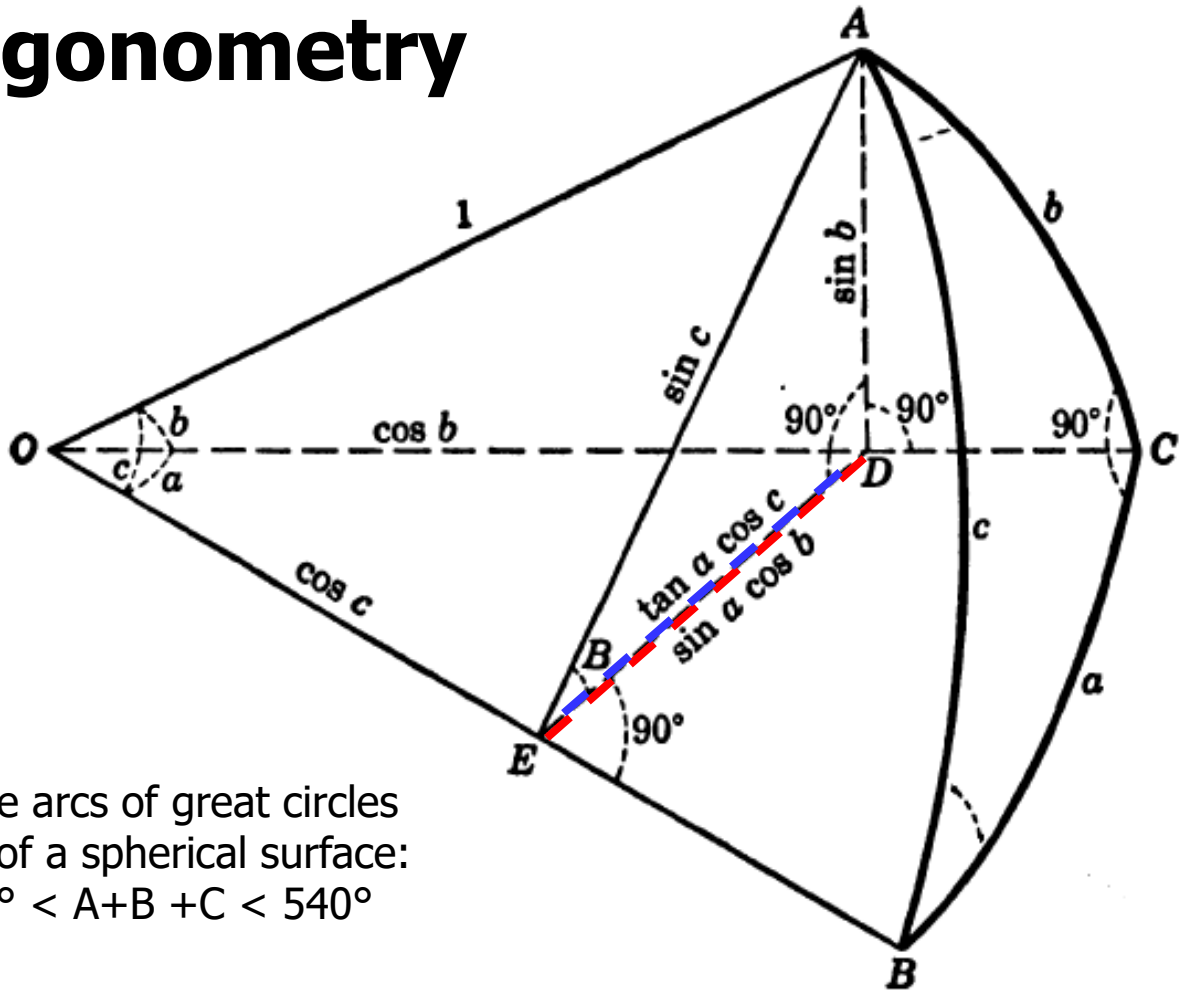
Source: Arianespace (2016) "Ariane 5 User's Manual Issue 5 Revision 2"

Spherical Trigonometry

Basic subject in navigation, geodesy,

Great circle: The locus of points in the sphere surface resulting from the intersection of a plane that passes through the sphere center O





A spherical triangle consists of three arcs of great circles that form the boundaries of a portion of a spherical surface:

- vertices: A, B, and C with 180° < A+B +C < 540°
- opposite sides: a, b, and c

For a **right-angle triangle** C=90º, there are 10 relations:

$$\cos A = \tan b \ \cot c \qquad \cos B = \tan a \ \cot c$$

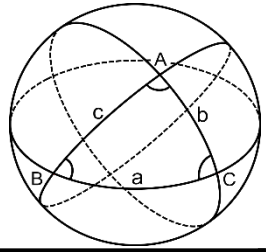
$$\mathbf{\cos A = \cos a \ \sin B} \qquad \mathbf{\cos B = \cos b \ \sin A}$$

$$\sin a = \sin c \ \sin A \qquad \mathbf{\sin b = \sin c \ \sin B} \qquad \cos c = \cos a \ \cos b$$

$$\sin a = \tan b \ \cot B \qquad \sin b = \tan a \ \cot A \qquad \cos c = \cot A \ \cot B$$

Fig. 2 of Kells ML, Kern WF, Bland JR (1940) "Plane and Spherical Trigonometry" 2nd Edition McGraw-Hill

Spherical Trigonometry



In the general case, we have a **oblique spherical triangle** $C \neq 90°$:
Given three parts of a spherical triangle, the other three can be computed:

Law	Sides	Angles
Cosine	$\cos a = \cos b \cos c + \sin b \sin c \cos A$	$\cos A = -\cos B \cos C + \sin B \sin C \cos a$
	$\cos b = \cos c \cos a + \sin c \sin a \cos B$	$\cos B = -\cos C \cos A + \sin C \sin A \cos b$
	$\cos c = \cos a \cos b + \sin a \sin b \cos C$	$\cos C = -\cos A \cos B + \sin A \sin B \cos c$
Four Parts	$\cot a \sin b = \sin C \cot A - \cos C \cos b$	$\cot A \sin B = \sin c \cot a - \cos c \cos B$
	$\cot b \sin c = \sin A \cot B - \cos A \cos c$	$\cot B \sin C = \sin a \cot b - \cos a \cos C$
	$\cot c \sin a = \sin B \cot C - \cos B \cos a$	$\cot C \sin A = \sin b \cot c - \cos b \cos A$
Sine	$\frac{\sin a}{\sin A} = \frac{\sin b}{\sin B} = \frac{\sin c}{\sin C}$	
Analogue to cosine	$\sin a \cos B = \cos b \sin c - \sin b \cos c \cos A$	$\sin A \cos b = \cos B \sin C + \sin B \cos C \cos a$
	$\sin b \cos C = \cos c \sin a - \sin c \cos a \cos B$	$\sin B \cos c = \cos C \sin A + \sin C \cos A \cos b$
	$\sin c \cos A = \cos a \sin b - \sin a \cos b \cos C$	$\sin C \cos a = \cos A \sin B + \sin A \cos B \cos c$